

Adaptive Design of Clustered Surveys and Experiments

Carlos Gonzalez-Perez*

[Click here for the most recent version](#)

September 3, 2026

Abstract

Researchers designing clustered surveys and experiments on a fixed budget must decide how many clusters to recruit and how many units to sample within each. The variance-minimizing allocation has been known since [Cochran \(1977\)](#), but it depends on the intra-cluster correlation (ICC)—a property of the very data the researcher has not yet collected. In practice, designers guess. I show that guessing is expensive: a design built on a misspecified ICC forfeits precision of the same order as the sampling variance the design exists to minimize. This paper makes the classical solution usable. I propose two pilot-based algorithms—one requiring a single design revision, one fully sequential—that learn the ICC while the experiment runs and reallocate the remaining budget optimally. Their excess variance vanishes a full order of the budget faster than that of any misspecified static design, maximin designs included, and no procedure can improve upon this rate. Adaptivity, moreover, does not compromise inference. Under a symmetry assumption on the errors, estimators remain asymptotically normal conditional on the realized design, with no data discarded and no correction. The underlying invariance principle is simple: designs driven by even functions of the data, such as sample variances, reveal nothing about the signs of the errors. Moreover, it extends beyond my algorithms and validates standard inference for popular procedures such as Neyman allocations computed from pilot data. Revisiting [Fink et al. \(2020b\)](#), I find that adaptive allocation would have reduced the variance of the study’s estimates by up to 41% at equal cost, cutting standard errors in the paper’s treatment-effect regressions by up to 19%. Finally, I demonstrate feasibility by fielding a live online survey whose design was chosen in real time by the algorithm.

1 Introduction

Consider a researcher planning a clustered survey or experiment on a fixed budget. She must choose two numbers: how many clusters to visit, and how many units to sample within each. Under the canonical model, she also knows how to choose them, as the solution has been available since [Cochran \(1977\)](#). There is a fixed cost associated with sampling a new cluster—travel, training additional staff, negotiating with local authorities—which is sunk before any unit is recruited, and a variable cost of sampling an additional unit within a cluster. The variance-minimizing split between *width* (the number of clusters, K) and *depth* (the number of units per cluster, N) then depends on a single statistical

*University of Oxford, Department of Economics. Email: carlos.gonzalezperez@economics.ox.ac.uk. I would like to thank Maximilian Kasy, Davide Viviano, Günther Fink, Rahul Singh and seminar participants at Harvard and Warwick for feedback on this manuscript.

parameter: the intra-cluster correlation (ICC), the share of total variation that is between rather than within clusters.¹ The formula is simple and the logic is settled.

She cannot use it. The ICC is a property of the data she has not yet collected, and it varies enormously across settings and outcomes: [Knox and Chondros \(2004\)](#) report estimates ranging from 0.055 to 0.451 on demographic covariates and from 0.005 to 0.059 on morbidity covariates—an order of magnitude of variation within a single population. Worse, the parameter is expensive to learn precisely when learning it matters. The difficult component entering the ICC is the between-cluster variance, which can only be estimated by sampling many clusters—the very expenditure the design is meant to economize. The designer thus faces a circularity: the optimal design requires knowing the ICC, and learning the ICC requires paying for a design. Because cluster-level costs are sunk once incurred, mistakes cannot be undone.

In practice, designers resolve the circularity by guessing. Some transport ICC estimates from prior studies in different populations, others run small pilots and treat the resulting estimate as the truth, many impose a conventional value and verify ex-post that the implied design effect is tolerable.² In summary, none of these practices features cost-aware learning, nor they claim to, as the tools for learning the ICC *within* the joint width-depth optimal design problem simply do not exist. The consequences are not innocuous. I show that any design built on a misspecified ICC incurs excess variance of order $\Theta(1/B)$, the same order as the oracle variance itself. In the application of Section 8, plausible misspecification costs up to 41% of the attainable precision. The researcher pays, in variance, an amount comparable to what her budget buys.

This paper makes the classical solution usable. I recast the width-versus-depth problem through a sequential learning framework: the experiment estimates its own ICC in early waves and reallocates the remaining budget accordingly. The contribution is threefold. First, I propose two algorithms—an explore-then-commit rule and a fully sequential one—that are simple enough to implement in the field and provably close to the oracle: their expected excess variance is of order $1/B^2$, a factor of B below the cost of guessing, and I show that no policy can do better in terms of rate (Theorems 5.4 and 5.5). Second, under a symmetry assumption on the errors, I show that inference can proceed as if the design had been fixed in advance, thus average estimators remain asymptotically normal with the usual variance—despite the design having been chosen by the data (Theorem 6.9). The result rests on a general principle which applies well beyond this problem: designs driven by *even* functions of the data, such as sample variances, reveal nothing about the signs of the errors, so conditioning on even designs does not compromise inference. This invariance extends well beyond clustered surveys, justifying standard inference for popular sampling methods such as Neyman allocations with estimated variances (Appendix D). Third, I demonstrate the method end to end: in simulations, in a semi-synthetic reconstruction of [Fink et al. \(2020b\)](#) where adaptive design recovers most of the oracle’s advantage at no additional cost, and in an actual online multi-cluster survey designed and fielded with the algorithm proposed in this paper.

¹The higher the ICC, the less an additional observation from an already-visited cluster contributes, and the more the designer should favor width over depth. In the extreme case where observations are perfectly correlated within clusters, the analyst extracts all of a cluster’s information from a single unit, hence she would sample one unit from as many clusters as the budget allows.

²The design in [Fink et al. \(2020b\)](#)—the application revisited in Section 8—used an ICC-informed power calculation estimated through a small pilot, leading to a selection of 175 villages with 22 households each. Yet, they did not optimize the allocation of sampling effort across villages and households against the relative costs of those margins. Their register only documents: “to be powered to detect an increase in farm production value of 10%” ([Fink et al., 2014](#)).

This is the first paper to establish formal optimality guarantees for an adaptive design under unknown ICC in the joint allocation problem. Starting from a minimal model with nested errors, linear costs and a hard budget, I first characterize the oracle design and the oracle variance—of order $\Theta(1/B)$ —under known ICC, following the canonical results in [Cochran \(1977\)](#). I then propose two algorithms that solve the problem with no prior knowledge of the ICC. Both share the same architecture: run a carefully-sized pilot, learn the ICC, allocate the remaining budget through plug-in rules. While Explore-then-Commit Cluster Selection (EtC-CS) implements a simple two-stage design by taking a worst-case approach in the pilot, Adaptive Cluster Selection (ACS) samples clusters sequentially, stops under a data-dependent rule, and commits to the final design. Both algorithms attain expected excess variance of order $O(1/B^2)$, a full factor of B below the oracle variance itself, and much better than any static design built on a misspecified ICC, which incur excess variance of $\Theta(1/B)$ —including the maximin design of [van Breukelen and Candel \(2015\)](#)—. By deriving matching lower bounds of $\Omega(1/B^2)$, I further show the adaptive algorithms to be rate unimprovable. Although both algorithms display bounds of the same rate, ACS outperforms EtC-CS in simulations at the expense of greater adaptivity requirements.

Data-dependent designs can compromise conventional inference. Most results in this literature therefore exclude pilot data ([Viviano, 2022](#)), re-weight the terminal estimator ([Yin and Shen, 2005](#)), rest on strong normality assumptions ([Rinott, 1978](#)), restrict conditioning to designs with degenerate limiting probabilities ([Hahn et al., 2011](#)), assess distortions through simulation ([Wittes et al., 1999](#)), or some combination of the above. My algorithms also select the design adaptively, so they could be subject to the same limitations. They are not. Under a symmetry assumption on the errors, inference on sample averages is invariant to adaptive data collection whenever the design depends only on (asymptotically) even functions of the data, such as sample variances. Asymptotic normality is then preserved with no correction and no data wastage. The result hinges on a simple principle. Under symmetry, error signs and error magnitudes are independent, so conditioning on an even design freezes the magnitudes while leaving the signs unrestricted, allowing a central limit theorem to run through the signs. The theory covers both of my algorithms, which are functions of ratios of within- to between-cluster variances, but extends well beyond the cluster setting, to popular procedures such as Neyman allocations computed from pilot variances. Unlike [Hahn et al. \(2011\)](#), it permits conditioning on design events with non-degenerate limiting probability, which is what makes conditional and unconditional inference meaningfully distinct. While the theory operates on exact evenness principles, feasible designs are never exactly even as sample variances depend on squared residuals rather than squared errors. Still, I show that the same conclusions extend to the feasible class of asymptotically even policies.

Simulations verify the inference results and delineate their limits: once symmetry or evenness fails, conditional inference can be distorted. The theoretical prescriptions on performance are also borne out in simulations and in a reconstruction of [Fink et al. \(2020b\)](#) (henceforth FJM). FJM ran a clustered experiment identifying the effects of relaxing liquidity constraints on rural Zambian households through market-competitive loans, randomly selecting 175 villages and 22 households per village. FJM approached the design with more care than is typical: they ran a small pilot survey, estimated the ICC from the sample, and implemented a design “sufficiently powered to detect an increase in farm production of 10%”. Yet neither the pilot nor the main experiment explicitly optimizes the width-depth allocation against its cost structure—and far from exceptional, this is the norm.³ As a

³If anything, by pre-estimating the ICC via a pilot, FJM display additional care relative to the large body of designs built on guesstimates borrowed from the literature.

consequence, under a plausible grid of fixed-to-variable cost ratios, FJM could have achieved substantially lower variance for the same money, or spent a substantially smaller budget at fixed precision. In particular, I show that ACS attains nearly zero excess variance, what entails a reduction in their estimator’s variance of up to 41% and a reduction in their standard errors of up to 19%. The application also illustrates how to test in practice for error symmetry—the load-bearing assumption for inference. Finally, I demonstrate that the methods are implementable in the field by actually designing and running an adaptive online survey with cluster-level fixed costs. The survey measures people’s ability to distinguish AI-generated from human-written text across languages. Each language forms a cluster with a genuine fixed cost, as outcomes are likely correlated within languages and materials must be produced and checked by paid native speakers. On top of that, each additional respondent also entails a variable cost. Given a fixed budget, the number of languages and respondents per language was chosen adaptively by ACS.

This paper connects several literatures. Survey theory has long studied cost-optimal clustered designs (Cochran, 1977; Bethel, 1989), but has rarely integrated ICC estimation within design optimization. Biostatistics, by contrast, has a long tradition on learning nuisance parameters through pilots dating back to Stein (1945). Within the internal-pilot literature (Wittes and Brittain, 1990), the closest papers to mine are Lake et al. (2002) and van Schie and Moerbeek (2014), where the authors re-estimate the ICC within a cluster trial, but each adjusts a single margin—width in the former, depth in the latter—and neither embeds a cost structure nor derives an optimality theory. In econometrics, my paper is closest to the two-wave tradition of Hahn et al. (2011) and, in structure, to Viviano (2022), from which it differs in decision space—clustered sampling rather than network experiments—and in requiring no sample splitting for inference. The performance criterion is the cluster-sampling analog of Neyman regret (Dai et al., 2023), and the inference results formalize and generalize an invariance tradition in sample-size re-estimation running from Stein (1945), where I replace normality by symmetry and extend fixed-stage adaptivity to arbitrary asymptotically even designs, in a spirit reminiscent of randomization inference under group invariance (Canay et al., 2017).

The remainder of the paper dives into the relevant literature, develops the design problem and adaptive algorithms, establishes their performance and inference guarantees, and evaluates them through simulations, the FJM counterfactual, and the prospective survey implementation. All proofs, including a standalone treatment of the general invariance result, are collected in the Appendix.

2 Literature Review

Clustered Designs in Survey Theory. Optimal multi-stage survey design under a budget constraint is a classical problem, dating back to Tschuprow (1923), Hansen and Hurwitz (1942) and Kish (1965).⁴ In its simplest form, a multi-stage survey partitions the population into non-overlapping clusters, samples K clusters at random, and then samples N units at random within each selected cluster. The objective of this literature is the one I adopt here: to minimize the variance of a population estimator through the allocation of sampling effort across clusters and units. My adaptive designs are benchmarked against the cost-optimal oracle allocation derived by Cochran (1977) under known variance components.⁵ According to Ballin and Barcaroli (2013), the literature divides into three strands: designs that take the clusters as given and optimize unit selection, designs that opti-

⁴See Biemer and Lyberg (2003) for a modern textbook treatment.

⁵Raudenbush (1997) and Moerbeek et al. (2000) obtain analogous oracle allocations for cluster designs.

mize the cluster partition, and designs that optimize both jointly. I take the partition as given and therefore belong to the first strand, whose canonical references are [Cochran \(1977\)](#) and [Bethel \(1989\)](#). Since Bethel’s algorithm, this sub-literature has extended his logic to multi-domain, multi-variable and multi-objective settings, with an emphasis on algorithmic performance ([Barcaroli et al., 2022](#)). Two features of this literature deserve emphasis. It has developed considerable technical machinery, and identifies the intra-cluster correlation as a central input to the optimal design.⁶ Yet it almost never estimates the ICC *inside* the design problem. [Barcaroli et al. \(2022\)](#) assume it can be recovered from auxiliary data, while applied guidance alternates between recruiting as many clusters as logistics permit ([Tam, 2005](#)) and fixing the number of clusters *ex ante* ([Sedgwick, 2015](#)). Neither prescription is optimal in general.

Adaptive Designs and ICC Re-estimation in Biostatistics. The idea of using an initial sample to learn a nuisance parameter that governs the design has a long history in statistics and biostatistics. It begins with [Stein \(1945\)](#), who proposes a two-stage rule for Gaussian data with unknown variance and preserves the t distribution of the second-stage statistic by choosing the total sample size as a function of the first-stage variance estimate. [Rinott \(1978\)](#) and [Dudewicz and Dalal \(1975\)](#) extend this logic to best-group identification, and [Yin and Shen \(2005\)](#) embed ICC re-estimation in a fully sequential GEE trial, where validity is restored through a martingale-weighted combination of block-wise statistics. Valid inference in this literature is typically recovered by discarding data, reweighting the estimator, or both. I show that, under symmetry, neither is necessary when the design depends on even functionals of the data such as sample variances.

An important literature revived this idea under the label *internal pilot*. [Wittes and Brittain \(1990\)](#) and [Wittes et al. \(1999\)](#) argue that nuisance parameters can be estimated mid-trial and the design revised accordingly, with the final analysis conducted on all data as if the design had been fixed *ex ante*. Much of this literature assesses performance and inference through simulation under Gaussianity, reporting type I error inflation that varies with the timing and size of the pilot, without formal guarantees. [Lake et al. \(2002\)](#) and [van Schie and Moerbeek \(2014\)](#) bring this idea to ICC estimation in cluster-randomized trials, and are the closest antecedents of my design problem. Two differences stand out. They optimize a single margin—width in [Lake et al. \(2002\)](#), depth in [van Schie and Moerbeek \(2014\)](#)—whereas I optimize (K, N) jointly. Also, they do not impose a cost structure and do not derive an optimality theory. A parallel literature acknowledges ICC uncertainty but holds the design fixed, either by placing Bayesian priors on the ICC ([Spiegelhalter, 2001](#); [Turner et al., 2005](#)) or by adopting maximin designs ([van Breukelen and Candel, 2015, 2018, 2021](#)). These approaches optimize against ICC uncertainty within a static design class, but do not benchmark the resulting loss against the known-ICC oracle. I establish worst-case excess variance bounds of $O(1/B^2)$ that cannot be improved in rate, and I show that my algorithms asymptotically dominate every misspecified static policy over the ICC range—including the maximin design of [van Breukelen and Candel \(2015\)](#), whose worst-case excess variance is $\Theta(1/B)$ (Lemma 5.2). To my knowledge, this is the first paper to treat the clusters-versus-units allocation as a sequential learning problem, and the first to provide formal optimality guarantees for an adaptive cluster design.

The Econometrics of Experimental Design. My work relates thematically to the econometric literature on optimal but non-adaptive design through stratification and covariate adjustment ([Cytryn-](#)

⁶See [Cicchitelli et al. \(1992\)](#) on the role of the ICC, [Knox and Chondros \(2004\)](#) for evidence on its variability across domains, and [Campbell et al. \(2004\)](#) on reporting practice.

baum, 2021; Bai, 2022; Cytrynbaum, 2024; Bai et al., 2024). That literature shares my variance-minimization objective, and occasionally my attention to sampling costs, but neither address the cluster-versus-unit margin nor the learning problem. Closer in structure is the two-wave tradition initiated by Hahn et al. (2011), in which a pilot estimates the parameters that enter the optimal design: treatment probabilities in Hahn et al. (2011), and optimal stratification in Chambaz et al. (2015) and Tabord-Meehan (2023). Cai and Rafi (2024) show that Neyman allocations computed from *small* pilots can be strictly worse than balanced randomization. Nearest to my setting is Viviano (2022), who designs two-wave experiments under network interference by minimizing estimator variance using pilot-estimated variance components, and derives regret guarantees, asymptotic normality, and a characterization of the regret-minimizing pilot size. I share the pilot-then-optimize architecture and the regret criterion, but my decision space is the clustered sampling design and my inference results require no data exclusion: Viviano (2022) must remove both the pilot units and their network neighbors from the main experiment to keep the design unconfounded, whereas I use the full sample. My work also connects to adaptive experimentation and policy learning in economics (Kasy and Sautmann, 2021; Adusumilli, 2026; Cesa-Bianchi et al., 2025; Li et al., 2026).

Neyman Regret in Sequential Experiments. My performance criterion—expected excess variance relative to the variance-minimizing oracle design—is the clustered-sampling analogue of the *Neyman regret* introduced by Dai et al. (2023) for adaptive treatment allocation. Working in a design-based framework, they attain $\tilde{O}(\sqrt{T})$ Neyman regret through clipped online gradient descent and show that explore-then-commit designs can suffer linear Neyman regret when early units are unrepresentative.⁷ Subsequent work attains logarithmic rates in stochastic environments (Neopane et al., 2024), and does so even under the stronger notion of Neyman regret in Neopane et al. (2025). Both rates are consistent with my guarantees.

Inference with Adaptive Designs. Adaptive data collection frequently invalidates conventional inference procedures: adaptivity can bias sample means (Nie et al., 2018) and destroy the asymptotic normality of standard estimators (Hadad et al., 2021).⁸ The remedies fall into two traditions: an *adjustment* tradition, which reweights the estimator to restore martingale central limit behavior (Hadad et al., 2021; Zhang et al., 2021; Shen and Fisher, 1999; Yin and Shen, 2005); and an *invariance* tradition, which shows that when the design depends on the data only through the sample variance, inference on the mean is exactly unaffected under normality assumptions (Stein, 1945; Rinott, 1978; Dudewicz and Dalal, 1975). The same invariance is used implicitly in the internal pilot and blinded sample size re-estimation literatures (Gould and Shih, 1992; Kieser and Friede, 2003). Proposition 6.7 and Theorem 6.9 in this paper belong to the latter, but I generalize this route in three directions: from normality to symmetry of the errors, from the sample variance to any asymptotically even functional of the data, and from finite-stage adaptivity to arbitrary designs. The resulting class contains my algorithms and, more broadly, Neyman allocations computed from pilot variances. For instance, the optimal second-stage propensity score of Hahn et al. (2011) is a ratio of pilot-estimated conditional standard deviations, and it is therefore even.⁹ In addition, the contrast with the class of conditioning events in Hahn et al. (2011) is instructive. While their validity rests on the second-stage design converging

⁷Their linear-regret example is specific to the design-based environment they study. I work in the stochastic *iid* setting, in which explore-then-commit designs such as EtC-CS do not incur linear regret.

⁸See Wager (2024) for a textbook treatment.

⁹Conditional-mean heterogeneity enters the semiparametric efficiency bound additively and does not depend on the assignment probability, so it drops out of the optimization. The optimal propensity depends on the pilot only through second moments.

to a deterministic limit, my theory allows for conditioning on events with non-degenerate limiting probability. The symmetry assumption plays a role reminiscent of group invariance in randomization inference (Canay et al., 2017). The boundary of the result is equally informative: adaptation through *odd* functionals of the data, such as estimated effects or futility rules, reintroduces the pathologies that motivate the adjustment tradition.

Other Connections. My work is more distantly related to the treatment of nested errors in panel data (Wooldridge, 2003), to standard error adjustment in clustered experiments (Abadie et al., 2023), and to the optimal-design tradition of Kiefer and Wolfowitz (1960), which underpins modern algorithms for linear bandits.¹⁰ These connections are conceptual rather than methodological.

3 Cluster Design Problem

3.1 Model

Consider a population with $\bar{K}$ equally sized clusters, each containing $\bar{N}$ units. The designer samples $K \ll \bar{K}$ clusters, indexed $k = 1, \dots, K$ and $N^k \ll \bar{N}$ units from cluster k indexed $i = 1, \dots, N^k$. Let the outcome of interest y_{ki} follow an additive model with nested errors,

$$y_{ki} = \mu + u_k + \varepsilon_{ki} \quad (1)$$

where μ is the population mean and estimand, u_k is the cluster-level error and ε_{ki} is the unit-level error.

Assumption 3.1. *Error Distributions.* $(u_k)_k$ is an iid sequence with $\mathbb{E}[u_k] = 0$, $\text{Var}(u_k) = \tau^2 > 0$; $(\varepsilon_{ki})_{k,i}$ is an iid sequence with $\mathbb{E}[\varepsilon_{ki}] = 0$, $\text{Var}(\varepsilon_{ki}) = \sigma^2 > 0$; and $(u_k)_k \perp\!\!\!\perp (\varepsilon_{ki})_{k,i}$.

The design is governed by the variance ratio $\lambda_0 = \sigma^2/\tau^2$, which maps one-to-one to the intra-cluster correlation, $\text{ICC} = \tau^2/(\tau^2 + \sigma^2) = 1/(1 + \lambda_0)$.¹¹ Throughout, λ_0 is unknown to the designer while the cost parameters are known.

Assumption 3.2. *Compact Parameter Space.* Let $\omega = (\tau^2, \lambda_0)$ be a strictly interior point of the compact set $\Omega = [\tau_m^2, \tau_M^2] \times [\lambda_m, \lambda_M]$ where $\tau_m^2 > 0, \lambda_m > 0$.

3.2 Variance Minimization in Survey Design

The designer chooses how many units N^k to sample from each cluster—and hence, implicitly, how many clusters to sample—subject to linear costs and a hard budget B . In the homogeneous-variance setting the natural estimator of μ is the average of averages

$$\hat{\mu} = \frac{1}{K} \sum_{k=1}^K \hat{\mu}_{N^k}^k, \quad \hat{\mu}_{N^k}^k = \frac{1}{N^k} \sum_{i=1}^{N^k} y_{ki}$$

where $K = \sum \mathbb{1}(N^k > 0)$ is the number of sampled clusters and $\hat{\mu}_{N^k}^k$ is the sample average of cluster k . Under multi-stage randomization $\hat{\mu}$ is unbiased for μ , so minimizing its mean squared

¹⁰See Lattimore and Szepesvári (2020) for a textbook treatment.

¹¹An upcoming extension introduces a more general model with heterogeneous cluster variances τ_k^2 .

error reduces to minimizing its variance. Appendix A shows that, for any fixed number of sampled clusters, the variance-minimizing allocation is balanced.¹² I therefore restrict attention to designs with $N^k \in \{0, N\}$ for all k . Under cross-cluster independence the designer’s problem becomes

$$\min_{K \geq 2, N \geq 2} W(K, N) = \frac{\tau^2}{K} \left(1 + \frac{\lambda_0}{N}\right) \quad \text{s.t. } K(F + VN) \leq B \quad (2)$$

where $W(K, N) = \text{Var}(\hat{\mu}_{K,N})$, F is the fixed cost of sampling an additional cluster, V is the variable cost of sampling an additional unit, and $B \geq 2(F + 2V)$ is the hard budget.¹³ For the performance analysis I work with the continuous relaxation of (2), treating (K, N) as real-valued.¹⁴ One further restriction rules out corner solutions,

Assumption 3.3. *Interior Solution.* $\lambda_m F/V \geq 4$ and $B \geq 2(F + \sqrt{\lambda_M FV})$.¹⁵

Under Assumptions 3.1, 3.2 and 3.3 the bound constraints never bind, and solving (2) from the first-order conditions delivers Cochran’s allocation (Cochran, 1977),

$$N^*(\lambda) = \sqrt{\lambda F/V}, \quad K^*(\lambda) = \frac{B}{F + VN^*(\lambda)}, \quad W_0 = \frac{\tau^2 \left(\sqrt{V\lambda_0} + \sqrt{F}\right)^2}{B} = \Theta(1/B) \quad (3)$$

I refer to $\lambda \mapsto (K^*(\lambda), N^*(\lambda))$ as the optimal map and to $W_0 = W(K^*(\lambda_0), N^*(\lambda_0))$ as the oracle variance—which depends on the unknown variance ratio λ_0 . The integer version of the oracle variance provides the natural benchmark for policies without access to λ_0 . Despite its name, the optimal map is *optimal* only when the designer inputs the true $\lambda = \lambda_0$. In general, applying the optimal map to a misspecified λ can inflate the resulting variance substantially. Figure 1 plots the integer optimal map and the excess variance it induces relative to the oracle over a grid of λ values. Even mild misspecification of λ can have large consequences for the induced design and the resulting excess variance.

¹²Balancedness is specific to the unweighted average of averages under homogeneous variances. It fails for the GLS-weighted average with heterogeneous cluster variances studied in an upcoming extension.

¹³The bounds $K \geq 2$, $N \geq 2$ are imposed for technical reasons that affect the estimation of unknown parameters, see Section 4. In practice the restriction is with little loss, as nearly all clustered surveys and experiments consider larger samples anyway.

¹⁴In simulations, all algorithms are mapped to integer allocations through $(\lfloor K^\pi \rfloor, \lfloor N^\pi \rfloor)$. This rounding criterion guarantees implementability, eases cross-algorithm comparison, and removes differences stemming from integer approximation. In practice, alternative terminal budget allocations may achieve superior performance.

¹⁵The assumption is stated in primitives and it is equivalent to requiring $N^*(\lambda) \geq 2$ and $K^*(\lambda) \geq 2$ for every $\lambda \in [\lambda_m, \lambda_M]$, with $K^*(\lambda), N^*(\lambda)$ defined in (3).

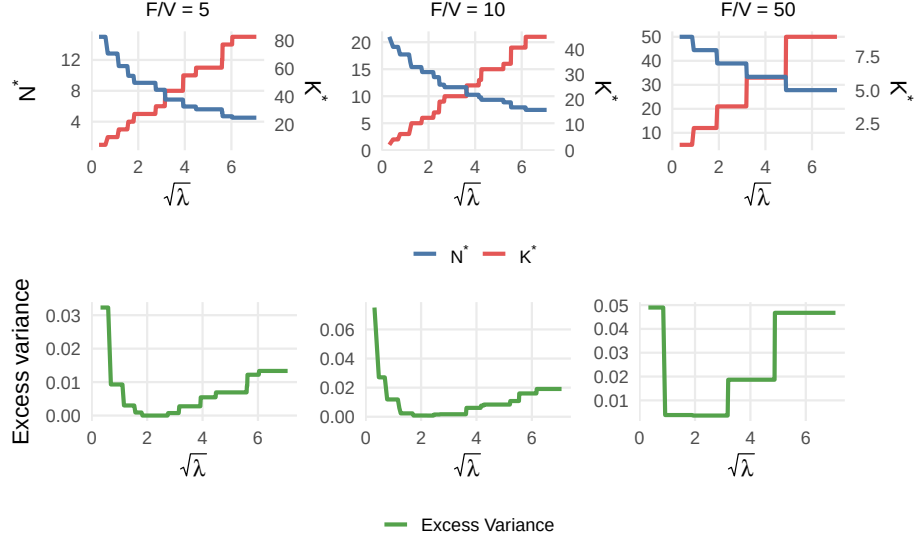


Figure 1: Designs and Excess Variance under λ misspecification

Notes: $\lfloor N^*(\lambda) \rfloor$ (top-left), $\lfloor K^*(\lambda) \rfloor$ (top-right) and Excess Variance (bottom) for different F/V ratios. $\mu = 0$, $\omega = (1, 5)$, $B = 1000$, $V = 2$.

Clustered Experimental Design. The survey design problem in (2) extends naturally to the two canonical clustered experimental designs: treatment assigned at the cluster level, where half the clusters are assigned to treatment and half to control; and clusters as blocks, where within each cluster half the units are assigned to treatment and half to control under no interference. To do so, the model in (1) is extended to the potential outcomes framework via $y_{ki}(d, \mathbf{d}_k) = \mu_d + u_k^d + \varepsilon_{ki}^d$, where $d \in \{0, 1\}$ is unit i 's own treatment and $\mathbf{d}_k$ collects the assignments of the remaining units in cluster k . In every case, the design variance takes the form $W(K, N) = \frac{c}{KN} (\tau^2 + \sigma^2/N)$ for some constant c , under the substitutions in Table 1. Appendix A introduces both settings in greater detail.

Problem	Estimand	Estimator	c	τ^2	σ^2
Survey	$\mathbb{E}[y_{ki}]$	$\frac{1}{KN} \sum_k^K \sum_i^N y_{ki}$	1	τ^2	σ^2
Cluster-Level Treatment	$\mathbb{E}[y_{ki}(1, \mathbf{1}) - y_{ki}(0, \mathbf{0})]$	$\frac{2}{K} (\sum_{k:D_k=1} \hat{\mu}_N^k - \sum_{k:D_k=0} \hat{\mu}_N^k)$	2	$\tau_1^2 + \tau_0^2$	$\sigma_1^2 + \sigma_0^2$
Clusters as Blocks	$\mathbb{E}[y_{ki}(1) - y_{ki}(0)]$	$\frac{2}{KN} \sum_k^K (\sum_{i:D_{ki}=1} y_{ki} - \sum_{i:D_{ki}=0} y_{ki})$	1	$\tau_1^2 + \tau_0^2 - 2\tau_{01}$	$2(\sigma_1^2 + \sigma_0^2)$

Table 1: Canonical Clustered Design Problems

Notes: Under these substitutions, all three problems share the optimal-map functional form in (3). (τ_d^2, σ_d^2) are the cluster- and unit-level variances of potential outcome d , and $\tau_{01} = \text{Cov}(u_k^1, u_k^0)$. Under cluster-level assignment $D_{ki} = D_k$ and the estimand is the TATE; under clusters as blocks the estimand is the ATE, identified under no interference. The blocks design admits a degenerate corner: when $u_k^1 = u_k^0$, within-cluster differencing removes the cluster-level noise entirely ($\tau^2 = 0$), and the variance is minimized by sampling a single cluster. Assumption 3.2 excludes this case by requiring $\tau_m^2 > 0$.

4 Adaptive Sampling under Unknown ICC

4.1 Estimating the Variance Ratio

Optimal design under an unknown λ_0 requires some form of estimation, which I now introduce. Fix $K \geq 2$ and $N \geq 2$, and define

$$S_{k,N}^2 = \frac{1}{N-1} \sum_{i=1}^N (y_{ki} - \hat{\mu}_N^k)^2, \quad \hat{w}_{K,N}^2 = \frac{1}{K} \sum_{k=1}^K S_{k,N}^2, \quad \hat{b}_{K,N}^2 = \frac{1}{K-1} \sum_{k=1}^K (\hat{\mu}_N^k - \hat{\mu})^2$$

$$\hat{\tau}_{K,N}^2 = \Pi_{[\tau_m^2, \tau_M^2]} \left(\hat{b}_{K,N}^2 - \frac{\hat{w}_{K,N}^2}{N} \right), \quad \hat{\lambda}_{K,N} = g(\hat{w}_{K,N}^2, \hat{b}_{K,N}^2) = \Pi_{[\lambda_m, \lambda_M]} \left(\frac{\hat{w}_{K,N}^2}{\hat{\tau}_{K,N}^2} \right)$$

where $\hat{w}^2$ is the within-cluster sample variance, $\hat{b}^2$ is the between-cluster sample variance, $\Pi_{[a,b]}(x) = \min\{b, \max\{a, x\}\}$ is the clip operator and $\hat{\lambda}$ is the (clipped) sample estimator of λ . Lemma 4.1 shows consistency of these estimators under different asymptotic regimes.

Lemma 4.1. *Consistency and Asymptotic Regimes.* Let $\theta_N^2 := \tau^2 + \frac{\sigma^2}{N}$ be the variance of a cluster sample mean, and let $\xrightarrow[T]{x}$ denote convergence in x under $T \rightarrow \infty$ asymptotics. Under Assumptions 3.1 and 3.2: (i) $\hat{w}_{K,N}^2 \xrightarrow[N]{a.s.} \sigma^2$ (with $K \geq 1$ fixed) and $\hat{w}_{K,N}^2 \xrightarrow[K]{a.s.} \sigma^2$ (with $N \geq 2$ fixed); (ii) $\hat{b}_{K,N}^2 \xrightarrow[K]{a.s.} \theta_N^2$ (with $N \geq 2$ fixed); and (iii) $\hat{\lambda}_{K,N} \xrightarrow[K]{a.s.} \lambda_0$ (with $N \geq 2$ fixed).

Width is the binding margin for design and for learning alike. Both learning and optimal design via variance minimization require asymptotic sampling of clusters, an observation that shapes the adaptive algorithms and explains their success. *Design.* For any $\tau^2 > 0$, $\hat{\mu}$ is consistent for μ if and only if $K \rightarrow \infty$. Intuitively, increasing N averages away the individual-level noise ε_{ki} , but only additional clusters average away the cluster-level noise u_k . The optimal map encodes this asymmetry, since $\lim_B K^*(\lambda) = \infty$ for any fixed λ while $N^*(\lambda)$ does not depend on B . *Learning* presents the same asymmetry. By Lemma 4.1, the within-cluster variance can be consistently learned under either N - or K -asymptotics, whereas the between-cluster variance—and therefore λ_0 —can be consistently learned only under K -asymptotics. Three consequences follow. First, any asymptotically optimal algorithm must send K to infinity while keeping N bounded; second, the alignment between the design and learning margins makes the $O(1/B^2)$ excess-variance rate attainable in a way that I make explicit in Section 5; and third, the alignment is only asymptotic: at finite budgets, learning and design compete, as sampling additional clusters improves estimation of λ_0 but risks committing resources to width beyond what the oracle allocation would prescribe. I expand on the latter consequence below.

4.2 Pilot-Based Policies and Implementability

Taking inspiration from the optimal map in (3), I focus on a policy class that I refer to as pilot-based policies.

Definition 4.2. *Pilot-Based Policies.* Let (K_{pB}, N_{pB}) be the possibly random number of clusters and units per cluster sampled in a first stage or pilot, and write $\hat{\lambda}_{pB} = \hat{\lambda}_{K_{pB}, N_{pB}}$. A policy π is pilot-based if its terminal design is the optimal map evaluated at the pilot estimate,

$$K_B^\pi = K^*(\hat{\lambda}_{pB}), \quad N_B^\pi = N^*(\hat{\lambda}_{pB})$$

Throughout, the pilot depth is held fixed at $N_{pB} = N_p \geq 2$, independent of B , and only the pilot width is allowed to vary with the budget or with the data. Because $K^*(\lambda)$ is proportional to the budget while $N^*(\lambda)$ is budget-free, the two design margins must be tracked on different scales. Define the scaled width $\nu_B^\pi = K_B^\pi/B$ and the oracle constants $\nu_0 = K^*(\lambda_0)/B$ and $N_0 = N^*(\lambda_0)$, neither of which depends on the budget, i.e. by (3) $\nu_0 = (F + \sqrt{FV\lambda_0})^{-1}$. It will also be convenient to define the two extreme widths and depths,

$$K_M = \lfloor K^*(\lambda_M) \rfloor, \quad K_m = \lfloor K^*(\lambda_m) \rfloor, \quad N_m = \lfloor N^*(\lambda_m) \rfloor, \quad N_M = \lfloor N^*(\lambda_M) \rfloor \quad (4)$$

Since $K^*(\cdot)$ is strictly decreasing and $N^*(\cdot)$ strictly increasing in λ , K_M is the narrowest and N_m the shallowest design the optimal map can ever prescribe on $[\lambda_m, \lambda_M]$, while K_m is the widest and N_M the deepest.

Crucially, *not every pilot-based policy can be run*. Clusters and units cannot be unsampled, and the pilot observations are retained as part of the terminal sample, so a pilot-based policy is realizable only if the pilot is nested within the design it prescribes. In the case of pilot-based policies, this property reunites the partial alignment between learning and variance minimization discussed in the previous section.

Definition 4.3. *Implementability.* A pilot-based policy π is implementable if its pilot is almost surely nested within its terminal design,

$$K_{pB} \leq K_B^\pi = K^*(\hat{\lambda}_{pB}), \quad N_p \leq N_B^\pi = N^*(\hat{\lambda}_{pB}) \quad a.s.$$

Lemma 4.4. *Implementable Policies with Diverging Pilots Converge to the Oracle Design.* Let π be an implementable pilot-based policy with pilot design (K_{pB}^π, N_p^π) , and suppose $K_{pB}^\pi \xrightarrow{p} \infty$. Then, under Assumptions 3.1, 3.2 and 3.3, $\hat{\lambda}_{pB} \xrightarrow{p} \lambda_0$, and therefore¹⁶

$$|\nu_B^\pi - \nu_0| \xrightarrow{p} 0, \quad |N_B^\pi - N_0| \xrightarrow{p} 0$$

Two remarks on the statement. First, the two margins of convergence in Lemma 4.4 are set on two different scales. The oracle depth N_0 is budget-free, so depth may indeed converge at the unit scale. For width, no unit-scale statement is available: $K^*(\lambda_0)$ is itself proportional to B , so the design and its target diverge together and only the budget-normalized ratio settles. I expand this issue in further detail in Section 6 where the scale of convergence will enter decisively. The takeaway from Lemma 4.4 is that implementable pilot-based policies which sample a diverging number of clusters within the pilot induce the optimal design in the limit (at the relevant scale). Second, Lemma 4.4 presents learning as an ally of optimal design, but at finite budgets the two collide. The lemma rewards wide pilots through sharper estimation $\hat{\lambda}_{pB} \rightarrow \lambda_0$, but the implementability requirement caps the pilot by the very design the pilot produces. Observe that the implementability constraint depends on the estimate $\hat{\lambda}_{pB}$, which is unavailable until the pilot has been sampled. So, a designer who explores aggressively sharpens the identification of λ_0 but risks committing resources that the terminal design cannot absorb. The two algorithms below resolve this circularity in different ways—the first by making the pilot conservative

¹⁶Write $N_0 = \lfloor N_0 \rfloor$. If the terminal depth is rounded to an integer and $N_0 \notin \mathbb{N}$, the second statement is equivalent to $\mathbb{P}(\lfloor N_B^\pi \rfloor = N_0) \rightarrow 1$.

enough that implementability can never bind, the second by monitoring the online estimator and stopping the moment the pilot is about to oversample.

4.3 Explore-then-Commit Cluster Selection

The simplest resolution is to choose a pilot small enough in both margins so that implementability holds regardless of what the pilot finds. As presented in (4), the conservative choice is given by the narrow-and-shallow pilot $(K_{pB}, N_p) = (K_M, N_m)$. Since the clipped estimator satisfies $\hat{\lambda}_{pB} \in [\lambda_m, \lambda_M]$ by construction, Assumptions 3.2 and 3.3 give

$$K_{pB} = K_M \leq K^*(\lambda_M) \leq K^*(\hat{\lambda}_{pB}) = K_B^\pi, \quad N_p = N_m \leq N^*(\lambda_m) \leq N^*(\hat{\lambda}_{pB}) = N_B^\pi$$

so implementability holds almost surely. I refer to this two-stage policy as Explore-then-Commit Cluster Selection (EtC-CS).

Algorithm 1 EtC-CS

Input B, F, V, Ω

Sample N_m units from each of K_M clusters

observe $(y_{ki})_{i \leq N_m, k \leq K_M}$

compute $\hat{\lambda}_p = \hat{\lambda}_{K_M, N_m}$

Set $(K_B^{\text{EtC-CS}}, N_B^{\text{EtC-CS}}) = (K^*(\hat{\lambda}_p), N^*(\hat{\lambda}_p))$

4.4 Adaptive Cluster Selection

While theorists emphasize the superior performance of adaptive designs, practitioners often face logistical obstacles that prevent their full deployment in the field (Chow and Corey, 2011). In that sense, the simplicity of EtC-CS is an asset: it attains rate-optimal performance guarantees (Theorem 5.4) with a single design update. Nevertheless, the reader may wonder what could be achieved were the designer given access to further adaptivity.

Example 4.5. Consider a researcher choosing a sampling width in $[20, 60]$, i.e. for the given budget, $K_M = 20$ and $K_m = 60$. EtC-CS would then collect a pilot of 20 clusters, compute $\hat{\lambda}_p = \hat{\lambda}_{20, N_m}$, and implement $(K^*(\hat{\lambda}_p), N^*(\hat{\lambda}_p))$. Suppose the design prescribed by the pilot is $K^*(\hat{\lambda}_p) = 50$. Committing immediately would be wasteful. The researcher knows she will recruit thirty more clusters in any case, so she could easily sharpen her estimate of λ_0 on some of them before fixing the terminal design, at negligible risk of oversampling.

Building on this heuristic, I propose an algorithm that samples clusters sequentially and commits according to a data-dependent stopping rule. I refer to this algorithm as Adaptive Cluster Selection (ACS), whose pilot width is determined by the stopping time

$$s = \inf \{k \in \mathbb{N} : k \geq K_M \text{ and } k + 1 > K^*(\hat{\lambda}_{k, N_m})\} \quad (5)$$

In words, the algorithm samples N_m units from one additional cluster at a time, until recruiting one more cluster would constitute oversampling according to the online estimate of the variance ratio. Holding N_p fixed and exploring only along the width margin keeps the rule simple while preserving the statistical guarantees of Lemma 4.4.

Two observations on s . First, $s \in [K_M, K_m]$ almost surely.¹⁷ Second, the designer could maximize the cost-adjusted marginal return to exploration by considering richer exploration paths; deciding at each step whether to sample the next unit from a previously visited cluster or from a fresh cluster. Such a path is not only computationally intractable but also impractical to administer in the field. More importantly, concentrating exploration along the width margin is the right simplification. Depth and width are not symmetric inputs to learning: increasing N_p lowers the asymptotic variance of $\hat{\lambda}$ only through a constant, whereas increasing K_p improves the rate at which $\hat{\lambda}$ concentrates ($K_p^{-1/2}$ for any fixed N_p).¹⁸ This makes width the natural margin along which to expand exploration asymptotically.

Algorithm 2 ACS

Input B, F, V, Ω

Sample N_m units from each of K_M clusters

observe $(y_{ki})_{i \leq N_m, k \leq K_M}$

set $k = K_M$ and **compute** $\hat{\lambda}_k = \hat{\lambda}_{k, N_m}$

while $k + 1 \leq K^*(\hat{\lambda}_k)$

sample N_m units from cluster $k + 1$ and **observe** $(y_{k+1,i})_{i \leq N_m}$

update $k = k + 1$ and **recompute** $\hat{\lambda}_k = \hat{\lambda}_{k, N_m}$

Set $s = k$ and **recover** $\hat{\lambda}_{\text{ACS}} = \hat{\lambda}_{\max\{s-1, K_M\}, N_m}$

Set $(K_B^{\text{ACS}}, N_B^{\text{ACS}}) = (K^*(\hat{\lambda}_{\text{ACS}}), N^*(\hat{\lambda}_{\text{ACS}}))$

ACS opens with the same worst-case pilot as EtC-CS, then recruits clusters one at a time and stops just before oversampling according to the online estimate $\hat{\lambda}_k$. It therefore uses weakly more information than EtC-CS without compromising implementability. The main difference between the two algorithms is that while EtC-CS may recruit a large number of clusters after committing, ACS concentrates nearly all cluster exploration within the pilot. Since λ_0 estimation depends critically on the width margin—which is also the expensive dimension in practice—moving clusters into the pilot improves estimation of the optimal design at no additional sampling cost. The gain does not change the order of the excess-variance bounds derived in Section 5, but it is visible at small budgets in the simulations of Section 7. The improvement in performance is, however, not free: ACS requires the designer to revise the sample size after every cluster, so the applied designer must weigh the logistic inconvenience against the expected gain when choosing between the two.

Remark 4.6. *Committing One Cluster Early.* ACS estimates λ_0 from $\max\{s-1, K_M\}$ clusters rather than all s . By definition of the stopping time, $s+1 > K^*(\hat{\lambda}_{s, N_p})$, and if cluster s was explored at all then $s \leq K^*(\hat{\lambda}_{s-1, N_p})$. Together these give $K^*(\hat{\lambda}_{s, N_p}) - 1 < s \leq K^*(\hat{\lambda}_{s-1, N_p})$, from which the implementability requirement $s \leq K^*(\hat{\lambda}_{s, N_p})$ cannot be inferred—except when $s = K_M$, where it holds under Assumption 3.3. Committing at $\max\{s-1, K_M\}$ instead delivers an almost surely implementable design at the cost of one cluster’s worth of information. For large pilots $\hat{\lambda}_{s, N_p} \approx \hat{\lambda}_{s-1, N_p}$, so the implications for the terminal design are immaterial by continuity of the optimal map, and at most one cluster is explored outside the pilot.

¹⁷The lower bound is immediate from the index set in (5). For the upper bound, suppose $s > K_M$, so that cluster s was actually explored; but then, the stopping condition must have failed at $k = s-1$, giving $s \leq K^*(\hat{\lambda}_{s-1, N_m}) \leq K^*(\lambda_m)$. Since s is an integer, $s \leq K_m$. When $s = K_M$ the upper bound holds trivially under Assumption 3.3.

¹⁸See Lemma E.14, where the asymptotic variance of $\sqrt{K}(\hat{\lambda}_{K, N_p} - \lambda_0)$ is finite and depends on N_p , but the rate does not.

4.5 Convergence to the Oracle Design

The two algorithms satisfy the requirements of Lemma 4.4, thus, inducing the optimal design in the limit.

Corollary 4.7. *EtC-CS and ACS Converge to the Oracle Design. Both algorithms are pilot-based with fixed pilot depth $N_p = N_m \geq 2$ and pilot widths $K_{pB} = K_M$ for EtC-CS and $K_{pB} = \max\{s - 1, K_M\}$ for ACS. Both are implementable—by construction for EtC-CS, and by committing one cluster early for ACS (Remark 4.6)—and both satisfy $K_{pB} \geq K_M \rightarrow \infty$ almost surely, since $K^*(\lambda)$ is monotonic in B for every fixed λ . The conditions of Lemma 4.4 therefore hold, and*

$$(\nu_B^\pi, N_B^\pi) \xrightarrow[B]{P} (\nu_0, N_0), \quad \pi \in \{\text{EtC-CS}, \text{ACS}\}$$

5 Excess Variance and Optimality

This section evaluates the sampling policies of Section 4 against the oracle. The performance criterion is the design excess variance of a policy, sometimes referred to in the literature as *Neyman regret* (Dai et al., 2023): the difference between the variance of the estimator induced by the policy’s design (K_B^π, N_B^π) and the oracle variance W_0 ,

$$\Delta_\pi = W(K_B^\pi, N_B^\pi) - W_0 \geq 0 \quad (6)$$

For adaptive policies the design is data-dependent, so Δ_π is a random variable. As is natural in the learning literature, I evaluate policies by its expectation $\mathbb{E}[\Delta_\pi]$.¹⁹ For static policies the design is non-random and the expectation is simply redundant. Every policy considered in this paper, adaptive or not, selects its terminal design by feeding a value of λ into the optimal map (3). Appendix C shows that the excess variance of any such policy can be written in closed form as

Lemma 5.1. *Excess Variance for Optimal Maps. Under Assumptions 3.1, 3.3, 3.2, any policy π which induces a terminal design $(K_B^\pi, N_B^\pi) = (K^*(\lambda), N^*(\lambda))$ for some input λ has an excess variance of the form*

$$\Delta(\lambda) = W(K^*(\lambda), N^*(\lambda)) - W_0 = \frac{\tau^2 \sqrt{FV}}{B} \cdot \frac{(\sqrt{\lambda} - \sqrt{\lambda_0})^2}{\sqrt{\lambda}} \quad \forall \lambda \in [\lambda_m, \lambda_M] \quad (7)$$

Two features of (7) are worth discussing. First, the cost of misspecification enters through the square of a deviation because the oracle satisfies the first-order condition at λ_0 . Misspecification errors therefore affect variance only to second order. Second, the entire expression carries a factor $1/B$, so any policy whose input is a fixed distance from λ_0 inherits the order of the oracle variance itself. It immediately follows that any policy which imports a misspecified guess from the literature, including the minimax optimal, will incur excess variance of order $\Theta(1/B)$. Next, I show that the learning algorithms EtC-CS and ACS, which input a converging estimator $\hat{\lambda}_p$ into the optimal map, improve substantially upon the static benchmark. In particular, I derive upper bounds on their expected excess variance of order $1/B^2$ —a magnitude of B below the oracle—, hence asymptotically negligible. Finally, I conclude that no policy can improve on the rate of the learning algorithms by deriving

¹⁹The expectation is taken over the filtration generated by the interaction between the policy and the environment. See Lattimore and Szepesvári (2020) for a textbook treatment.

matching lower bounds of $\Omega(1/B^2)$ on a generalized problem, so the algorithms are also rate optimal. Table 2 summarizes these rates.

	Rate	Result
Oracle variance	$\Theta(1/B)$	Equation (3)
Misspecified static policies incl. vBC (2015) minimax design	$\Theta(1/B)$	Lemma 5.2
EtC-CS and ACS	$O(1/B^2)$	Theorem 5.4
Minimax lower bound	$\Omega(1/B^2)$	Theorem 5.5

Table 2: Performance Rates

Notes: Results hold under the usual Assumptions 3.1, 3.2 and 3.3, and the additional subgaussian Assumption 5.3.

5.1 Static Policies and the Cost of Guessing

It is common practice to insert an ICC guess from the literature or a previous field survey into the optimal map. Figure 1 already cautioned against blind imputation, and now Lemma 5.2 makes the warning precise by locating the excess variance of these policies at the rate $\Theta(1/B)$. A second approach, developed by van Breukelen and Candel,²⁰ replaces the guess with a minimax compromise over the parameter space which also incurs excess variance of order $\Theta(1/B)$. In this setting, misspecifying the ICC is as costly as the sampling noise the design sets out to minimize.

Lemma 5.2. *Misspecified and Minimax-Optimal Static Policies.* Let $\lambda_0, \lambda_b \in [\lambda_m, \lambda_M]$ with $\lambda_b = \lambda_0 + b$. Under Assumptions 3.1, 3.2 and 3.3:

- (i) Any static input. Define the static policy $\pi_b : (K_B^{\pi_b}, N_B^{\pi_b}) = (K^*(\lambda_b), N^*(\lambda_b))$. Then, the excess variance of π_b satisfies

$$\frac{\tau^2 \sqrt{FV}}{4\lambda_M^{3/2}} \cdot \frac{b^2}{B} \leq \Delta_{\pi_b} \leq \frac{\tau^2 \sqrt{FV}}{4\lambda_m^{3/2}} \cdot \frac{b^2}{B} \quad \Delta_{\pi_b} = \Theta(1/B) \quad \forall b \neq 0$$

- (ii) The minimax-optimal input. Let

$$\lambda_{mM} = \arg \min_{\lambda \in [\lambda_m, \lambda_M]} \max_{\omega = (\tau^2, \lambda_0) \in \Omega} \left[W(K^*(\lambda), N^*(\lambda); \omega) - W(K^*(\lambda_0), N^*(\lambda_0); \omega) \right]$$

Define the minimax-optimal policy $\pi_{mM} : (K^*(\lambda_{mM}), N^*(\lambda_{mM}))$. Then, the minimax input and its worst-case excess variance are given by

$$\lambda_{mM} = \left(\frac{\sqrt{\lambda_m} + \sqrt{\lambda_M}}{2} \right)^2, \quad \max_{\omega \in \Omega} \Delta_{\pi_{mM}} = \frac{\tau_M^2 \sqrt{FV}}{2B} \cdot \frac{(\sqrt{\lambda_M} - \sqrt{\lambda_m})^2}{\sqrt{\lambda_m} + \sqrt{\lambda_M}} = \Theta(1/B)$$

Part (ii) is worth emphasizing. Optimizing over the static class does not change the order of the penalty, it only pins down the constant. The constant is governed by $\sqrt{\lambda_M} - \sqrt{\lambda_m}$, which is the width of the designer's prior uncertainty about the ICC, so more uncertain designers pay a higher price for their lack of adaptivity. $\Theta(1/B)$ is the benchmark rate the learning algorithms improve upon.

²⁰See van Breukelen and Candel (2015, 2018, 2021).

5.2 Upper Bounds for the Learning Algorithms

I now show that learning improves on the static benchmark by a full order of B . Applying (7) at the pilot estimate and bounding the denominator by compactness,

$$\Delta_\pi = \Delta(\hat{\lambda}_{pB}) = \frac{\tau^2 \sqrt{FV}}{B} \cdot \frac{(\sqrt{\hat{\lambda}_{pB}} - \sqrt{\lambda_0})^2}{\sqrt{\hat{\lambda}_{pB}}} \leq \frac{\tau_M^2 \sqrt{FV}}{4\lambda_m^{3/2}} \cdot \frac{(\hat{\lambda}_{pB} - \lambda_0)^2}{B} \quad (8)$$

so the problem reduces to bounding $\mathbb{E}[(\hat{\lambda}_{pB} - \lambda_0)^2]$. To that end, I impose a standard subgaussianity condition, which delivers the concentration of the estimators entering pilot-based policies.²¹

Assumption 5.3. *Errors Subgaussianity.* u_k is κ_u -subgaussian and ε_{ki} is κ_ε -subgaussian.

Theorem 5.4. *Expected Excess Variance of EtC-CS and ACS.* Under Assumptions 3.1, 3.2, 3.3 and 5.3, there exist finite constants $C_{\text{EtC-CS}}$ and C_{ACS} , independent of B , such that

$$\mathbb{E}[\Delta_{\text{EtC-CS}}] \leq \frac{C_{\text{EtC-CS}}}{B^2}, \quad \mathbb{E}[\Delta_{\text{ACS}}] \leq \frac{C_{\text{ACS}}}{B^2}$$

In practice, the Theorem implies that a designer who fixes her design according to EtC-CS or ACS incurs a variance penalty that vanishes relative to the oracle variance itself. As anticipated in Section 4, ACS delivers superior average performance in simulations, since it exploits a weakly larger set of clusters in the estimation of λ_0 . However, this does not translate into an improved rate because the two pilot widths K_{pB} are of the same order. They both lie in $[K_M, K_m]$, an interval whose endpoints are each $\Theta(B)$, so $K_{pB}^{\text{ACS}}/K_{pB}^{\text{EtC-CS}} = O(1)$ almost surely.²²

Why $1/B^2$? The result rests on a $1/B \times 1/B$ mechanism. The first term follows directly from the order of the oracle stationarity itself, discussed in (7). The second term is nonetheless surprising and relates directly to the partial alignment argument between learning and optimal design implementation presented earlier. Mathematically, Lemma C.5 in Appendix C shows that both algorithms satisfy $\mathbb{E}[(\hat{\lambda}_{pB} - \lambda_0)^2] \leq C/K_M$ for some constant C . Since $K_M = \Theta(B)$, the result follows, but this final observation requires further attention. The designer surely wants $K_p = \Theta(B)$ clusters for estimation, but, surprisingly, she can have them because the oracle design is itself $\Theta(B)$ clusters wide. This feature is somewhat fortuitous. Were the oracle design to require only $\Theta(1)$ clusters, a pilot of width $\Theta(B)$ would be infeasible, consistent learning of the between cluster variance could fail, and the B^{-2} rate would no longer follow.

5.3 Lower Bound

Finally, I show that the two algorithms are essentially unimprovable, by deriving a matching lower bound of order $\Omega(1/B^2)$, implying that for every sampling policy there exists an environment $\omega \in \Omega$ in which it incurs excess variance of that order. EtC-CS and ACS are therefore minimax rate optimal in the joint (K, N) design problem, and can be improved by no more than a constant factor. The bound

²¹Analogous results hold under weaker concentration conditions such as subexponential tails.

²²Recovering the rate is technically more demanding under sequential sampling, since the deviations must be bounded for a pilot of random width, rather than deterministic. Appendix C circumvents this technicality by noting that $\hat{\lambda}_k - \lambda_0$ can be decomposed asymptotically as a function of martingales—which are amenable to a maximal version of Bernstein’s inequality—and that K_m/K_M is bounded uniformly in B .

is derived for the generalized problem of Appendix A, in which the learner may sample a different number of units N^k from each cluster and revisit clusters at will.

Theorem 5.5. *Lower Bound on Expected Excess Variance. There exists a constants $C > 0$ and $B_0 < \infty$, independent of B and of the policy, such that for every $\pi \in \Pi$ and every $B \geq B_0$,²³ there exists a data generating process indexed by $\omega \in \Omega$ such that*

$$\mathbb{E}_\pi[\Delta_\pi] \geq C/B^2$$

The proof proceeds in three steps. *First*, I work with an easier version of the problem in which the cluster error u_k is revealed to the learner the first time she visits cluster k . The “augmented experiment” gives the learner weakly more information, so any lower bound established there carries over to the original problem. This simplification restores the conditional independence on which standard divergence-decomposition arguments rely.²⁴ In the clustered setting, observations within a cluster are correlated, so once a cluster has been visited its subsequent observations are not independent conditional on the history. Revealing u_k on the first visit avoids this nuisance. *Second*, within the augmented environment I consider two data generating processes y_- and y_+ of the form (1), indexed by

$$\omega_- = (\tau_-^2, \lambda_-), \quad \lambda_- = \sigma^2/\tau_-^2, \quad \omega_+ = (\tau_+^2, \lambda_+), \quad \lambda_+ = \sigma^2/\tau_+^2$$

The two environments share the same mean μ and the same unit-level variance σ^2 , so they only differ in the cluster-level variance τ^2 , what forces the learner to sample many clusters to distinguish between them. For an interior $\omega \in \Omega$ and a small $\delta_B > 0$, the cluster variances are chosen so that $\lambda_- = \lambda - \delta_B$ and $\lambda_+ = \lambda + \delta_B$. Recall from Lemma 5.2 that optimizing against the wrong instance is costly, for instance if implementing $(K^*(\lambda_+), N^*(\lambda_+))$ when the truth is ω_- . Figure 2 illustrates this dynamic.

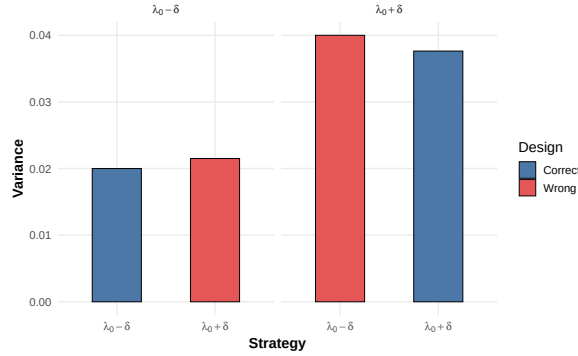


Figure 2: Optimal Policies across Environments (y_+, y_-) .

Notes: True Variance $W(K^*(\cdot), N^*(\cdot))$ under ω_- (columns 1, 2) and ω_+ (columns 3, 4) for policy $(K^*(\lambda_-), N^*(\lambda_-))$ (columns 1, 3) and policy $(K^*(\lambda_+), N^*(\lambda_+))$ (columns 2, 4). $\mu = 0$, $\omega = (1 \pm \delta, 5/(1 \pm \delta))$, $\delta = 0.5$, $B = 2000$, $F = 20$, $V = 2$.

Third, δ_B is chosen to balance this trade-off between identification and punishment. A large δ_B makes the two environments easy to tell apart, so the learner optimizes correctly most of the times,

²³The policy class Π is defined in Appendix C. The restriction excludes only policies that fail to terminate in finite time or violate the budget constraint with positive probability. Policies in Π may sample a different number of units from different clusters, revisit clusters freely, and sample a different number of units on each visit.

²⁴See Chapter 15 in Lattimore and Szepesvári (2020).

but is penalized heavily when she does not. Conversely, a small δ_B makes them hard to distinguish, but shrinks the cost of choosing the wrong design. The two forces are balanced at $\delta_B \asymp B^{-1/2}$, as is customary in learning problems, at which point the resulting bound is maximized at order $1/B^2$.

6 Inference

Adaptive data collection generally invalidates conventional inference. There is therefore a risk that the variance reduction above may be of limited use if it cannot be paired with standard inference. This section shows that, under a symmetry restriction on the errors, there is no trade-off between variance reduction and inference. Design-conditional inference on population means is invariant to designs induced by *even* policies, and asymptotically invariant to designs induced by *asymptotically even* policies. Informally, a policy is even if it depends on the data only through even functionals of the errors, such as squared errors; and asymptotically even if the design it induces is close to one generated by an even policy—as happens when the policy is driven by sample variances rather than by true squared errors—. When the design is of either kind, group averages remain asymptotically normal under symmetry, with no correction and no exclusion of pilot data. Since EtC-CS and ACS are functions of sequences of sample variance ratios, the general principle covers them.

The Section proceeds in three steps. Section 6.1 first discusses the Gaussian benchmark, frequently invoked in the sample size re-estimation literature, where the result is exact and finite-sample. Section 6.2 weakens Gaussianity into symmetry and establishes an asymptotic result for exactly even designs at every design scale. Section 6.3 extends the results to the feasible class, and identifies the unique scale at which conditioning events are both well defined and non-degenerate. The exposition is mostly confined to the cluster design problem, but connections to the general theory of Appendix D, which applies well beyond clustered sampling, are made throughout.

6.1 Exact Design Independence under Gaussian Errors

That sample averages and the vector of demeaned observations are independent under exact Gaussianity has been exploited repeatedly in the sample size re-estimation literature (Stein, 1945; Dudewicz and Dalal, 1975; Rinott, 1978) and in the internal pilot literature (Wittes and Brittain, 1990; Wittes et al., 1999; Lake et al., 2002; van Schie and Moerbeek, 2014). As a result, any design that depends on the data only through residuals is independent of the sample average, reducing conditional inference to unconditional inference at any finite sample size.²⁵ Policies depending on sample variances are simply a subset of the class of policies which do not compromise inference on the sample average. Both EtC-CS and ACS are functions of sequences of sample variance ratios, hence of residuals, so the argument covers them. I characterize the relevant policy class as the set of policies which are measurable with respect to the σ -algebra generated by the residuals.

Definition 6.1. *Residual σ -algebras.* For a fixed pilot depth N_p , an iid auxiliary sequence $(U_k)_k \sim \text{Unif}[0, 1]$ independent of the errors, and $j \geq 2$, define the residual σ -algebra of the first j clusters,

$$\mathcal{R}_j := \sigma\left((\varepsilon_{ki} - \bar{\varepsilon}_{k, N_p})_{i \leq N_p, k \leq j}, (M_{k, N_p} - \bar{M}_{j, N_p})_{k \leq j}, (U_k)_{k \geq 1}\right)$$

²⁵Equivalently, the sample average is sufficient for μ while the residual vector is ancillary, so independence also follows from Basu's theorem. Joint Gaussianity delivers it directly through zero covariance.

In words, $\mathcal{R}_j$ collects every functional of the first j cluster residuals, both within-cluster and between-cluster, which do not depend on the level of the data. Demeaning removes μ , so all generators are observable. Additionally, it includes $(U_k)_k$ as an auxiliary randomization for tie-breaking. The adaptive algorithms are measurable with respect to the filtration $(\mathcal{R}_j)_j$. To see this, note that for every $k \leq j$, $\hat{\lambda}_{k,N_p}$ is a function of the within- and between-cluster variances, which are functions, respectively, of the within- and between-cluster residuals. In particular, the relevant σ -algebra for each policy is the one generated by the residuals it observes up to commitment (Lemma E.1 in Appendix E). **EtC-CS** commits at the deterministic stage K_M , so I set $\mathcal{P}^{\text{EtC-CS}} := \mathcal{R}_{K_M}$. **ACS** however commits at the random stage s , with s being decided by the residuals of the first j clusters alone, so s is a stopping time for $(\mathcal{R}_j)_j$, and $\mathcal{P}^{\text{ACS}} := \mathcal{R}_s$ is the associated stopped σ -algebra. In both cases the terminal design (K_B^π, N_B^π) is $\mathcal{P}^\pi$ -measurable, so every design event $A_B = \{(K_B^\pi, N_B^\pi) \in C\}$ with C Borel belongs to $\mathcal{P}^\pi$.

Proposition 6.2. *Exact Conditional Normality under Gaussian Errors. Under Assumptions 3.1, 3.2 and 3.3, let $u_k \sim N(0, \tau^2)$ and $\varepsilon_{ki} \sim N(0, \sigma^2)$. Then, for $\pi \in \{\text{EtC-CS}, \text{ACS}\}$, every budget B , and every design event $A_B \in \mathcal{P}^\pi$ with $\mathbb{P}(A_B) > 0$,*

$$\mathcal{L} \left(\frac{\sqrt{K_B^\pi} (\hat{\mu}_{K_B^\pi, N_B^\pi} - \mu)}{\theta_{N_B^\pi}} \middle| A_B \right) = N(0, 1) \quad \text{exactly}$$

Three notes on Proposition 6.2. First, exactness holds only for the infeasible standardization by $\theta_{N_B^\pi}$: replacing it by the sample standard deviation $\hat{b}_{K_B^\pi, N_B^\pi}$ requires an asymptotic argument, because numerator and denominator are computed from the same cluster means. Second, the argument relies on the pilot being nested within the terminal design— $K_{pB} \leq K_B^\pi$ and $N_p \leq N_B^\pi$ —, guaranteed for both algorithms by the implementability condition. Nesting makes the terminal average orthogonal to every residual generating $\mathcal{P}^\pi$, so the pilot observations need not be discarded.

Third, the proposition delineates both what Gaussianity can and cannot buy. Under Gaussianity the invariant class is large. The design may depend arbitrarily on the residuals—not just through the sample variances but also through any *odd* functional of the residuals—and still deliver exact conditional inference at every budget. When Gaussianity is weakened to symmetry in the next two subsections, the class must shrink. The conditioning σ -algebra contracts from the residuals to their magnitudes, and the invariant policy class contracts with it, from any residual-driven policy to (asymptotically) even policies. What no assumption rescues are policies that condition on the *level* of the data rather than on its contrasts. A rule that reads the pilot mean—such as the sign policy of Section 7—conditions on precisely the direction that demeaning removes; its design events lie outside $\mathcal{P}^\pi$, and conditional normality fails even under Gaussian errors (Figure 8).

6.2 Inference for Even Designs under Symmetry

Exact Gaussianity is a strong restriction to impose on the errors. I weaken this assumption and extend the invariance tradition in three directions: from Gaussianity to symmetry, from the sample variance to any (asymptotically) even functional of the data, and from finite-stage adaptivity to arbitrary designs.

Assumption 6.3. *Error Symmetry. Let $C_k = (u_k, \varepsilon_{k1}, \varepsilon_{k2}, \dots)$. Then $C_k \stackrel{d}{=} -C_k$.*

Because of independence of u_k and ε_{ki} , a simple sufficient alternative for Assumption 3.1 is

that u_k and ε_{ki} are each symmetric about zero. The symmetry assumption is familiar and comparatively benign, as it underlies most of sign and permutation tests, as well as symmetry-based randomization inference (Canay et al., 2017). Additionally, it is far weaker than the parametric assumptions practitioners routinely impose in power calculations, and it is partially testable.²⁶ The restriction is nonetheless not innocuous: it excludes important error distribution families such as binary and count outcomes, as well as censored and one-sided distributions.

As hinted in the introduction, Assumption 6.3 permits a decomposition of the errors into independent signs and magnitudes. Conditioning on any event that depends on the magnitudes alone—such as events generated by even functionals of the data—“freezes” both the design and the magnitudes while leaving the signs unrestricted, so that a central limit theorem can run through the signs. Formally,

Lemma 6.4. *Cluster Sign Factorization. Fix N and let $(\zeta_k)_k$ be an iid Rademacher sequence independent of $((C_k)_k, (U_k)_k)$. Define*

$$\chi_{k,N} = \mathbb{1}(M_{k,N} \neq 0) \text{sign}(M_{k,N}) + \mathbb{1}(M_{k,N} = 0) \zeta_k, \quad R_{k,N} = |M_{k,N}|,$$

so that $M_{k,N} = \chi_{k,N} R_{k,N}$. Under Assumptions 3.1 and 6.3, $(\chi_{k,N})_k$ is iid Rademacher and $(\chi_{k,N})_k \perp\!\!\!\perp (R_{k,N})_k$.

Corollary 6.5. *Signs are Independent of the Even σ -algebra. For a fixed N_p and an iid sequence $(U_k)_k \sim \text{Unif}[0, 1]$ independent of $((C_k)_k, (\zeta_k)_k)$, define the even σ -algebra*

$$\mathcal{G}_\infty := \sigma\left((S_{k,N_p}^2)_{k \geq 1}, (M_{k,N}^2)_{k \geq 1, N \geq 1}, (U_k)_{k \geq 1}\right)$$

Then, under Assumptions 3.1 and 6.3, for every fixed $N \geq 1$ the sign sequence $(\chi_{k,N})_{k \geq 1}$ is iid Rademacher and independent of $\mathcal{G}_\infty$, hence independent of every event $A \in \mathcal{G}_\infty$.

Next, I introduce the class of policies which are measurable with respect to the even σ -algebra.

Definition 6.6. *A policy π is*

- (i) **Even** if the terminal design (K_B^π, N_B^π) is $\mathcal{G}_\infty$ -measurable;
- (ii) **Non-starving** if $\mathbb{P}(\nu_B^\pi \geq c) \rightarrow 1$ for some constant $c > 0$;
- (iii) **N -consistent** if $N_B^\pi \xrightarrow{P} N_0 = \lfloor N^*(\lambda_0) \rfloor \in \mathbb{N}$.

Evenness is the substantive requirement: an even policy induces design events in $\mathcal{G}_\infty$, so conditioning on them freezes the design and the magnitudes but leaves the signs unrestricted. The other two conditions are technical but do real work. Any asymptotic argument will require a divergent sequence, which is guaranteed by non-starvation. As such, it is also present in the general theory of Appendix D. The linear form of the divergence is immaterial for exactly even policies, but becomes necessary when transferring results to the feasible class. N -consistency is specific to the cluster problem: it stabilizes the depth margin so that the analysis can proceed along K -asymptotics alone. By Corollary 4.7, EtC-CS and ACS are both non-starving and N -consistent—but neither is exactly even—.

The comparison between the even σ -algebra $\mathcal{G}_\infty$ and the residuals σ -algebra $\mathcal{R}_j$ is quite

²⁶Section 8 implements a symmetry test on the average cluster errors $M_{k,N} = u_k + \frac{1}{N} \sum_{i=1}^N \varepsilon_{ki}$ using the data of Fink et al. (2020b).

telling. By replacing Gaussianity with the weaker symmetry assumption, the designer reduces the class of design policies that do not interfere with inference from any functional of the residuals to any functional of the squared residuals, i.e. (asymptotically even policies).

Proposition 6.7. *Uniform Mixing for Even Designs.* Let $((K_B^\pi, N_B^\pi), (A_B)_B)$ be a design and a sequence of design events induced by an even, non-starving, N -consistent policy. Define the scaled sum

$$Z_B^\pi = \sqrt{K_B^\pi} (\hat{\mu}_{K_B^\pi, N_B^\pi} - \mu) = \frac{1}{\sqrt{K_B^\pi}} \sum_{k=1}^{K_B^\pi} M_{k, N_B^\pi}$$

and the conditional characteristic function as $\phi_B(t) = \mathbb{E}[\exp(itZ_B^\pi) \mid \mathcal{G}_\infty]$. Write $\phi(t) = \exp(-\theta_{N_0}^2 t^2/2)$ for the Gaussian characteristic function with oracle depth N_0 . Then, under Assumptions 3.1, 3.2, 3.3 and 6.3,

$$\left| \mathbb{E}[\phi_B(t) \mathbb{1}_{A_B}] - \phi(t) \mathbb{P}(A_B) \right| \rightarrow 0 \quad \forall t \in \mathbb{R},$$

uniformly over $A_B \in \mathcal{G}_\infty$. Moreover, for any such sequence with $\liminf_B \mathbb{P}(A_B) > 0$,²⁷

$$\mathcal{L}(Z_B^\pi \mid A_B) \Rightarrow N(0, \theta_{N_0}^2), \quad \mathcal{L}\left(\frac{Z_B^\pi}{\hat{b}_{K_B^\pi, N_B^\pi}} \mid A_B\right) \Rightarrow N(0, 1)$$

Proposition 6.7 displays a very strong mode of convergence: the scaled sums converge $\mathcal{G}_\infty$ -uniformly mixing. The implication is that asymptotic normality survives conditioning on any $\mathcal{G}_\infty$ -measurable event, at any scale, terminal or not. Nonetheless, this result is of limited use on its own as the policy class—policies which have access to the true squared errors—is very thin. At best, policies may condition on sequences of nearly even statistics such as sample variances. Recall that sample variances demean the observations by the sample average, so a policy depending on a sequence of sample variances could induce designs depending on the relative signs, hence not in $\mathcal{G}_\infty$.²⁸ The next section shows that the results extend to the feasible class, once the set of conditioning events is restricted to the appropriate scale.

6.3 Inference for Asymptotically Even Designs under Symmetry

Definition 6.8. *Asymptotically Even Policies.* Let $K_{0B} = K^*(\lambda_0)$ be the deterministic oracle width and define $\Xi_B^\pi = (K_B^\pi - K_{0B})/\sqrt{B}$ where $K_{0B} = K^*(\lambda_0)$ is a deterministic centering. A policy π is asymptotically even if there exists a $\mathcal{G}_\infty$ -measurable pair $(\tilde{K}_B, \tilde{N}_B)$, with $\tilde{\Xi}_B = (\tilde{K}_B - K_{0B})/\sqrt{B}$, such that

$$|\Xi_B^\pi - \tilde{\Xi}_B| \xrightarrow[B]{P} 0, \quad |N_B^\pi - \tilde{N}_B| \xrightarrow[B]{P} 0$$

In essence, a policy is asymptotically even if its design is close in probability to one induced by an even policy, with closeness measured on the $\sqrt{B}$ scale for width and on the unit scale for depth. Observe that the definition places no restriction on the surrogate. The even design $(\tilde{K}_B, \tilde{N}_B)$ need

²⁷For exactly even designs, symmetry buys more than asymptotic normality. Conditional on $\mathcal{G}_\infty$ the signs $(\chi_{k,N})_k$ are iid Rademacher (Corollary 6.5), so sign-flip randomization on the centered cluster means yields an exact finite-sample test of $H_0: \mu = \mu_0$, in the spirit of symmetry-based randomization inference (Canay et al., 2017). Exactness does not extend to feasible policies: EtC-CS and ACS are only asymptotically even, so for them the sign-flip test is asymptotically valid on the same footing as the normal approximation pursued here.

²⁸Concretely, for a simple model $x_i = \mu + \varepsilon_i$ with $(\varepsilon_i^2)_s \in \mathcal{G}_\infty$, construct a sequence of sample variances $(\hat{s}_i^2)_i$. The policy may then recover $\hat{\varepsilon}_i^2$ for every i , and hence $\hat{\varepsilon}_i \varepsilon_{i+1} = [(i+1)^2 \hat{\varepsilon}_{i+1}^2 - i^2 \hat{\varepsilon}_i^2 - \varepsilon_{i+1}^2]/(2i)$ —a relative sign, which by Corollary 6.5 is not $\mathcal{G}_\infty$ -measurable. See Appendix D for the general statement. The leakage is nonetheless asymptotically negligible relative to the even component, which is what licenses the notion of asymptotic evenness below.

be neither feasible nor related to π in any substantive way. However, in practice, the analyst may construct the even surrogate by replacing the squared residuals entering π with the true squared errors $M_{k,N}^2$, and then establishing convergence of the two designs at the relevant scale.

The transition from exact to asymptotic evenness restricts the class of admissible conditioning events to the set of events that agree asymptotically under the two policies—in the sense that $\mathbb{P}(A_B \triangle \tilde{A}_B) \rightarrow 0$ —. When events are expressed on the same scale as the convergence of the designs, survival is warranted under a mild margin condition. The natural conditioning class is then given by

$$(A_B)_B : A_B = \{(\Xi_B^\pi, N_B^\pi) \in C\}, \quad C \subseteq \mathbb{R} \times \mathbb{N} \text{ Borel} \quad (9)$$

Which scale should the analyst condition on? Three scales are *a priori* available in the width dimension (unit, $1/\sqrt{B}$ and $1/B$), and only one works. For exactly even policies the choice is immaterial: by Proposition 6.7, any $\mathcal{G}_\infty$ event at any scale leaves the limiting law of Z_B^π unchanged. For asymptotically even policies the choice is decisive. K_B^π is proportional to B and Lemma E.10 shows that it fluctuates over a window of width $|K_B^\pi - K_{0B}| = O_p(\sqrt{B})$ around the oracle, therefore the feasible and surrogate designs cannot couple at the unit scale: $\mathbb{P}(A_B \triangle \tilde{A}_B) \not\rightarrow 0$, and inference results do not transfer. At the scale of $1/B$, conditioning is asymptotically vacuous. By Lemma 4.4, $\nu_B^\pi \xrightarrow{P} \nu_0$, so $\mathbb{P}(\nu_B^\pi \in C) \rightarrow \mathbb{1}\{\nu_0 \in C\} \in \{0, 1\}$ for every C whose boundary ν_0 avoids, and the conditional law degenerates to the unconditional one. At that scale a random-index central limit theorem in the spirit of Anscombe (1952) and Chow and Robbins (1965) suffices, and symmetry is never needed.

The intermediate $1/\sqrt{B}$ scale is the only one on which the coupling between feasible and surrogate designs survives and the conditioning events remain non-degenerate. Specifically, this permits non-trivial conditioning on quantiles of the realized width, the events actually of interest to an analyst reporting design-conditional inference. These events lie outside the reach of Anscombe (1952), requiring both symmetry and policy evenness. Table 3 summarizes the scale discussion.

Scale	Does Surrogate Coupling Survive?	Non-degenerate Events?	Tools
Unit	No	—	—
$1/B$	Yes	No	Anscombe
$1/\sqrt{B}$	Yes	Yes	Symmetry and Evenness

Table 3: Scale of Conditioning Events under Symmetry

Theorem 6.9. *Inference for Asymptotically Even Designs.* Let π be an asymptotically even, non-starving, N -consistent policy, and let $(A_B)_B$ be a sequence of events as in (9) satisfying the margin condition

$$\lim_{\eta \downarrow 0} \limsup_B \mathbb{P}\left(\text{dist}((\Xi_B^\pi, N_B^\pi), \partial C) \leq \eta\right) = 0 \quad (10)$$

Then, under Assumptions 3.1, 3.2, 3.3 and 6.3,²⁹

$$|\mathbb{E}[\exp(itZ_B^\pi) \mathbb{1}_{A_B}] - \phi(t)\mathbb{P}(A_B)| \rightarrow 0 \quad \forall t \in \mathbb{R}$$

²⁹Clearly, the uniform mixing statement cannot be established for Z_B^π , as this scaled sum is not even $\mathcal{G}_\infty$ measurable. Yet, a qualitatively similar result holds pointwise along the given sequence of events. In particular, the two conditional limit laws, which are what an applied analyst uses, carry over unchanged. See Appendix E for a complete treatment.

If in addition $\liminf_B \mathbb{P}(A_B) > 0$, then

$$\mathcal{L}(Z_B^\pi | A_B) \implies N(0, \theta_{N_0}^2), \quad \mathcal{L}\left(\frac{Z_B^\pi}{\hat{b}_{K_B^\pi, N_B^\pi}} \middle| A_B\right) \implies N(0, 1)$$

Condition (10) is an anti-concentration requirement on the choice of C . The scaled design must not pile up on the boundary of the conditioning set, since it is there that the feasible and surrogate designs can disagree. In practice this condition is mild, and follows from a central limit theorem—requiring finite fourth moments of the errors—for the statistic governing the design, $\hat{\lambda}$ in the context of the cluster design problem.

It remains to verify that the two algorithms are asymptotically even. Propositions E.11 and E.12 do so constructively, building surrogates $\text{Et}\tilde{\text{C}}\text{-CS}$ and $\tilde{\text{A}}\text{CS}$ that replace the nearly even statistic $\hat{b}_{K,N}^2$ with its exactly even counterpart $K^{-1} \sum_{k \leq K} M_{k,N}^2$.³⁰ For EtC-CS the coupling is immediate: the design depends on the single statistic $\hat{\lambda}_p$, the two versions of which differ by $O_p(1/B)$, so the induced widths differ by $O_p(1)$ —an order below the required $o_p(\sqrt{B})$ —. ACS is more delicate. Replacing residuals by errors would still leave the analyst with the difficult task of coupling two λ estimates evaluated at two different random stopping times. I avoid the difficulty by giving the surrogate a deterministic stopping rule, $\tilde{s} = K_{0B}$, and localizing the ACS stopping time within an $O_p(\sqrt{B})$ window of it.

Theorem 6.10. *Design-Robust Inference for EtC-CS and ACS. Let (K_B^π, N_B^π) be the design induced by $\pi \in \{\text{EtC-CS}, \text{ACS}\}$ and let $A_B = \{(\Xi_B^\pi, N_B^\pi) \in C\}$ for any C satisfying (10) and $\liminf_B \mathbb{P}(A_B) > 0$. Then, under Assumptions 3.1, 3.2, 3.3 and 6.3,*

$$\mathcal{L}\left(\frac{Z_B^\pi}{\hat{b}_{K_B^\pi, N_B^\pi}} \middle| A_B\right) \implies N(0, 1)$$

The takeaway from Theorem 6.10 is that *nothing needs to change*. The estimator is the ordinary average of cluster averages computed on the full sample—pilot observations included—the standard error is the usual clustered one, and the normal approximation is valid conditional on the realized design. No reweighting, no sample splitting, and no discarding of exploration data is required. A full array of simulations verify this prescription.

The boundary of the result. Policy evenness and error symmetry are not merely sufficient. The simulation evidence shows that relaxing either can destroy the invariance. A policy that adapts through odd functionals of the data—estimated treatment effects, interim averages, futility rules—breaks the sign–magnitude factorization of Lemma 6.4, because conditioning on the design then carries information about the signs themselves. For such policies the design-conditional mean of Z_B^π need not vanish, and the pathologies documented by Nie et al. (2018) and Hadad et al. (2021) reappear, along with the necessity to introduce the reweighting corrections developed in those papers. Similarly, error asymmetry allows exactly even policies to pick up information about the error signs, breaking, once again, the sign–magnitude factorization with comparable effects. Section 7 makes the boundary concrete by assessing Theorem 6.9 in a grid which interacts policy evenness and error symmetry. Only the joint symmetry–evenness pair does satisfy asymptotic normality.

³⁰The within-cluster variance $\hat{w}_{K,N}^2$ is both even and feasible, so it requires no adjustment.

6.4 Beyond Clustered Sampling

Nothing in the preceding argument is specific to clusters. Stripped to its essentials, the mechanism requires only that units be sampled adaptively from a collection of populations, that the errors be symmetric, and that the design reads the data through (asymptotically) even functionals. Conditioning on such a design then freezes the error magnitudes while leaving the signs untouched, and a central limit theorem runs through the signs exactly as before. Appendix D develops this as a standalone theory exactly as expected: with an arbitrary adaptive sampling protocol over K sub-populations, a horizon of N observations, and design events at the $\sqrt{N}$ scale. Proposition 6.7 and Theorem 6.9 are simple specializations of this broad theory to the cluster setting.³¹

The leading application is the oldest design rule in the book, a Neyman allocation with plug-in variances. At a high level, a designer would like to estimate a set of populations means, where each mean has a different dispersion, by first running a uniform pilot to estimate each arm’s variance, and then allocating the remaining observations proportionally to the estimated standard deviations. This design is driven by sample variances, which differ from even functionals of the errors only through squared pilot means—a gap of order $1/N$ —so the policy is asymptotically even, and the general theory applies off the shelf.

Corollary 6.11. *Two-Stage Neyman Allocations (Informal). Let K arms have symmetric iid errors with strictly positive variances and finite fourth moments. Consider the two-stage implementable design that samples all arms uniformly over a fixed share of the horizon and then sets terminal counts proportional to the pilot standard deviations. Conditional on any design event with a non-degenerate limit—quantile cells of the realized counts included—the studentized arm means are jointly asymptotically standard normal: computed on the full sample, with the usual standard errors, and with no correction. Proposition D.35 and Corollary D.38 contain the formal statements.*

The same reasoning covers other second-moment-driven rules. The optimal second-stage propensity score of Hahn et al. (2011), for instance, is a ratio of pilot-estimated conditional standard deviations, and is therefore even. The boundary, however, travels with the principle: adaptation through odd functionals of the data—interim means, estimated treatment effects, futility rules—remains uncovered under symmetry alone. Whenever adaptation reads only magnitudes, inference on means proceeds as if the design had been fixed in advance; the moment it reads levels or signs, the corrections of the adjustment tradition become necessary.

7 Simulations

This Section confronts the adaptive algorithms with the theoretical prescriptions above: Section 7.1 with the excess-variance bounds of Section 5 and Section 7.2 with the conditional normality results of Section 6. Data are generated from the survey model (1), $y_{ki} = \mu + u_k + \varepsilon_{ki}$, with $\mu = 0$. Every reported statistic is computed over 10^5 independent replications of the complete design-and-estimation cycle, so that the randomness of the pilot, the induced design, and the terminal estimator, are all accounted for. The cost structure is $(F, V) = (20, 2)$ throughout and the designer’s parameter space is $\Omega = [\tau_m^2, \tau_M^2] \times [\lambda_m, \lambda_M] = [0.25, 4] \times [1, 100]$.³² The baseline environment is $\omega = (\tau^2, \lambda_0) = (1, 5)$

³¹The specialization is not mechanical: the cluster problem carries two error layers and a depth margin that does not diverge, which is what Appendix E handle.

³²These bounds enter the algorithms directly, and not only through Assumption 3.2. The pilot width is $K_M = \lfloor K^*(\lambda_M) \rfloor$ and the pilot depth is $N_p = \lfloor N^*(\lambda_m) \rfloor$.

at a budget of $B = 2,000$, which places the oracle design at $(K^*(\lambda_0), N^*(\lambda_0)) = (58.6, 7.1)$ and the pilot at $(K_M, N_p) = (24, 3)$. Assumption 3.3 holds at every point of the grid reported below. Errors are Gaussian in the performance exercises, and take a range of symmetric and asymmetric shapes in the inference exercises.

7.1 Performance

Figure 3 reports the distribution of the terminal design (K_B^π, N_B^π) under each algorithm, together with the pilot estimate $\hat{\lambda}_{pB}$ that generates the design through the optimal map (3). Both algorithms concentrate around the oracle, as prescribed by Corollary 4.7, with ACS inducing tighter designs as it commits later and therefore estimates λ_0 from a weakly larger set of clusters (Remark 4.6). The distribution of $\hat{\lambda}_{pB}$ in the right-hand panel is sharp around the truth but visibly right-skewed. The reason is that $\hat{\tau}^2$ is recovered as a difference of two random objects $\hat{b}^2 - \hat{w}^2/N_p$, so when τ^2 is small relative to σ^2 , $\hat{\tau}^2$ is estimated as a noisy residual between two large and separately noisy quantities. Because $\hat{\tau}^2$ enters the ratio through the convex map $\hat{\tau}^2 \mapsto 1/\hat{\tau}^2$, any symmetric noise in the denominator becomes a heavy right tail in $\hat{\lambda}_{pB}$. The $\hat{\lambda}_{pB}$ skew, a finite-sample feature, is partially however partially absorbed by the design as $\hat{\lambda}_{pB}$ enters the optimal map through a square root which compresses the upper tail. Additionally, the excess variance (7) is flat to the right of λ_0 and steep to its left, so mass in the upper tail is comparatively cheap. Figure 13 in Appendix F shows that both algorithms track the oracle design relative well with minor depth oversampling and width undersampling.

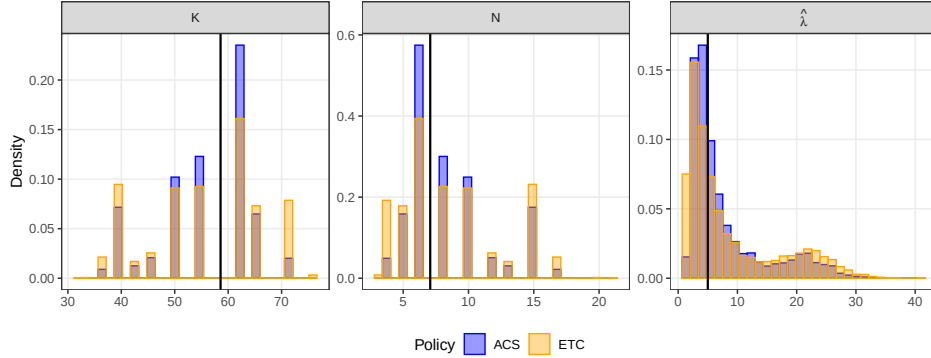


Figure 3: Design (K_B^π, N_B^π) and Pilot Estimate $\hat{\lambda}_{pB}$ for EtC-CS and ACS

Notes: Solid black lines mark the oracle values $\lfloor K^*(\lambda_0) \rfloor$, $\lfloor N^*(\lambda_0) \rfloor$ and λ_0 . 10^5 replications. $\omega = (1, 5)$, $B = 2,000$, $F = 20$, $V = 2$.

Next, I test the main performance theoretical prediction. Figure 4 plots excess variance (6) against λ_0 for the two adaptive algorithms and a set of five static policies, each indexed by the imputed variance ratio. Static policies do not incur sampling risk, so their excess variance is deterministic and given exactly by (7). In particular, the curve vanishes where the imputed value is correct and rises on either side. By contrast, the adaptive algorithms stay close to the oracle over the whole grid. The right panel reports the mean as in Theorem 5.4, and the left panel, the median, whose performance is empirically better since an adaptive design occasionally draws an unlucky pilot and pays for it. Performance is best at small λ_0 , where τ^2 is large relative to σ^2 and the pilot estimate is well conditioned.

The red markers give each static policy's worst case over the grid, which fall at the endpoints

by convexity of the excess variance map for optimal map policies. The minimax policy minimizes the worst case excess variance by equalizing the two ends. In this parameter space, that input is $\lambda_{mM} = 30.25$, so the policy imputed at $\lambda = 30$ is the [van Breukelen and Candel \(2015\)](#) maximin design up to rounding. The comparison becomes clear. A static policy can beat the algorithms in a neighbourhood of its own imputed value, but the worst case of either adaptive algorithm improves on the worst case of every static competitor, minimax design included. Adaptation buys robustness against the misspecification of the intra-cluster correlation.

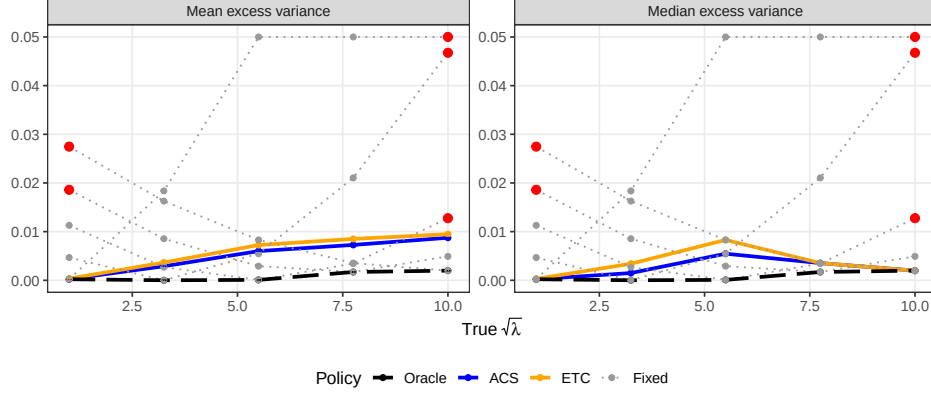


Figure 4: Excess variance across λ_0

Notes: Mean (left) and Median (right). Static policies imputed at $\lambda \in \{1, 10, 30, 60, 100\}$. 10^5 replications. $\mu = 0$, $\tau^2 = 1$, $B = 2,000$, $F = 20$, $V = 2$.

Finally, Figure 5 fixes the environment and moves along the budget line. The figure illustrates the performance guarantees both in absolute terms (left) and in terms of rates (right), with static policies incurring $\Theta(1/B)$ by Lemma 5.2(i), and the algorithms $O(1/B^2)$ by Theorem 5.4. The left panel shows that for this parameter combination the excess variance of EtC-CS and ACS is already small at modest budgets and falls sharply until converging with oracle variance. In the right panel the axes are logarithmic, so the prediction is a gap in slopes, approximately -1 for the static policies and -2 for the adaptive algorithms. This figure shows that the excess variance of the algorithms falls at roughly twice the rate of any static comparison. ACS dominates EtC-CS throughout, and the gap is widest at small budgets. Recall from Remark 4.6 that both pilot widths are of the same order, so the commitment advantage manifests only through a constant and not a rate, hence, it is more salient at small budgets.

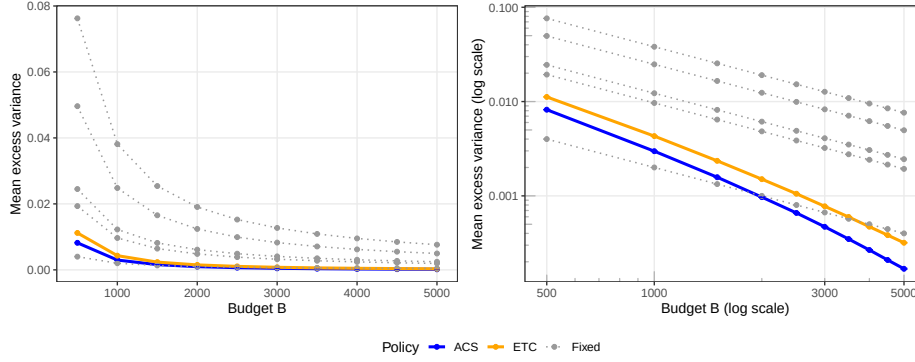


Figure 5: Mean excess variance across budget

Notes: Linear scale (left) and Log Scale (right). Static policies imputed at $\lambda \in \{1, 10, 30, 60, 100\}$. 10^5 replications. $\mu = 0$, $\tau^2 = 1$, $B = 2,000$, $F = 20$, $V = 2$.

7.2 Conditional Inference

I now turn to inference, with an emphasis on ACS.³³ Figure 6 reports, for a grid of symmetric error distributions including bimodal distributions and Laplace (heavy-tailed) laws, the unconditional density of $Z_B^\pi/\theta_{N_B^\pi}$ together with its density conditional on the modal design.³⁴ As per Theorem 6.9, conditioning on the realized design leaves the limiting law unchanged, hence the two coincide throughout.

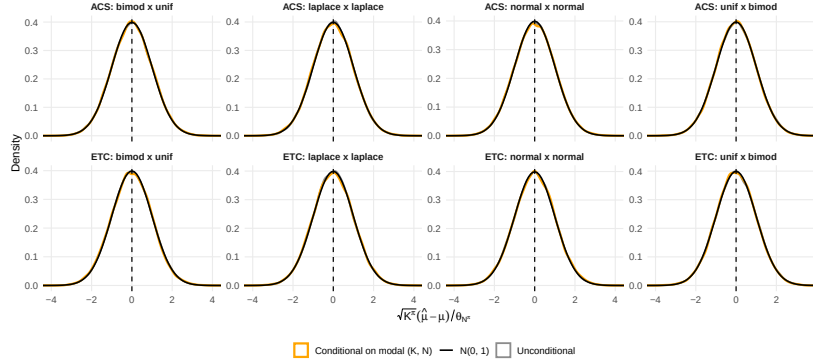


Figure 6: Empirical density of $Z_B^\pi/\theta_{N_B^\pi}$ at the modal design and unconditionally under symmetry

Notes: Distribution of u_k (left), of ε_{ki} (right). Distributions considered: Bimodal-Uniform, Laplace-Laplace, Normal-Normal, Uniform-Bimodal. 10^5 replications. $\mu = 0$, $\omega = (1, 5)$, $B = 2,000$, $F = 20$, $V = 2$.

Figure 6 uses the infeasible standardization $\theta_{N_B^\pi}$. Figure 7 repeats the exercise with the statistic an analyst would actually compute, reporting the share of replications in which the interval $\hat{\mu}_{K^\pi, N^\pi} \pm 1.96 \hat{b}_{K^\pi, N^\pi}/\sqrt{K^\pi}$ covers μ , cell by cell in the realized width. As prescribed by Theorem 6.10, the coverage is close to nominal in every cell.³⁵

³³ACS interacts with the data far more than EtC-CS. It conducts nearly all cluster exploration inside the pilot, so the design not only contains more data, but depends on the whole sequence $(\hat{\lambda}_{k, N_p})_{k \leq s}$ rather than on a single statistic $\hat{\lambda}_p$. It is therefore the more demanding test of the results in Section 6.

³⁴Figure 14 in the Appendix displays the full grid of combinations.

³⁵Coverage tilts mildly with the realized width. The tilt is a studentization effect rather than a design effect, as it is absent from Figure 6, which standardizes by $\theta_{N_B^\pi}$. Recall that the studentized result relies on convergence in probability

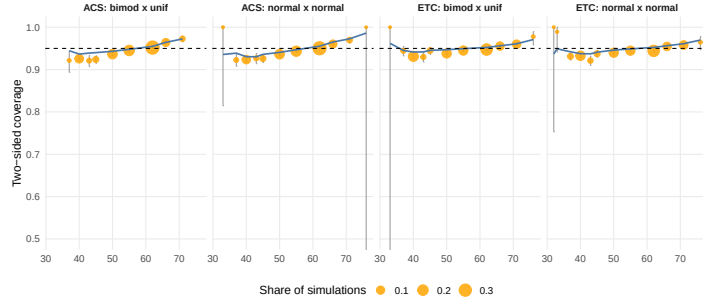


Figure 7: Design-conditional coverage of $\hat{\mu}_{K_B^\pi, N_B^\pi} \pm 1.96 \hat{b}_{K_B^\pi, N_B^\pi} / \sqrt{K_B^\pi}$

Notes: Normal-normal and bimodal-Laplace errors. Grey lines are Clopper-Pearson 95% intervals at each realized K_B^π . 10^5 replications. $\mu = 0$, $\omega = (1, 5)$, $B = 2,000$, $F = 20$, $V = 2$.

I next document the role of evenness within the inference framework. Theorem 6.9 implies that $\mathbb{E}[Z_B^\pi | A_B] = \mathbb{E}[Z_B^\pi] = 0$ for an asymptotically even policy, hence, to show that the requirement actually bites, I compare three asymptotically even policies—a naive even policy, **EtC-CS** and **ACS**—against a naive odd policy that reads levels, hence the signs. The naive odd and even policies follow the same structure. They first run the same pilot as **EtC-CS**, sampling K_M clusters at depth N_m . The odd policy sets $K^o = K_M$ if the pilot cluster mean $\bar{M}_{K_p, N_p}$ is positive and $K^o = K_m$ otherwise, with depth N^o chosen to exhaust the budget. Similarly, the even policy sets $K^e = K_M$ if the pilot's average magnitude $K_p^{-1} \sum_k |M_{k, N_p}|$ is greater than $\mathbb{E}|M_{k, N_p}| = \theta_{N_p} \sqrt{2/\pi}$ and $K^e = K_m$ otherwise.³⁶ In this example, the odd policy reads the sign of the pilot mean, so conditioning on it carries information about the signs $(\chi_{k, N})_k$ themselves, and the factorization of Lemma 6.4 fails. Conversely, the even policy depends only on magnitudes, and the inference principle applies. Figure 8 partitions replications into quintiles $(C_j)_j$ of the realized width and plots $\mathbb{E}[Z_B^\pi | C_j]$ for each. The conditional expectations of **EtC-CS** and **ACS** are indistinguishable from zero across cells, just like the naive even policy ones, while those of the odd policy live visibly away from zero and change sign as predicted.³⁷

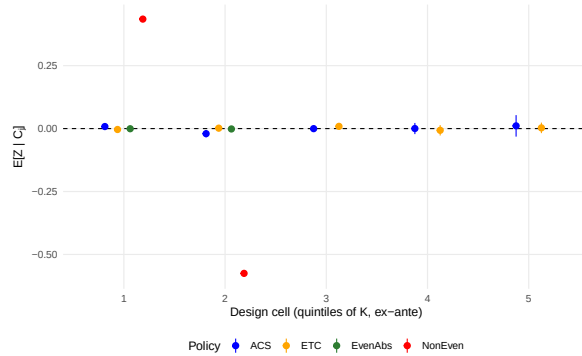


Figure 8: Design-conditional expectation $\mathbb{E}[Z_B^\pi | C_j]$ over width quintiles

Notes: $(C_j)_j$ are quintiles of the realized width. Gaussian errors. The naive even and naive odd policies are two-valued, so only two quintiles carry positive mass. 10^5 replications. $\mu = 0$, $\omega = (1, 5)$, $B = 2,000$, $F = 20$, $V = 2$.

of $\hat{b}_{K_B^\pi, N_B^\pi}^2 \rightarrow \theta_{N_0}^2$, which only happens consistently at large budgets as $\hat{b}_{K_B^\pi, N_B^\pi}^2$ is not even. Still, the tilt balances across cells, leaving unconditional coverage unaffected according to the Anscombe route outlined in the text, which requires no symmetry.

³⁶ $\mathbb{E}|M_{k, N}| = \theta_{N_p} \sqrt{2/\pi}$ is the exact mean of the half-normal under Gaussian errors. Hence, by a CLT argument, the designer should expect roughly 1/2 of the designs below the cutoff and 1/2 above.

³⁷Observe that the odd policy does not only read the signs but also the levels. As a result, it is not measurable with respect to the filtration $(\mathcal{R}_j)_j$, so, even if dealing with Gaussian errors, the policy is not covered by Proposition 6.2.

In what follows I examine the asymptotic behavior of the conditional laws. Figure 9 examines Theorem 6.9 by plotting the maximum cell-frequency discrepancy

$$D_B = \sup_{t \in \mathcal{T}} \max_j \left| \mathbb{E}[e^{itZ_B^\pi} \mathbb{1}_{C_j}] - \phi(t) \mathbb{P}(C_j) \right|$$

against B . The Theorem claims that the discrepancy should uniformly vanish for asymptotically even policies. Four combinations are shown: the naive odd policy under Gaussian errors, ACS under asymmetric errors— $u_k \sim \Gamma(\alpha, \beta)$ with $\alpha = 1$ so that $\text{Skew}(u_k) = 2/\sqrt{\alpha} = 2$ —, ACS under Laplace errors, and ACS under Gaussian errors.

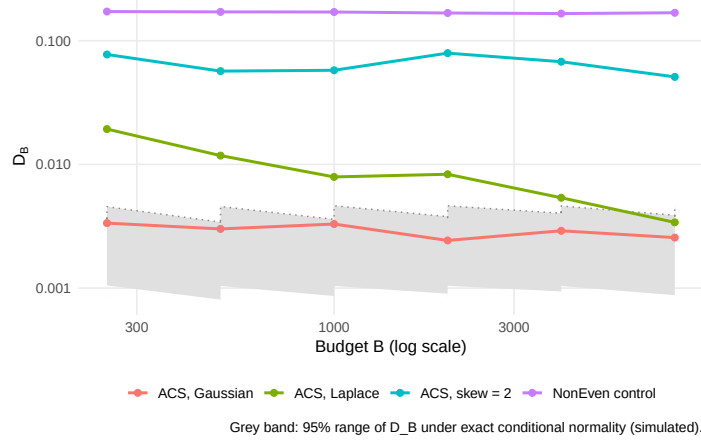


Figure 9: Maximal discrepancy between the conditional and normal characteristic functions, as a function of B

Notes: $(C_j)_j$ are quintiles of the realized width. The naive odd policy is two-valued, so only two quintiles carry positive mass. Symmetric errors are $N(0, 1)$, asymmetric errors are centered $\Gamma(1, 1)$. 10^5 replications. $\mu = 0$, $\omega = (1, 5)$, $F = 20$, $V = 2$.

The figure separates the two ingredients entering the inference theory cleanly. Dropping evenness breaks the result: D_B for the odd policy reading levels with Gaussian errors is bounded away from zero. Dropping symmetry breaks the it too: D_B for ACS under $\Gamma(1, 1)$ errors does not decrease with the budget. As a result, the simulation evidence suggests that in general neither condition can be dispensed with. By contrast, where both hold, the theory is confirmed. D_B for ACS under Laplace errors declines steadily in B , exactly as predicted by asymptotic evenness and Theorem 6.10. For the Gaussian case, D_B sits at the simulation noise floor at every budget as predicted by Proposition 6.2.³⁸

Finally, Figure 10 pins down the monotonicity within the symmetry assumption by plotting the cell-conditional expectations $\mathbb{E}[Z_B^\pi | C_j]$ against $\text{Skew}(u_k)$. At exact symmetry all cells are centered at zero, but as skewness rises the conditional expectations fan out. The result is expected. Under asymmetry the design cell becomes informative about the signs themselves, and the more extreme the cell, the more it reveals—which is why the departures are largest in cells 1 and 5—. ACS departs more than EtC-CS at every level of skewness, consistent with its design depending on a longer sequence of statistics, therefore leaking weakly more information into the terminal average when conditioning.

³⁸The grey band gives the 95% range of D_B under an exact null, obtained by drawing designs from the empirical design distribution of ACS and generating the corresponding statistics as independent standard normals, so that D_B is zero by construction and the band measures Monte Carlo error alone.

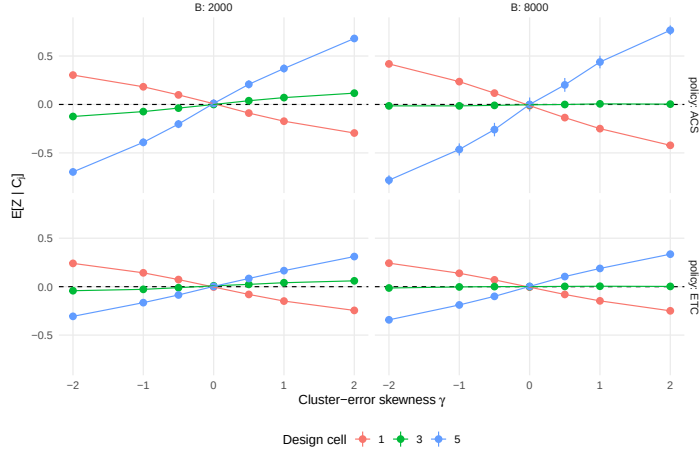


Figure 10: Design-conditional expectation $\mathbb{E}[Z_B^\pi | C_j]$ as a function of $\text{Skew}(u_k)$

Notes: $(C_j)_j$ are quintiles of the realized width, $\pi \in \{\text{ACS}, \text{EtC-CS}\}$. 10^5 replications. $\mu = 0$, $\omega = (1, 5)$, $F = 20$, $V = 2$.

8 Empirical Application: Fink et al. (2020b)

8.1 Setting and Design Problem

FJM study whether relaxing liquidity constraints during the agricultural lean season changes labor allocation and harvest outcomes among smallholder farmers in Chipata District, Zambia. Farmers were offered favorable loans, in cash or in maize, repayable after the harvest. The experiment was implemented through a cluster design with $K = 175$ villages and a target of $N = 22$ households per village. The treatment was assigned at the village level in three equal arms (cash loan, maize loan and control) after a baseline survey.³⁹ The experiment is a natural setting for implementing the adaptive sampling procedure described in this paper: cluster design with treatment at the village level, ex-ante fixed budget, and relevant role of fixed costs.

Importantly, unlike many other studies, FJM are not inattentive to the intra-cluster correlation. The authors run a small pilot and estimate the ICC, which they later use to power the study appropriately but not in a cost-aware fashion.⁴⁰ Conventional power calculation fix a detectable effect and solve for a sample size, but they do not interact the statistical requirements with the cost structure, specially the F/V ratio governing the split between units and clusters.⁴¹ The realized design appears instead to have been driven by logistical concerns as the 22 household target per village is simply the size of the smallest village in the sample.⁴² These practices are not unique to FJM, but prevalent in field experiments and medical trials. In fact, this is the *status quo* that the manuscript seeks to replace.

³⁹Realized depth effectively ranged from 5 to 22 households, averaging 17.9, leading to a baseline sample of 3,139 households, of which 3,028 appear in the year-one survey. Appendix G.1 describes the sampling frame, the outcome variables, and the residualization process entering my analysis in detail.

⁴⁰The experiment register indicates that the study is powered to detect a ten percent increase in farm production value (see Fink et al. (2024)). Further design details can be found in Fink et al. (2020a) and the pre-analysis plan Fink et al. (2014).

⁴¹There is also no evidence that the size of the pilot was selected according to a meaningful criterion.

⁴²Other features of the design point in the same direction. The initial design considered sampling a round five villages per administrative block, and villages were excluded from the initial list of 250 for reasons such as the impossibility of reaching them by vehicle during the rainy season. Fixed costs and logistics are then plausibly first-order, as they are in most field experiments. Yet, none of this implies the realized design ended up being cost-suboptimal.

There are two caveats in the counterfactual exercise below. First, I do not observe the cost parameters (F, V) , so every result below is stated across a grid of ratios $F/V \in [1, 100]$. Hence, none of my claims should suggest that the realized design was suboptimal at the ratio FJM actually faced, even if the design was not intentionally selected in a cost-aware fashion. Second, the exercise uses baseline data alone to identify the design. Post-treatment outcomes would in principle be available (Kasy and Sautmann, 2021), but the one-year horizon of this experiment leaves no room for adaptation within it. I therefore assume the variance ratio is stable across survey rounds and arms, so that $\tau_d^2 = \tau^2$ and $\sigma_d^2 = \sigma^2$. Appendix G.2 tests both restrictions and cannot reject either.

The designer’s problem. With three arms of equal size, the intention-to-treat (ITT) estimator for arm d specializes the cluster-level treatment estimator of Section 3 and Appendix A.2.1 to

$$\hat{\text{ITT}}_{K,N}^{(d)} = \hat{\mu}_d - \hat{\mu}_0 = \frac{3}{K} \left(\sum_{k: D_k=d} \frac{1}{N} \sum_{i=1}^N y_{ki} - \sum_{k: D_k=0} \frac{1}{N} \sum_{i=1}^N y_{ki} \right)$$

Under the homogeneity restriction the two treated arms have identical variance, so the designer’s minimax problem across arms collapses to a single objective,

$$\min_{K,N} \max_{d \in \{1,2\}} \text{Var}(\hat{\text{ITT}}_{K,N}^{(d)}) = \min_{K,N} \frac{6\tau^2}{K} \left(1 + \frac{\lambda}{N} \right) \quad \text{s.t.} \quad K(F + NV) \leq B$$

This is the objective of Section 3 scaled by a constant, so the optimal map $(K^*(\lambda), N^*(\lambda))$ of (3) remains unchanged. The two arms cost approximately the same per household—about 60 ZMW excluding administration—so V should be read as the average per-household cost across arms.

Design inputs. I estimate λ_0 from the full baseline sample and calibrate the designer’s interval $[\lambda_m, \lambda_M]$ through a cluster bootstrap of $\hat{\lambda}$, taking λ_m just below the 2.5th percentile and λ_M just above the 97.5th.⁴³ Table 4 reports the results for the two outcomes I study: $\log(1 + \text{harvest value})$, the register’s primary outcome, and $\log(1 + \text{ganyu mandays})$, the informal-labor measure the published paper emphasizes. Both are residualized according to the covariates included in the paper’s specification.

Outcome	$\lambda_0 (= \hat{\lambda}_{K,N})$	ICC	$Q_{0.025}(\hat{\lambda}_{K,N}^{(b)})$	$Q_{0.975}(\hat{\lambda}_{K,N}^{(b)})$	λ_m	λ_M
1_harvest_value	9.658	0.093	7.072	14.295	7	15
1_ganyu_mandays	27.682	0.034	18.337	53.187	18	54

Table 4: Design Inputs

Notes: The design inputs are estimated from the full baseline sample. Percentiles from a 2,000-replication cluster bootstrap holding village sizes fixed. Covariate-residualized data.

Is inference valid here? The results of Section 6 rest on Assumption 6.3, which requires the whole error block $C_k = (u_k, \varepsilon_{k1}, \varepsilon_{k2}, \dots)$ to satisfy symmetry. That restriction is not directly testable, since the analyst observes cluster averages rather than the block itself. The analyst is therefore limited to testing symmetry on the cluster average residuals $M_{k,N} = u_k + \bar{\varepsilon}_{k,N}$, which is implied

⁴³This λ box is possibly too wide in a way that hurts the implementation of the adaptive algorithms. In practice, the authors probably had access to prior information that could narrow the interval. The intervals for both variables satisfy Assumption 3.3 over the full F/V grid.

at every depth by the symmetry of C_k . I use the symmetry test of Miao et al. (2006) to test this implication on the centered residuals $\hat{M}_{k,N}$ at $N \in \{4, 8, 12, 16, 20\}$. The results are calibrated against a Rademacher sign-flip null and aggregated across subsampling draws of size N by the dependence-robust rule of Meinshausen et al. (2009). Symmetry is not rejected for either outcome at any depth, at any conventional level. While this evidence cannot be considered final towards testing Assumption 6.3, the results are consistent with block symmetry. Appendix G.3 develops the point in greater detail and reports the full results. Under the symmetry null, standard errors and significance levels below can be computed in the ordinary way: on the full sample, with no correction, and under Gaussian asymptotics.

8.2 Was the Realized Design Cost-Optimal?

I first inspect if the realized design can be rationalized by some cost structure, and how much is lost if the true one differs. Normalizing $V = 1$ and varying F , each ratio implies a budget $B = 175 \times (F + 17.9)$ and an oracle design $(K^*(\lambda_0), N^*(\lambda_0))$. Panel (a) of Figure 11 plots the oracle depth against the realized $N = 17.9$, and Panel (b) plots the excess variance of the realized design relative to the oracle.

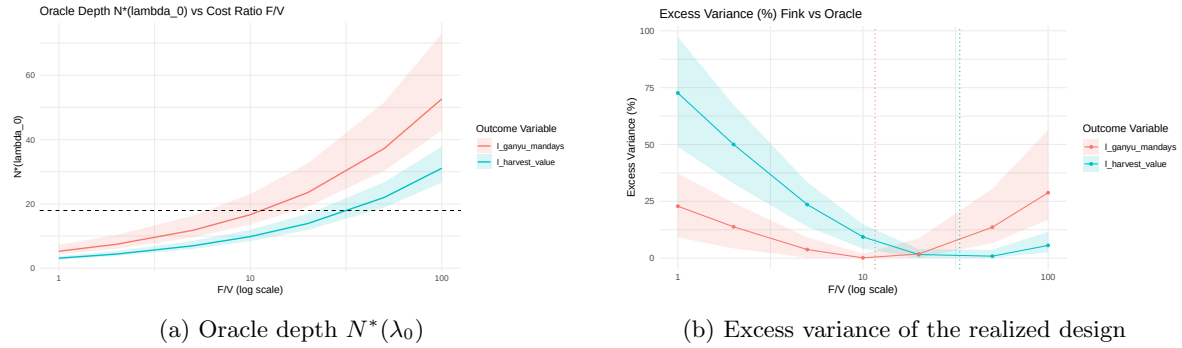


Figure 11: Realized FJM design against the oracle across F/V cost ratios

Notes: V normalized to one. Realized average depth $N = 17.9$ (dashed, left), rationalizing ratios (dotted, right). $B/V = 175 \times (F/V + 17.9)$. Ribbons are 95% cluster-bootstrap intervals, obtained by plugging the bootstrap percentiles of $\hat{\lambda}$ from Table 4 into the optimal map. Covariate-residualized data.

The realized design is rationalized at a cost ratio of 33.2 for harvest value and 11.6 for ganyu mandays, meaning that only at those ratios $N^*(\lambda_0)$ coincides with 17.9 and so the FJM design achieves zero excess variance. Away from the rationalized costs ratios the excess variance grows quickly, with oracle depths ranging from 3 to 52 units per village. At the low end of the cost grid the realized design carries an excess variance of about 70 percent for harvest value, and, at the high end, of about 30 percent for ganyu mandays. An excess variance of 70 percent corresponds to a variance reduction of roughly 40 percent, and a standard error reduction of roughly 25 percent. I emphasize once again that the current analysis remains silent on whether the realized FJM design is or is not cost optimal, yet, the broader point prevails. Without proper factorization of the statistical features of the data and cost structure, no design is safe from landing on the steep part of the excess variance curve.⁴⁴

⁴⁴Cluster availability is not the binding constraint at any point on the grid. The village list from which FJM sampled contains 733 villages with ten or more smallholder households, 853 with five or more, and 973 with three or more. Additional districts around Chipata District within Eastern Province were also available had they been needed.

8.3 What an Adaptive Design Would Have Delivered

Counterfactual designs cannot be run on the realized data, since any design at the same budget requires either more villages or more households per village than those surveyed by the authors. I therefore build a semi-synthetic population under the additive model $y_{ki} - \mathbf{x}_{ki}\boldsymbol{\beta} = \mu + u_k + \varepsilon_{ki}$, resampling the rescaled empirical cluster- and household-level residuals to match $(\hat{\mu}, \hat{\tau}^2, \hat{\sigma}^2)$ exactly. Appendix G.4 documents the construction and validates it against the realized quantile distribution. Figure 12 reports the excess variance of EtC-CS and ACS on this population, against the realized design, across the cost grid.

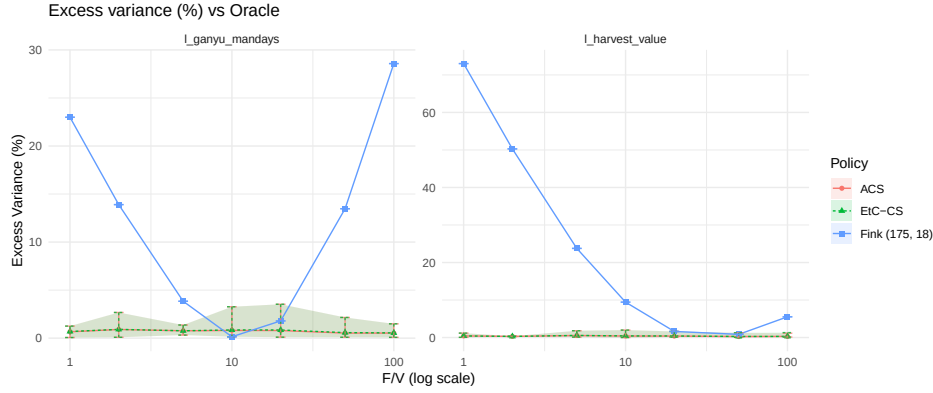


Figure 12: Excess variance across cost ratios

Notes: Excess variance is presented relative to the oracle design (%). Lines are means over 10,000 simulations, bands are 95% intervals across simulations.

Three features stand out. First, both algorithms are essentially optimal at every cost ratio, with mean excess variance close to zero and narrow confidence bands, especially for the harvest value outcome. The policies exhibit negligible excess variance relative to the oracle variance. Second, both algorithms display a similar performance. In this application, the budget supports several hundred villages, so even the worst case conservative pilot (K_M, N_m) entering EtC-CS identifies λ_0 with high precision, lowering the value of the extra clusters in ACS. Third, both algorithms dominate the realized design across the grid except at the rationalized ratio—which is unknown to the designer, hence the rationale for the exercise—.

Consequences for the reported estimates. Finally, I ask what the design would have meant for the paper’s own results. Applying the same construction to year-one outcomes, I re-estimate the treatment-effect specification of Fink et al. (2020b), Table 4 Panel B, on data generated under each counterfactual design. Tables 5 and 6 report the results. The synthetic replication of the realized design (column 3) tracks the real data (column 2) closely. The gap in standard errors comes from the synthetic design being balanced while the realized design is not, an imbalance that is pronounced for ganyu mandays.⁴⁵ Point estimates are stable across designs, as they should be—the design affects precision, not identification—while standard errors fall by up to 17 percent for harvest value and 19 percent for ganyu mandays relative to the balanced replication of the realized design (column 2). The reduction is even more significant when compared to the unbalanced design actually implemented.

⁴⁵My estimates differ slightly from those in Fink et al. (2020a) (column 1) because the published table pools year one and year two, whereas I use year one alone. Generating year-two data would have required modeling treatment rotation, differential attrition, and post-default exclusions, none of which the additive model accommodates cleanly.

	Paper (Real)	FJM (Real)	FJM (Synthetic)	F/V 1	F/V 2	F/V 5	F/V 10	F/V 20	F/V 50	F/V 100
Cash	0.105* (0.041)	0.112** (0.046)	0.112*** (0.042)	0.109*** (0.035)	0.111*** (0.038)	0.114*** (0.039)	0.113*** (0.041)	0.112*** (0.042)	0.111*** (0.042)	0.111*** (0.040)
Maize	0.056 (0.054)	0.037 (0.047)	0.036 (0.042)	0.034 (0.035)	0.037 (0.038)	0.037 (0.039)	0.036 (0.041)	0.037 (0.042)	0.035 (0.041)	0.038 (0.040)
K/N	175/17.9	175/17.9	175/18	774/3	543/4	332/7	245/10	194/14	165/22	157/32

Table 5: Treatment-effect estimates for `l.harvest_value`, year-one outcomes.

Notes: Column 1 is reproduced from [Fink et al. \(2020a\)](#), Table 4 Panel B (years one and two pooled). Counterfactual designs are generated by ACS; coefficients and standard errors are means over 1,000 simulations, and K/N reports the mean realized design. CR1 cluster standard errors, as in the original. Stars from the normal approximation: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

	FJM (Real)	FJM (Synthetic)	F/V 1	F/V 2	F/V 5	F/V 10	F/V 20	F/V 50	F/V 100
Cash	-0.182* (0.096)	-0.180** (0.080)	-0.186*** (0.065)	-0.180** (0.071)	-0.184** (0.076)	-0.186** (0.079)	-0.182** (0.080)	-0.182** (0.078)	-0.178** (0.074)
Maize	-0.210** (0.101)	-0.209*** (0.080)	-0.211*** (0.065)	-0.209*** (0.071)	-0.213*** (0.077)	-0.212*** (0.079)	-0.208*** (0.080)	-0.209*** (0.078)	-0.209*** (0.075)
K/N	175/17.9	175/18	521/6	357/8	236/12	181/17	151/24	135/38	135/53

Table 6: Treatment-effect estimates for `l.ganyu_mandays`, year-one outcomes.

Notes: This specification is not reported in [Fink et al. \(2020b\)](#). Counterfactual designs are generated by ACS; coefficients and standard errors are means over 1,000 simulations, and K/N reports the mean realized design. CR1 cluster standard errors. Stars from the normal approximation: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

9 Extension: Heterogeneous Cluster Variances

The baseline model explicitly imposes variance homogeneity across clusters. This section relaxes the restriction by allowing clusters to carry a set of observable covariates that shift the dispersion of its cluster-level error,

$$y_{ki} = \mu + u_k + \varepsilon_{ki}, \quad \mathbb{E}[u_k^2 | X_k] = \tau_k^2 = \ell(X_k, \gamma), \quad (X_k)_k \text{ iid} \in [x_m, x_M], \quad (11)$$

where the link ℓ is known up to a finite-dimensional parameter $\gamma \in \Gamma$, and the remaining error structure is unchanged: cross-cluster independence, homogeneous unit-level variance σ^2 with $\varepsilon_{ki} \perp\!\!\!\perp X_k$, and cross-error terms independence $u_k \perp\!\!\!\perp \varepsilon_{ki}$.⁴⁶ I further introduce two new population objects, namely the arithmetic and harmonic cluster variance means, and their corresponding variance ratios. Expectations are taken with respect to the distribution of covariates which is assumed to be known to the designer.

$$\bar{\tau}^2 := \mathbb{E}[\tau_k^2], \quad \tau_h^2 := \frac{1}{\mathbb{E}[1/\tau_k^2]}, \quad \bar{\lambda} := \frac{\sigma^2}{\bar{\tau}^2}, \quad \lambda_h := \frac{\sigma^2}{\tau_h^2}$$

⁴⁶Appendix H collects the formal assumptions (H1)–(H6): conditional location, unit-error homogeneity, the variance link, compactness—which bounds $\tau_k^2 \in [\tau_m^2, \tau_M^2]$ away from zero and degeneracy—interiority, and subgaussianity.

Timeline. The designer first commits to an estimator by selecting a set of non-negative weights entering a weighted average of cluster means $\hat{\mu}_\omega = \sum_k \omega_k \hat{\mu}_{N_k}^k / \sum_k \omega_k$. When the designer recruits a new cluster, she first pays the fixed cost F , then she observes X_k , and finally she selects N^k possibly as a function of X_k and the history. Therefore, cluster recruitment cannot depend on its own covariates and cannot be rejected ex post. The designer’s problem can be written as

$$\min_{K, \{N^k\}} \mathbb{E}[\text{Var}(\hat{\mu}_\omega \mid X^K)] = \min_{K, \{N^k\}} \mathbb{E} \left[\frac{1}{\left(\sum_k \omega_k\right)^2} \sum_k \omega_k^2 \left(\tau_k^2 + \frac{\sigma^2}{N^k} \right) \right] \quad \text{s.t.} \quad KF + V \sum_k N_k \leq B \quad \text{a.s.} \quad (12)$$

where $X^K = (X_1, \dots, X_K)$. I study (12) for the two leading estimators: the unweighted average with $\omega_k = 1/K$, and the GLS average with $\omega_k = 1/(\tau_k^2 + \sigma^2/N_k)$.⁴⁷

Heterogeneity is irrelevant for the unweighted average. Replacing $\omega_k = 1/K$ in (12) shows that cluster variances $(\tau_k^2)_k$ does not interact with depth. Intuitively, once cluster k is recruited, τ_k^2 enters the conditional variance additively and sunk, so no choice of N^k touches it. Depth only buys a reduction of σ^2/N^k , whose marginal value does not depend on X_k . The variance minimizing solution then yields a balanced covariate-free allocation, and the problem reduces to the original formulation. Lemma 9.1 formalizes this result.

Lemma 9.1. *Design Irrelevance under Equal Weights.* Under (H1)–(H4), random recruitment and no cluster rejection, the design problem for the unweighted average collapses to the homogeneous problem of Section 3 evaluated at $\bar{\tau}^2$. The optimal map equals $(K^*(\bar{\lambda}), N^*(\bar{\lambda}))$, hence it is constant at every realization of X^K . Similarly, the oracle variance is given by $W_u^* = \bar{\tau}^2(\sqrt{V\bar{\lambda}} + \sqrt{F})^2/B$.

Corollary 9.2. *EtC-CS is Robust to Heterogeneity.* Under (H1)–(H6), the sequence of cluster errors is unconditionally iid, so EtC-CS, run unchanged with inputs $(\bar{\tau}^2, \bar{\lambda})$, retains its $O(1/B^2)$ excess-variance guarantee against the oracle W_u^* .

The takeaway is clear. Heterogeneity on its own creates no motive for unbalanced sampling, so the baseline algorithm is robust to heterogeneity and requires no modification. The motive for unbalanced sampling emerges only when heterogeneity is interacted with weighted estimation.

Clairvoyant Designer and GLS weights. By construction, GLS weights deliver $\text{Var}(\hat{\mu}_\omega \mid X^K) = 1/\sum_k \omega_k$ with $\omega_k = (\tau_k^2 + \sigma^2/N^k)^{-1}$, so depth choices now interact with the heterogeneous variances τ_k^2 . The optimal solution is adapted, as N^k can only depend on the covariates of observed clusters, and not future ones. As a result, no closed-form solution exists for (12), but it can be approximated efficiently. In particular, consider a *clairvoyant* designer who observes the full covariate vector X^K before allocating. Her optimal allocation equalizes $\tau_k^2 N_k$ across clusters, so depth is inversely proportional to the cluster variance—the opposite of a Neyman allocation—. The reason is that from the depth’s perspective, τ_k^2 operates as an irreducible noise floor. Noisy clusters are downweighted anyway through the GLS weights, so depth efforts are focused on lowering the individual variance σ^2/N^k of quiet clusters. Substituting the optimal profile back, the clairvoyant variance collapses exactly to the baseline form,

$$\text{Var}(\hat{\mu}_\omega \mid K, X^K) = \frac{1}{K} \left(\tau_{h,K}^2 + \frac{\sigma^2}{N} \right), \quad \tau_{h,K}^2 := \frac{K}{\sum_k 1/\tau_k^2} \quad (13)$$

⁴⁷GLS weights are given by the inverse of the cluster variance conditional on the realized X_k .

with the realized harmonic mean in place of τ^2 and the average depth $\bar{N}$ in place of N . The only clairvoyant object in (13) is given by $\tau_{h,K}^2$. By Jensen, replacing this object by the population counterpart τ_h^2 delivers a feasible lower bound for the variance at negligible cost (provided $\tau_{h,K}^2 \rightarrow \tau_h^2$).

Proposition 9.3. *The Weighted Oracle. Under (H1)–(H5), every feasible policy with deterministic width satisfies*

$$\mathbb{E}[\text{Var}(\hat{\mu}_\omega \mid X^K)] \geq W_\omega^* := \frac{(\sqrt{F\tau_h^2} + \sqrt{V\sigma^2})^2}{B}$$

which is attained up to $O(1/B^2)$ by the pseudo-oracle maps

$$K_\omega^* = \frac{B}{F + \sqrt{\lambda_h F V}}, \quad N_{k,\omega}^* = \frac{\tau_h^2}{\tau_k^2} \sqrt{\lambda_h \frac{F}{V}} \quad (14)$$

Once again, the optimal coincides with the optimal maps of Section 3 evaluated at (τ_h^2, λ_h) , with an additional depth scaling term of τ_h^2/τ_k^2 . All the machinery developed for the homogeneous optimal maps therefore transfers nearly verbatim.

Proposition 9.4. *Ignoring Heterogeneity incurs First Order Loss. Under (H1)–(H4),*

$$W_u^* - W_\omega^* = \frac{F(\bar{\tau}^2 - \tau_h^2) + 2\sqrt{VF}\sigma^2(\bar{\tau} - \tau_h)}{B} = \Theta(1/B) \geq 0 \quad (15)$$

with equality if and only if τ_k^2 is degenerate. Additionally, the gap is increasing in the dispersion of τ_k^2 through the two Jensen gaps.

Assuming that learning will only cost $O(1/B^2)$ as in the main text, misspecification to heterogeneity under weighted averages becomes a first order concern. The excess variance is order equivalent to λ missimpuation in static policies and of the same order than the oracle variance itself.

Estimation and algorithm. The main obstacle to input identification concerns τ_k^2 . Each cluster contributes a single draw of u_k , so τ_k^2 is not identified cluster by cluster, but only identified by pooling across clusters through the link. I now briefly discuss the estimation procedure and leave the details for Appendix H. I first construct pseudo-cluster errors $q_{k,K}$ by re-centering squared demeaned cluster means, which are conditionally unbiased for τ_k^2 up to a uniform $O(1/K)$ term. I then estimate the γ vector via nonlinear least squares of $q_{k,K}$ on $\ell(X_k, \cdot)$. The resulting estimates, which are $\sqrt{K}$ -consistent under standard conditions and which are computed on a carefully sized pilot, enter the optimal map as in the benchmark algorithms. In particular, the weighted analogue of EtC-CS, denoted wEtC-CS, runs a conservative pilot, estimates $(\hat{\gamma}, \hat{\sigma}^2)$, and commits to a depth profile that is explored sequentially until exhausting the budget instead of committing to a fixed depth and terminal width. The reason is that cluster depth is unknown ex-ante, and so they are the cluster costs $F + V \sum_k N^k$, thus the terminal width cannot be determined ex-ante.

Algorithm 3 wEtC-CS

Input $B, F, V, \Gamma, [x_m, x_M], \ell(\cdot, \cdot), [\sigma_m^2, \sigma_M^2]$
derive $\tau_m^2, \tau_M^2, \lambda_m, \lambda_M, K_M^\omega = \lfloor K_\omega^*(\lambda_M) \rfloor, N_{\max} = \left\lfloor \frac{\tau_M^2}{\tau_m^2} \sqrt{\lambda_M F/V} \right\rfloor$
for clusters $1 \leq k \leq K_M^\omega$
 visit cluster k , **observe** X_k , **sample** $N_p^k = \left\lfloor \frac{\tau_m^2}{\tau_{k,M}^2} \sqrt{\lambda_m F/V} \right\rfloor$ units, with $\tau_{k,M}^2 = \sup_{\gamma \in \Gamma} \ell(X_k, \gamma)$
 estimate $\hat{w}^2, \hat{\gamma}, \hat{\tau}_k^2 = \ell(X_k, \hat{\gamma}), \hat{\tau}_h^2, \hat{\lambda}_h$
 complete every pilot cluster $k \leq K_M^\omega$ up to $N^k = \left\lfloor \frac{\hat{\tau}_h^2}{\hat{\tau}_k^2} \sqrt{\hat{\lambda}_h F/V} \right\rfloor$
while remaining budget $\geq F + V N_{\max}$
 visit a new cluster k , **observe** X_k , **sample** $N^k = \left\lfloor \frac{\hat{\tau}_h^2}{\hat{\tau}_k^2} \sqrt{\hat{\lambda}_h F/V} \right\rfloor$ units
Output the design $\{(X_k, N^k)\}_k$ and $\hat{\mu}_{\hat{w}}$ with weights $\hat{w}_k = (\hat{\tau}_k^2 + \hat{w}^2/N^k)^{-1}$

The Appendix shows that pilot sampling and the top-up of pilot clusters are implementable pathwise under the stated assumptions, so no if/else guardrails are imposed.

Conjecture 9.5. *Performance of wEtC-CS.* Under (H1)–(H6), $\mathbb{E}[\text{Var}(\hat{\mu}_{\hat{w}}^{GLS} \mid X^K)] - W_\omega^* = O(1/B^2)$.

Formalizing the conjecture on performance requires localizing the random terminal width around $K_\omega^*(\lambda_h)$, which I expect to follow the arguments of Appendix C. The pilot is $\Theta(B)$ clusters wide, so $(\hat{\gamma}, \hat{\tau}_h^2, \hat{\lambda}_h)$ concentrate at the parametric rate; the terminal width then deviates from $K_\omega^*(\lambda_h)$ by $O_p(\sqrt{B})$. Finally, the collapsed objective (13) is smooth and first-order stationary along the budget line at the oracle, so deviations of this size cost $O(1/B^2)$.

10 Conclusion

This paper turns the classical width-versus-depth allocation problem into a sequential learning problem and provides its first solution with formal guarantees. Two algorithms—EtC-CS, which requires a single design revision, and ACS, which recruits clusters one at a time under a data-dependent stopping rule—learn the intra-cluster correlation from a carefully sized pilot and reallocate the remaining budget through plug-in optimal rules. Their expected excess variance is of order $1/B^2$, a full order of the budget below the $\Theta(1/B)$ cost of any misspecified static design, maximin designs included. A matching lower bound shows the rate cannot be improved.

These variance gains do not come at the expense of inference. Under a symmetry restriction on the errors, conditioning on designs driven by asymptotically even functionals of the data—such as sample variances—leaves the estimator asymptotically normal with the usual clustered standard errors: no reweighting, no sample splitting, no discarded pilot data. The principle reaches beyond clustered sampling, validating standard inference for Neyman allocations computed from pilot variances, and it permits conditioning on design events with non-degenerate limiting probability, where earlier two-stage arguments do not apply. Its boundary is equally sharp: simulations show that asymmetric errors, or adaptation through odd functionals such as interim means or futility rules, restore the distortions that motivate the reweighting literature.

The empirical results indicate that the stakes are economically meaningful: in the reconstruction of Fink et al. (2020b), adaptive allocation delivers up to 41% lower variance for the same money, and the live survey shows the procedure runs in the field in real time. The framework’s limits chart

the path forward. Symmetry excludes binary and censored outcomes, costs are assumed known and linear, and cluster variances are homogeneous⁴⁸. Within these limits, the results provide a foundation for cost-aware adaptive data collection in multistage surveys and clustered randomized trials.

⁴⁸An extension to heterogeneous variances under GLS weighting is developed in an upcoming extension

References

- Abadie, A., Athey, S., Imbens, G. W., and Wooldridge, J. M. (2023). When should you adjust standard errors for clustering? *The Quarterly Journal of Economics*, 138(1):1–35.
- Adusumilli, K. (2026). How to sample and when to stop sampling: The generalised wald problem and minimax policies. *Review of Economic Studies*, 93(1):1–34.
- Anscombe, F. J. (1952). Large-sample theory of sequential estimation. In *Mathematical Proceedings of the Cambridge Philosophical Society*, volume 48, pages 600–607. Cambridge University Press.
- Bai, Y. (2022). Optimality of matched-pair designs in randomized controlled trials. *American Economic Review*, 112(12):3911–3940.
- Bai, Y., Jiang, L., Romano, J. P., Shaikh, A. M., and Zhang, Y. (2024). Covariate adjustment in experiments with matched pairs. *Journal of Econometrics*, 241(1):105740.
- Ballin, M. and Barcaroli, G. (2013). Joint determination of optimal stratification and sample allocation using genetic algorithm. *Survey Methodology*, 39(2):369–393.
- Barcaroli, G., Fasulo, A., Guandalini, A., and Terribili, M. D. (2022). Two-stage sampling design and sample selection with the r package r2beat. *arXiv preprint arXiv:2209.06532*.
- Bethel, J. (1989). Sample allocation in multivariate surveys. *Survey methodology*, 15(1):47–57.
- Biemer, P. P. and Lyberg, L. E. (2003). *Introduction to survey quality*. John Wiley & Sons.
- Bruhn, M. and McKenzie, D. (2009). In pursuit of balance: Randomization in practice in development field experiments. *American economic journal: applied economics*, 1(4):200–232.
- Cai, Y. and Rafi, A. (2024). On the performance of the Neyman allocation with small pilots. *Journal of Econometrics*, 242(1):105793.
- Campbell, M. K., Grimshaw, J. M., and Elbourne, D. R. (2004). Intraclass correlation coefficients in cluster randomized trials: empirical insights into how should they be reported. *BMC Medical Research Methodology*, 4(1):9.
- Canay, I. A., Romano, J. P., and Shaikh, A. M. (2017). Randomization tests under an approximate symmetry assumption. *Econometrica*, 85(3):1013–1030.
- Cesa-Bianchi, N., Colomboni, R., and Kasy, M. (2025). Adaptive maximization of social welfare. *Econometrica*, 93(3):1073–1104.
- Chambaz, A., van Der Laan, M. J., and Zheng, W. (2015). Targeted covariate-adjusted response-adaptive lasso-based randomized controlled trials. *Modern adaptive randomized clinical trials: statistical, operational, and regulatory aspects*, pages 345–368.
- Chow, S.-C. and Corey, R. (2011). Benefits, challenges and obstacles of adaptive clinical trial designs. *Orphanet journal of rare diseases*, 6(1):79.

- Chow, Y. S. and Robbins, H. (1965). On the asymptotic theory of fixed-width sequential confidence intervals for the mean. *The Annals of Mathematical Statistics*, 36(2):457–462.
- Cicchitelli, G., Herzel, A., and Montanari, G. E. (1992). *Il campionamento statistico*. il Mulino.
- Cochran, W. G. (1977). *Sampling techniques*, volume 3. wiley New York.
- Cytrynbaum, M. (2021). Optimal stratification of survey experiments. *arXiv preprint arXiv:2111.08157*.
- Cytrynbaum, M. (2024). Covariate adjustment in stratified experiments. *Quantitative Economics*, 15(4):971–998.
- Dai, J., Gradu, P., and Harshaw, C. (2023). Clip-OGD: An experimental design for adaptive Neyman allocation in sequential experiments. In *Advances in Neural Information Processing Systems*, volume 36, pages 32235–32269.
- Dudewicz, E. J. and Dalal, S. R. (1975). Allocation of observations in ranking and selection with unequal variances. *Sankhyā: The Indian Journal of Statistics, Series B*, pages 28–78.
- Fink, G., Jack, B. K., and Masiye, F. (2014). Food constraints, yield uncertainty and “ganyu” labour: Pre-analysis plan. Pre-analysis plan. AEA RCT Registry ID: AEARCTR-0000130.
- Fink, G., Jack, B. K., and Masiye, F. (2020a). Online appendix to seasonal liquidity, rural labor markets, and agricultural production. Online appendix. Appendix to American Economic Review, 110(11): 3351–3392.
- Fink, G., Jack, B. K., and Masiye, F. (2020b). Seasonal liquidity, rural labor markets, and agricultural production. *American Economic Review*, 110(11):3351–3392.
- Fink, G., Jack, B. K., and Masiye, F. (2024). Food constraints, yield uncertainty and “ganyu” labour. *AEA RCT Registry*.
- Gould, A. L. and Shih, W. J. (1992). Sample size re-estimation without unblinding for normally distributed outcomes with unknown variance. *Communications in Statistics – Theory and Methods*, 21(10):2833–2853.
- Hadad, V., Hirshberg, D. A., Zhan, R., Wager, S., and Athey, S. (2021). Confidence intervals for policy evaluation in adaptive experiments. *Proceedings of the national academy of sciences*, 118(15):e2014602118.
- Hahn, J., Hirano, K., and Karlan, D. (2011). Adaptive experimental design using the propensity score. *Journal of Business & Economic Statistics*, 29(1):96–108.
- Hansen, M. H. and Hurwitz, W. N. (1942). Relative efficiencies of various sampling units in population inquiries. *Journal of the American Statistical Association*, 37(217):89–94.
- Kasy, M. and Sautmann, A. (2021). Adaptive treatment assignment in experiments for policy choice. *Econometrica*, 89(1):113–132.

- Kiefer, J. and Wolfowitz, J. (1960). The equivalence of two extremum problems. *Canadian Journal of Mathematics*, 12:363–366.
- Kieser, M. and Friede, T. (2003). Simple procedures for blinded sample size adjustment that do not affect the type I error rate. *Statistics in Medicine*, 22(23):3571–3581.
- Kish, L. (1965). Survey sampling. new york: John wesley & sons. *Am Polit Sci Rev*, 59(4):1025.
- Knox, S. A. and Chondros, P. (2004). Observed intra-cluster correlation coefficients in a cluster survey sample of patient encounters in general practice in australia. *BMC Medical Research Methodology*, 4(1):30.
- Lake, S., Kammann, E., Klar, N., and Betensky, R. (2002). Sample size re-estimation in cluster randomization trials. *Statistics in Medicine*, 21(10):1337–1350.
- Lattimore, T. and Szepesvári, C. (2020). *Bandit algorithms*. Cambridge University Press.
- Li, D., Raymond, L., and Bergman, P. (2026). Hiring as exploration. *Review of Economic Studies*, 93(2):1200–1240.
- Lu, J. (2025). Big data analysis. *Springer Books*.
- Meinshausen, N., Meier, L., and Bühlmann, P. (2009). P-values for high-dimensional regression. *Journal of the American Statistical Association*, 104(488):1671–1681.
- Miao, W., Gel, Y. R., and Gastwirth, J. L. (2006). A new test of symmetry about an unknown median. In *Random walk, sequential analysis and related topics: A festschrift in honor of Yuan-Shih Chow*, pages 199–214. World Scientific.
- Moerbeek, M., van Breukelen, G. J. P., and Berger, M. P. F. (2000). Design issues for experiments in multilevel populations. *Journal of Educational and Behavioral Statistics*, 25(3):271–284.
- Neopane, O., Ramdas, A., and Singh, A. (2024). Logarithmic neyman regret for adaptive estimation of the average treatment effect. *arXiv preprint arXiv:2411.14341*.
- Neopane, O., Ramdas, A., and Singh, A. (2025). Optimistic algorithms for adaptive estimation of the average treatment effect. *arXiv preprint arXiv:2502.04673*.
- Nie, X., Tian, X., Taylor, J., and Zou, J. (2018). Why adaptively collected data have negative bias and how to correct for it. In *International Conference on Artificial Intelligence and Statistics*, pages 1261–1269. PMLR.
- Raudenbush, S. W. (1997). Statistical analysis and optimal design for cluster randomized trials. *Psychological Methods*, 2(2):173–185.
- Rinott, Y. (1978). On two-stage selection procedures and related probability-inequalities. *Communications in Statistics-Theory and methods*, 7(8):799–811.
- Rudelson, M. and Vershynin, R. (2013). Hanson-wright inequality and sub-gaussian concentration.
- Sedgwick, P. (2015). Multistage sampling. *Bmj*, 351.

- Shen, Y. and Fisher, L. (1999). Statistical inference for self-designing clinical trials with a one-sided hypothesis. *Biometrics*, 55(1):190–197.
- Spiegelhalter, D. J. (2001). Bayesian methods for cluster randomized trials with continuous responses. *Statistics in Medicine*, 20(3):435–452.
- Stein, C. (1945). A two-sample test for a linear hypothesis whose power is independent of the variance. *Annals of Mathematical Statistics*, 16(3):243–258.
- Tabord-Meehan, M. (2023). Stratification trees for adaptive randomisation in randomised controlled trials. *Review of Economic Studies*, 90(5):2646–2673.
- Tam, B. (2005). Household sample surveys in developing and transition countries.
- Tschuprow, A. (1923). On the two different aspects of the representative method: the method of stratified son the mathematical expectation of the moments of frequency distributions in the case of correlated observation sampling and the method of purposive selection. *Metron*, 2(646-683):192.
- Turner, R. M., Prevost, A. T., and Thompson, S. G. (2005). Prior distributions for the intraclass correlation coefficient, based on multiple previous estimates, and their application in cluster randomized trials. *Clinical Trials*, 2(2):108–118.
- van Breukelen, G. J. P. and Candel, M. J. J. M. (2015). Efficient design of cluster randomized and multicentre trials with unknown intraclass correlation. *Statistical Methods in Medical Research*, 24(5):540–556.
- van Breukelen, G. J. P. and Candel, M. J. J. M. (2018). Efficient design of cluster randomized trials with treatment-dependent costs and treatment-dependent unknown variances. *Statistics in Medicine*, 37:3027–3046.
- van Breukelen, G. J. P. and Candel, M. J. J. M. (2021). Maximin design of cluster randomized trials with heterogeneous costs and variances. *Biometrical Journal*, 63(7):1444–1463.
- van Schie, S. and Moerbeek, M. (2014). Re-estimating sample size in cluster randomised trials with active recruitment within clusters. *Statistics in Medicine*, 33(19):3253–3268.
- Viviano, D. (2022). Experimental design under network interference. *arXiv preprint arXiv:2003.08421*.
- Wager, S. (2024). Causal inference: A statistical learning approach. URL https://web.stanford.edu/swager/causal-inf_book.pdf.
- Wittes, J. and Brittain, E. (1990). The role of internal pilot studies in increasing the efficiency of clinical trials. *Statistics in Medicine*, 9(1-2):65–72.
- Wittes, J., Schabenberger, O., Zucker, D. M., Brittain, E., and Proschan, M. (1999). Internal pilot studies I: Type I error rate of the naive t -test. *Statistics in Medicine*, 18(24):3481–3491.
- Wooldridge, J. M. (2003). Cluster-sample methods in applied econometrics. *American Economic Review*, 93(2):133–138.

- Yin, G. and Shen, Y. (2005). Adaptive design and estimation in randomized clinical trials with correlated observations. *Biometrics*, 61(2):362–369.
- Zhang, K., Janson, L., and Murphy, S. (2021). Statistical inference with m-estimators on adaptively collected data. *Advances in neural information processing systems*, 34:7460–7471.

A Proofs and Appendix to Section 3

A.1 Generalized Problem

Let Assumptions 3.1, 3.2 and 3.3 hold throughout. Consider the general problem in which the designer samples an arbitrary number of units $N^k \geq 0$ from each cluster k subject to the budget constraint. Let $\mathbf{N} = (N^k)_k$, and define $\mathcal{K} = \{k : N^k > 0\}$ with $K = |\mathcal{K}|$. The variance of the average-of-averages estimator is

$$\begin{aligned} W(\mathbf{N}) &= \text{Var}(\hat{\mu}_{\mathbf{N}}) = \text{Var}\left(\frac{1}{K} \sum_{k \in \mathcal{K}} \frac{1}{N^k} \sum_{i=1}^{N^k} y_{ki}\right) = \text{Var}\left(\mu + \frac{1}{K} \sum_{k \in \mathcal{K}} u_k + \frac{1}{K} \sum_{k \in \mathcal{K}} \frac{1}{N^k} \sum_{i=1}^{N^k} \varepsilon_{ki}\right) \\ &= \frac{K}{K^2} \tau^2 + \frac{1}{K^2} \sum_{k \in \mathcal{K}} \frac{\sigma^2}{N^k} = \frac{\tau^2}{K} \left(1 + \frac{\lambda_0}{K} \sum_{k \in \mathcal{K}} \frac{1}{N^k}\right) = \frac{\tau^2}{K} \left(1 + \frac{\lambda_0}{N^H(\mathbf{N})}\right) \end{aligned}$$

where the second equality uses cross-cluster independence and $u_k \perp\!\!\!\perp \varepsilon_{ki}$ from Assumption 3.1, and

$$N^H(\mathbf{N}) = \left(\frac{1}{K} \sum_{k \in \mathcal{K}} \frac{1}{N^k}\right)^{-1}$$

is the harmonic mean of the sampled depths. As expected, for $N^k = N$ for all $k \in \mathcal{K}$ the expression reduces to (2). The generalized problem is

$$\min_{\mathbf{N}} W(\mathbf{N}) \quad \text{s.t.} \quad KF + V \sum_{k \in \mathcal{K}} N^k \leq B. \quad (16)$$

Conditional on $\mathcal{K}$, $W(\mathbf{N})$ is strictly decreasing in each N^k , so the unit-level budget constraint bind. Next, write $S = \sum_{k \in \mathcal{K}} N^k$, so that at the optimum, conditional on K , the total number of units is fixed, $S = (B - KF)/V$. By (16), W minimization then reduces to minimizing $\sum_{k \in \mathcal{K}} 1/N^k$. Since $x \mapsto 1/x$ is strictly convex on $(0, \infty)$, Jensen's inequality gives

$$\frac{1}{K} \sum_{k \in \mathcal{K}} \frac{1}{N^k} \geq \frac{1}{\frac{1}{K} \sum_{k \in \mathcal{K}} N^k} = \frac{K}{S}$$

equivalently $N^H(\mathbf{N}) \leq S/K$, with equality if and only if $N^k = S/K = N$ for all $k \in \mathcal{K}$. Hence $\sum_{k \in \mathcal{K}} 1/N^k \geq K^2/S$, and for any fixed K the variance is minimized by the balanced allocation.

By restricting to balanced allocations and substituting the binding budget constraint $K(N) = B/(F + VN)$,

$$W(K(N), N) = \frac{\tau^2(F + VN)}{B} \left(1 + \frac{\lambda_0}{N}\right) = \frac{\tau^2}{B} \left[F + V\lambda_0 + VN + \frac{F\lambda_0}{N}\right]$$

The expanded form makes both optimality conditions (FOC and SOC) immediate,

$$W_N = \frac{\tau^2}{B} \left[V - \frac{F\lambda_0}{N^2}\right] = 0 \implies N^*(\lambda_0) = \sqrt{\lambda_0 F/V}, \quad W_{NN} = \frac{2F\lambda_0\tau^2}{BN^3} > 0 \text{ for } N > 0$$

Strict convexity on $(0, \infty)$ makes the stationary point the global minimum, and Assumption 3.3 places

it in the interior of the feasible region. Finally, evaluating the objective at $N^*(\lambda_0)$ gives

$$W_0 = \frac{\tau^2}{B} \left[F + V\lambda_0 + 2\sqrt{FV\lambda_0} \right] = \frac{\tau^2 \left(\sqrt{F} + \sqrt{V\lambda_0} \right)^2}{B}$$

The optimal map and the oracle variance of the generalized problem therefore coincide with those of the balanced problem in (3). $\square$

A.2 Experimental Design

I first extend Assumption 3.1 to the potential outcomes framework and impose it throughout.

Assumption A.1. *Potential-Outcome Errors.* For $d \in \{0, 1\}$, $\{(u_k^0, u_k^1)\}_k$ is an iid sequence with $\mathbb{E}[u_k^d] = 0$, $\mathbb{E}[(u_k^d)^2] = \tau_d^2$ and $\text{Cov}(u_k^0, u_k^1) = \tau_{01}$; $\{(\varepsilon_{ki}^0, \varepsilon_{ki}^1)\}_{k,i}$ is an iid sequence with $\mathbb{E}[\varepsilon_{ki}^d] = 0$ and $\mathbb{E}[(\varepsilon_{ki}^d)^2] = \sigma_d^2$; $(u_k^d)_{k,d} \perp (\varepsilon_{ki}^d)_{k,i,d}$. Moreover, treatment assignment is independent of the full collection of potential outcomes.

A.2.1 Clusters as Treatment Level

The designer samples $K \ll \bar{K}$ clusters and $N \ll \bar{N}$ units per cluster, and randomly allocates $K/2$ clusters to treatment and $K/2$ to control. Let $D_{ki} \in \{0, 1\}$ denote the treatment status of unit i in cluster k . Because all units in a cluster share a treatment status, write $D_k = D_{ki}$. Let $y_{ki}(d, \mathbf{d})$ denote the potential outcome of unit i in cluster k when $D_{ki} = d$ and $(D_{kj})_{j \neq i} = \mathbf{d}$. Under cluster-level assignment $\mathbf{d} = (d, \dots, d)$, so the potential outcome admits the simplified representation $y_{ki}(d, \mathbf{d}) = y_{ki}(d)$, which is assumed to follow the nested-error model of (1),

$$y_{ki}(d) = \mu_d + u_k^d + \varepsilon_{ki}^d$$

The estimand is the total average treatment effect $\text{TATE} = \mathbb{E}[y_{ki}(1, \mathbf{1}) - y_{ki}(0, \mathbf{0})] = \mu_1 - \mu_0$, and the natural estimator is the difference in cluster means

$$\widehat{\text{TATE}}_{K,N} = \frac{2}{K} \left(\sum_{k:D_k=1} \hat{\mu}_N^k - \sum_{k:D_k=0} \hat{\mu}_N^k \right).$$

Under Assumption A.1, $\widehat{\text{TATE}}_{K,N}$ is unbiased for the TATE. Because a cluster contributes $u_k^{D_k}$ for exactly one d , the arms draw on disjoint sets of clusters and no covariance term $\text{Cov}(u_k^1, u_k^0)$ arises,

$$\text{Var}(\widehat{\text{TATE}}_{K,N}) = \frac{4}{K^2} \cdot \frac{K}{2} \left(\tau_1^2 + \frac{\sigma_1^2}{N} \right) + \frac{4}{K^2} \cdot \frac{K}{2} \left(\tau_0^2 + \frac{\sigma_0^2}{N} \right) = \frac{2}{K} \left(\tau_1^2 + \tau_0^2 + \frac{\sigma_1^2 + \sigma_0^2}{N} \right)$$

so that, in the notation of Table 1,

$$W(K, N) = \frac{2\tau^2}{K} \left(1 + \frac{\lambda}{N} \right), \quad \tau^2 = \tau_1^2 + \tau_0^2, \quad \sigma^2 = \sigma_0^2 + \sigma_1^2, \quad \lambda = \sigma^2 / \tau^2$$

A.2.2 Clusters as Blocks under No Interference

The designer samples $K \ll \bar{K}$ clusters and $N \ll \bar{N}$ units per cluster, and randomly allocates treatment to $N/2$ units within each cluster. Under no interference $y_{ki}(d, \mathbf{d}) = y_{ki}(d)$, and the estimand is the average treatment effect $\text{ATE} = \mathbb{E}[y_{ki}(1) - y_{ki}(0)] = \mu_1 - \mu_0$. The natural estimator is the average of

within-cluster differences in means,

$$\text{ATE}_{K,N} = \frac{2}{KN} \sum_{k=1}^K \left(\sum_{i:D_{ki}=1} y_{ki} - \sum_{i:D_{ki}=0} y_{ki} \right)$$

which again, under Assumption A.1, $\text{ATE}_{K,N}$ is unbiased for the ATE. Moreover, the term τ_{01} now appears in the estimator's variance as both potential cluster errors are active within the same cluster. In particular, $\text{Var}(\text{ATE}_{K,N}) = \frac{(\tau_1^2 + \tau_0^2 - 2\tau_{01})}{K} + \frac{2(\sigma_1^2 + \sigma_0^2)}{KN}$. Re-writing

$$W(K, N) = \frac{\tau^2}{K} \left(1 + \frac{\lambda}{N} \right), \quad \tau^2 = \tau_1^2 + \tau_0^2 - 2\tau_{01}, \quad \sigma^2 = 2(\sigma_0^2 + \sigma_1^2), \quad \lambda = \sigma^2/\tau^2$$

Remark A.2. *Degenerate Corner.* When $u_k^1 = u_k^0$ almost surely, $\tau_{01} = \tau_1^2 = \tau_0^2$ and hence $\tau^2 = 0$. In other words, within-cluster differencing removes the cluster-level noise entirely, so the variance is minimized by sampling a single cluster as deeply as the budget allows. Assumption 3.2 excludes this case by imposing $\tau_m^2 > 0$.

B Proofs of Section 4

Proof of Lemma 4.1. Throughout, write $M_{k,N} = \hat{\mu}_N^k - \mu = u_k + \bar{\varepsilon}_{k,N}$ with $\bar{\varepsilon}_{k,N} = \frac{1}{N} \sum_{i=1}^N \varepsilon_{ki}$. Define as well $\bar{M}_{K,N} = \frac{1}{K} \sum_{k=1}^K M_{k,N}$. Under Assumption 3.1, $(M_{k,N})_k$ is *iid*, mean zero, with $\mathbb{E}[M_{k,N}^2] = \tau^2 + \sigma^2/N = \theta_N^2 < \infty$.

Unbiasedness. Since each $y_{ki} - \hat{\mu}_N^k = \varepsilon_{ki} - \bar{\varepsilon}_{k,N}$ has mean zero and variance $\frac{N-1}{N}\sigma^2$,

$$\mathbb{E}[S_{k,N}^2] = \frac{1}{N-1} \sum_{i=1}^N \text{Var}(\varepsilon_{ki} - \bar{\varepsilon}_{k,N}) = \frac{1}{N-1} \cdot N \cdot \frac{N-1}{N} \sigma^2 = \sigma^2 \implies \mathbb{E}[\hat{w}_{K,N}^2] = \sigma^2$$

By the sample-variance identity applied to the *iid* sequence $(\hat{\mu}_N^k)_k$,

$$\mathbb{E}[\hat{b}_{K,N}^2] = \frac{1}{K-1} \mathbb{E} \left[\sum_{k=1}^K (M_{k,N} - \bar{M}_{K,N})^2 \right] = \text{Var}(M_{k,N}) = \tau^2 + \frac{\sigma^2}{N}$$

(i) Under K -asymptotics with $N \geq 2$ fixed, $\hat{w}_{K,N}^2 = \frac{1}{K} \sum_{k=1}^K S_{k,N}^2$ is an average of *iid* summands: cross-cluster independence (Assumption 3.1) gives $S_{k,N}^2 \perp\!\!\!\perp S_{l,N}^2$ for $k \neq l$, and $\mathbb{E}[S_{k,N}^2] = \sigma^2 < \infty$. The SLLN yields $\hat{w}_{K,N}^2 \xrightarrow[N]{\text{a.s.}} \sigma^2$.

Under N -asymptotics with $K \geq 1$ fixed, exchange the order of summation and expand each cluster's term:

$$\hat{w}_{K,N}^2 = \frac{1}{K} \sum_{k=1}^K \frac{1}{N-1} \sum_{i=1}^N (\varepsilon_{ki} - \bar{\varepsilon}_{k,N})^2 = \frac{1}{K} \sum_{k=1}^K \frac{N}{N-1} \left[\frac{1}{N} \sum_{i=1}^N \varepsilon_{ki}^2 - \bar{\varepsilon}_{k,N}^2 \right]$$

For each fixed k , the SLLN gives $\frac{1}{N} \sum_i \varepsilon_{ki}^2 \xrightarrow[N]{\text{a.s.}} \sigma^2$ and $\bar{\varepsilon}_{k,N} \xrightarrow[N]{\text{a.s.}} 0$, so by the CMT the bracket converges a.s. to σ^2 . Since $N/(N-1) \rightarrow 1$ and the outer average is over finitely many k , $\hat{w}_{K,N}^2 \xrightarrow[N]{\text{a.s.}} \sigma^2$.

(ii) With $N \geq 2$ fixed,

$$\hat{b}_{K,N}^2 = \frac{1}{K-1} \sum_{k=1}^K (M_{k,N} - \bar{M}_{K,N})^2 = \frac{K}{K-1} \left[\frac{1}{K} \sum_{k=1}^K M_{k,N}^2 - \bar{M}_{K,N}^2 \right]$$

The SLLN applied to the *iid* sequences $(M_{k,N}^2)_k$ and $(M_{k,N})_k$ gives $\frac{1}{K} \sum_k M_{k,N}^2 \xrightarrow{\text{a.s.}} \tau^2 + \sigma^2/N$ and $\bar{M}_{K,N} \xrightarrow{\text{a.s.}} 0$. Since $K/(K-1) \rightarrow 1$, the CMT delivers $\hat{b}_{K,N}^2 \xrightarrow{\text{a.s.}} \tau^2 + \sigma^2/N$.

(iii) By parts (i) and (ii) and the CMT, the unclipped cluster-variance estimator $\hat{\tau}_{K,N}^2$ satisfies

$$\hat{\tau}_{K,N}^2 = \hat{b}_{K,N}^2 - \frac{1}{N} \hat{w}_{K,N}^2 \xrightarrow{\text{a.s.}} \left(\tau^2 + \frac{\sigma^2}{N} \right) - \frac{\sigma^2}{N} = \tau^2$$

The projection $\Pi_{[a,b]}$ onto a closed interval is 1-Lipschitz, hence continuous, so $\hat{\tau}_{K,N}^2 = \Pi_{[\tau_m^2, \tau_M^2]}(\hat{\tau}_{K,N}^2) \xrightarrow{\text{a.s.}} \Pi_{[\tau_m^2, \tau_M^2]}(\tau^2) = \tau^2$, where the final equality uses that $\tau^2 \in [\tau_m^2, \tau_M^2]$ by Assumption 3.2. Because $\tau_m^2 > 0$, the denominator is bounded away from zero, so a further application of the CMT gives $\hat{w}_{K,N}^2 / \hat{\tau}_{K,N}^2 \xrightarrow{\text{a.s.}} \sigma^2 / \tau^2 = \lambda_0$. Applying the outer projection and using that $\lambda_0 \in [\lambda_m, \lambda_M]$,

$$\hat{\lambda}_{K,N} = \Pi_{[\lambda_m, \lambda_M]} \left(\frac{\hat{w}_{K,N}^2}{\hat{\tau}_{K,N}^2} \right) \xrightarrow{\text{a.s.}} \Pi_{[\lambda_m, \lambda_M]}(\lambda_0) = \lambda_0$$

Strict interiority of (τ^2, λ) was never invoked, yet this assumption will play an important role in future derivations. $\square$

Lemma B.1. *Random-Index Composition.* Let $(X_M)_{M \geq 1}$ be a sequence of random variables with $X_M \xrightarrow[M]{\text{a.s.}} X$, and let $(M_B)_B$ be a sequence of $\mathbb{N}$ -valued random variables where $\mathbb{P}(M_B \geq m) \rightarrow 1$ for every fixed $m \in \mathbb{N}$. Then $X_{M_B} \xrightarrow[B]{p} X$.

Proof of Lemma B.1. Fix $\epsilon, \delta > 0$ and set $T_M = \sup_{j \geq M} |X_j - X|$, which is measurable and non-increasing in M . Almost sure convergence gives $T_M \downarrow 0$ a.s., so there exists a finite M_0 such that $\mathbb{P}(T_{M_0} > \epsilon) < \delta/2$. Since $M_B \xrightarrow[B]{p} \infty$, there exists a finite (budget) B_0 such that $\mathbb{P}(M_B < M_0) < \delta/2$ for all $B \geq B_0$. On the event $\{M_B \geq M_0\}$ we then have $|X_{M_B} - X| \leq T_{M_0}$, hence for every $B \geq B_0$,

$$\mathbb{P}(|X_{M_B} - X| > \epsilon) \leq \mathbb{P}(M_B < M_0) + \mathbb{P}(T_{M_0} > \epsilon) < \delta \quad \square$$

Remark B.2. Lemma B.1 is used repeatedly to jump between a.s. convergence over a fixed divergent sequence and convergence in probability over a random divergent sequence, both in performance and inference results. Moreover, observe that the lemma operates without any restriction on the dependence between the random index $(M_B)_B$ and the sequence $(X_M)_M$.

Proof of Lemma 4.4. By Lemma 4.1(iii), $\hat{\lambda}_{K,N_p} \xrightarrow[K]{\text{a.s.}} \lambda_0$ along any deterministic diverging sequence of widths, with $N_p \geq 2$ fixed. Since $K_{pB} \rightarrow_p \infty$, Lemma B.1 applies with $X_K = \hat{\lambda}_{K,N_p}$ and $M_B = K_{pB}$, giving $\hat{\lambda}_{pB} \rightarrow_p \lambda_0$. Define the map $h(\lambda) = (F + \sqrt{FV\lambda})^{-1}$, which will be invoked repeatedly in the derivations. The map $h(\cdot)$ collects everything about the optimal width that does not scale with the budget. It is strictly decreasing, and, because $\lambda_m > 0$ under Assumption 3.2, it is continuously differentiable and Lipschitz on $[\lambda_m, \lambda_M]$ with constant $L_h = \sup_{\lambda \in [\lambda_m, \lambda_M]} |h'(\lambda)| < \infty$.

Under Assumption 3.3 the map $\lambda \mapsto (h(\lambda), N^*(\lambda))$ is continuous on $[\lambda_m, \lambda_M]$, so under the CMT $\nu_B^\pi = h(\hat{\lambda}_{pB}) \rightarrow_p h(\lambda_0) = \nu_0$ and $N_B^\pi = N^*(\hat{\lambda}_{pB}) \rightarrow_p N_0$. For the integer statement in the footnote, $\lfloor \cdot \rfloor$ is continuous at every non-integer point, so when $N_0 \notin \mathbb{N}$, $\lfloor N_B^\pi \rfloor \rightarrow_p \lfloor N_0 \rfloor$ by the CMT. Finally, convergence in probability of integer-valued sequences to an integer constant is equivalent to $\mathbb{P}(\lfloor N_B^\pi \rfloor = \lfloor N_0 \rfloor) \rightarrow 1$.

C Proofs of Section 5

Proof of Equation (7). Let Assumptions 3.1, 3.2, 3.3 hold throughout. Starting from the equality derived in Appendix A, for any design on the budget line $K(F + VN) = B$

$$W(K(N), N) = \frac{\tau^2}{B} \left[F + V\lambda_0 + VN + \frac{F\lambda_0}{N} \right]$$

Recall $N^*(\lambda) = \sqrt{\lambda F/V}$, hence $VN^*(\lambda) = \sqrt{FV\lambda}$ and $F\lambda_0/N^*(\lambda) = \sqrt{FV} \lambda_0/\sqrt{\lambda}$. Therefore

$$W(K^*(\lambda), N^*(\lambda); (\tau^2, \lambda_0)) = \frac{\tau^2}{B} \left[F + V\lambda_0 + \sqrt{FV} \left(\sqrt{\lambda} + \frac{\lambda_0}{\sqrt{\lambda}} \right) \right]$$

Differencing at λ and λ_0 ,

$$\Delta(\lambda) = \frac{\tau^2 \sqrt{FV}}{B} \left(\sqrt{\lambda} - 2\sqrt{\lambda_0} + \frac{\lambda_0}{\sqrt{\lambda}} \right) = \frac{\tau^2 \sqrt{FV}}{B} \cdot \frac{\lambda - 2\sqrt{\lambda\lambda_0} + \lambda_0}{\sqrt{\lambda}} = \frac{\tau^2 \sqrt{FV}}{B} \cdot \frac{(\sqrt{\lambda} - \sqrt{\lambda_0})^2}{\sqrt{\lambda}}$$

■

C.1 Static Policies

Proof of Lemma 5.2. Part (i). Observe

$$\frac{(\sqrt{\lambda_b} - \sqrt{\lambda_0})^2}{\sqrt{\lambda_b}} = \frac{(\lambda_b - \lambda_0)^2}{(\sqrt{\lambda_b} + \sqrt{\lambda_0})^2 \sqrt{\lambda_b}} \in \left[\frac{(\lambda_b - \lambda_0)^2}{4\lambda_M^{3/2}}, \frac{(\lambda_b - \lambda_0)^2}{4\lambda_m^{3/2}} \right]$$

where I used $\lambda_b, \lambda_0 \in [\lambda_m, \lambda_M]$. Bring this result into (7) and substitute $(\lambda_b - \lambda_0)^2 = b^2$. The resulting bounds are then $\Theta(1/B)$ for $b \neq 0$ as desired. For part (ii), first observe that $\Delta(\lambda) = \frac{\tau^2 \sqrt{FV}}{B} \cdot \frac{(\sqrt{\lambda} - \sqrt{\lambda_0})^2}{\sqrt{\lambda}}$ is monotonically increasing in $\tau^2 > 0$, so the maximum is achieved at $\tau^2 = \tau_M^2$. Additionally, for some fixed λ , the map $\sqrt{\lambda_0} \mapsto (\sqrt{\lambda} - \sqrt{\lambda_0})^2$ is the square of an affine function, hence convex, and attains its maximum at the end points, i.e. $\lambda_0 \in \{\lambda_m, \lambda_M\}$, whichever yields a higher loss. The minimax policy is then given by

$$\min_{\lambda} \max \left\{ \frac{(\sqrt{\lambda} - \sqrt{\lambda_M})^2}{\sqrt{\lambda}}, \frac{(\sqrt{\lambda} - \sqrt{\lambda_m})^2}{\sqrt{\lambda}} \right\}$$

which yields $\lambda_{mM} = \left(\frac{\sqrt{\lambda_m} + \sqrt{\lambda_M}}{2} \right)^2$. At λ_{mM} , the losses at λ_m and λ_M are equal, then,

evaluating either endpoint, say $\lambda_0 = \lambda_M$ and plug-in λ_{mM} into (7) gives

$$\max_{\omega \in \Omega} \Delta_{\pi_{mM}} = \max_{\omega \in \Omega} \Delta(\lambda_{mM}) = \frac{\tau_M^2 \sqrt{FV}}{B} \cdot \frac{\left(\frac{\sqrt{\lambda_m} + \sqrt{\lambda_M}}{2} - \sqrt{\lambda_M} \right)^2}{\frac{\sqrt{\lambda_m} + \sqrt{\lambda_M}}{2}} = \frac{\tau_M^2 \sqrt{FV}}{2B} \cdot \frac{(\sqrt{\lambda_m} - \sqrt{\lambda_M})^2}{\sqrt{\lambda_m} + \sqrt{\lambda_M}} \quad \square$$

C.2 Upper Bounds

Throughout this section, I will invoke repeatedly the notion of centered squared errors. The same objects will play a central role in the inference results of Section 6 and their proofs. As such, I first characterize them properly and discuss their concentration properties. Finally, I leverage these results as part of an intermediate lemma upper bounding the expected square deviation $\mathbb{E}[(\hat{\lambda}_p - \lambda_0)^2]$, as hinted in the main text. The proof of Theorem 5.4 will follow as a simple corollary.

Definition C.1. *Centered Squared Errors* Let $\varepsilon_k = (\varepsilon_{k1}, \varepsilon_{k2}, \dots)$. Recall the definitions of $\bar{\varepsilon}_{k,N}$, $S_{k,N}^2$, $\hat{w}_{K,N}^2$, $M_{k,N}$ and $\bar{M}_{K,N}$ in Section 3 and Appendix B. Define

$$\alpha_{k,N} = S_{k,N}^2 - \sigma^2, \quad \beta_{k,N} = M_{k,N}^2 - \theta_N^2$$

where $\mathbb{E}[\alpha_{k,N}] = \mathbb{E}[\beta_{k,N}] = 0$.

Remark C.2. *Between-Variance Decomposition.* The different proof techniques will exploit repeatedly that the within- and between-cluster variances are near-martingales. The mapping is exact for $\hat{w}_{K,N}^2$ as it is a sum of iid zero-mean errors,

$$\hat{w}_{K,N}^2 - \sigma^2 = \frac{1}{K} \sum_{k=1}^K \alpha_{k,N}$$

$\hat{b}_{K,N}^2$ however cannot be written as a sum of iid zero-mean errors as each of the individual terms are correlated through the de-meaning object $\bar{M}_{K,N}$ (the grand mean). An approximate decomposition remains possible

$$\begin{aligned} \hat{b}_{K,N}^2 - \theta_N^2 &= \frac{1}{K-1} \sum_{k=1}^K (M_{k,N} - \bar{M}_{K,N})^2 - \theta_N^2 \\ &= \frac{1}{K-1} \left(\sum_{k=1}^K M_{k,N}^2 - K \bar{M}_{K,N}^2 \right) - \theta_N^2 \\ &= \frac{1}{K-1} \left(K \theta_N^2 + \sum_{k=1}^K (M_{k,N}^2 - \theta_N^2) - K \bar{M}_{K,N}^2 \right) - \theta_N^2 \\ &= \frac{1}{K-1} \sum_{k=1}^K \beta_{k,N} + \frac{\theta_N^2}{K-1} - \frac{K}{K-1} \bar{M}_{K,N}^2 \end{aligned}$$

Lemma C.3. *Concentration of the Centered Errors.* Under the usual Assumptions and Assumption 5.3, $M_{k,N}, \alpha_{k,N}, \beta_{k,N}$ are subexponential with suitable parameters (κ, b) .

Proof of Lemma C.3. Let $\{c_j\}_{j \leq 4}$ represent a set of strictly positive finite constants. Let $\kappa_M = \sqrt{\kappa_u^2 + \kappa_\varepsilon^2/N}$. By independence of $u_k \perp \bar{\varepsilon}_{k,N}$, $M_{k,N}$ is κ_M -subgaussian. Hence, it is subexponential with parameters $\kappa = \kappa_M^2, b_M = 0$. $\beta_{k,N}$ is the centered square of $M_{k,N}$, a κ_M -

subgaussian variable, therefore it is subexponential with parameters $\kappa_B = c_1 \kappa_M^4$, $b_B = c_2 \kappa_M^2$. Next, define the centering matrices

$$\mathbf{P}_N = I_N - \frac{1}{N} \mathbf{1}\mathbf{1}', \quad \tilde{\mathbf{P}}_N = \frac{1}{N-1} \mathbf{P}_N$$

then

$$\alpha_{k,N} = \frac{1}{N-1} (\varepsilon_k' \mathbf{P}_N \varepsilon_k - (N-1)\sigma^2) = \varepsilon_k' \tilde{\mathbf{P}}_N \varepsilon_k - \sigma^2$$

such that $\mathbb{E}[\varepsilon_k' \mathbf{P}_N \varepsilon_k] = \sigma^2 \text{tr}(\mathbf{P}_N) = \sigma^2(N-1)$ and $\alpha_{k,N}$ is centered. By Hanson-Wright inequality (Rudelson and Vershynin, 2013), $\alpha_{k,N}$ is subexponential with parameters $\kappa_A = c_3 \kappa_\varepsilon^4 \|\tilde{\mathbf{P}}_N\|_F^2$ and $b_A = c_4 \kappa_\varepsilon^2 \|\tilde{\mathbf{P}}_N\|$, where $\|\cdot\|_F$ is the Frobenius norm and $\|\cdot\|$ is the operator norm. Because $\mathbf{P}_N$ is an orthogonal projection of rank $N-1$, $\|\mathbf{P}_N\|_F^2 = (N-1)$ and $\|\mathbf{P}_N\| = 1$. After scaling, $\alpha_{k,N}$ is subexponential with parameters $\kappa_A = c_3 \frac{\kappa_\varepsilon^4}{N-1}$ and $b_A = c_4 \frac{\kappa_\varepsilon^2}{N-1}$.

Lemma C.4. *Maximal Bernstein Inequality.* Let $\{\xi_k\}$ be a sequence of iid, mean-zero, (κ, b) -subexponential random variables. Then, for every $K_+ \geq 1$ and $t > 0$,

$$\mathbb{P} \left(\max_{1 \leq k \leq K_+} \left| \sum_{k=1}^K \xi_k \right| \geq t \right) \leq 2 \exp \left(-\frac{1}{2} \min \left(\frac{t^2}{\kappa \cdot K_+}, \frac{t}{b} \right) \right)$$

Proof of Lemma C.4. This result is standard, therefore omitted from the present manuscript.⁴⁹

Lemma C.5. *$\hat{\lambda}$ Concentration for K -asymptotic Pilot Policies.* Let π be a pilot policy with random pilot size (K_{pB}^π, N_p^π) with $N_p^\pi = N_p$ and $K_{pB}^\pi \in \mathcal{G}_B = [K_{B-}, K_{B+}] \cap \mathbb{N}$ a.s., where $K_{B+}/K_{B-} \leq C_0$ for some finite positive constant C_0 not depending on B and $K_{B-} \xrightarrow{B} \infty$. Under Assumptions 3.1, 3.2, 3.3 and 5.3, there exist positive finite constants C_1 and B_0 such that for all $B \geq B_0$ and every $\varepsilon > 0$,

$$\mathbb{P} \left(|\hat{\lambda}_{K_{pB}^\pi, N_p} - \lambda_0| \geq \varepsilon \right) \leq 6 \exp(-C_1 K_{B-} \varepsilon^2)$$

Consequently, for all $B \geq B_0$

$$\mathbb{E} \left[\left(\hat{\lambda}_{K_{pB}^\pi, N_p} - \lambda_0 \right)^2 \right] \leq \frac{6}{C_1 K_{B-}}$$

Proof of Lemma C.5.

Let $D_w := \sup_{K \in \mathcal{G}_B} |\hat{w}_{K, N_p}^2 - \sigma^2|$. For some $u > 0$

$$\begin{aligned} \mathbb{P}(D_w \geq u) &\leq \mathbb{P} \left(\frac{1}{K_{B-}} \max_{K_{B-} \leq K \leq K_{B+}} \left| \sum_{k=1}^K \alpha_{k, N_p} \right| \geq u \right) \leq 2 \exp \left(-\frac{1}{2} \min \left(\frac{u^2 K_{B-}^2}{K_{B+} \kappa_A}, \frac{u K_{B-}}{b_A} \right) \right) \\ &\leq 2 \exp \left(-\frac{K_{B-}}{2} \min \left(\frac{u^2}{C_0 \kappa_A}, \frac{u}{b_A} \right) \right) \end{aligned}$$

where the second inequality follows from maximal Bernstein's inequality (Lemma C.4) over $t = u K_{B-}$, given α_{k, N_p} is sub-exponential with constants (κ_A, b_A) by Lemma C.3. Similarly, define

⁴⁹See for instance Chapter 6 in Lu (2025).

$D_b := \sup_{K \in \mathcal{G}_B} |\hat{b}_{K, N_p}^2 - \theta_{N_p}^2|$, and recall the decomposition in Remark C.2

$$\hat{b}_{K, N_p}^2 - \theta_{N_p}^2 = \frac{K}{K-1} \cdot \frac{1}{K} \sum_{k=1}^K \beta_{k, N_p} + \frac{\theta_{N_p}^2}{K-1} - \frac{K}{K-1} \left(\frac{1}{K} \sum_{k=1}^K M_{k, N_p} \right)^2$$

Then, $D_b \leq 2D_B + 2D_M^2 + r_B$ where $D_B := \sup_{K \in \mathcal{G}_B} |\frac{1}{K} \sum_{k \leq K} \beta_{k, N_p}|$, $D_M := \sup_{K \in \mathcal{G}_B} |\frac{1}{K} \sum_{k \leq K} M_{k, N_p}|$, and $r_B := \frac{\tau_M^2(1+\lambda_M)}{K_B-1} < \infty$ (by Assumption 3.2). The inequality uses the fact that $K/(K-1) \leq 2$ for $K \geq 2$. I continue by arguing that for every $u \geq 3r_B$,

$$\begin{aligned} \mathbb{P}(D_b \geq u) &\leq \mathbb{P}\left(D_B \geq \frac{u}{6}\right) + \mathbb{P}\left(D_M \geq \sqrt{\frac{u}{6}}\right) + \mathbb{P}\left(r_B \geq \frac{u}{3}\right) \\ &\leq 2 \exp\left(-\frac{K_B-}{2} \min\left(\frac{u^2}{36C_0\kappa_B}, \frac{u}{6b_B}\right)\right) + 2 \exp\left(-\frac{K_B-}{2} \frac{u}{6C_0\kappa_M}\right) \end{aligned}$$

where the third term is zero by selection of u and the main inequality follows from maximal Bernstein over $(\beta_{k, N_p}, M_{k, N_p})$: the bound for M_{k, N_p} does not include the subexponential refinement because $b_M = 0$ — M_{k, N_p} is subgaussian. Next, I apply the triangle inequality twice to translate bounds over the primitives $\hat{w}_{K, N_p}^2$ and $\hat{b}_{K, N_p}^2$ into bounds over $\hat{\tau}_{K, N_p}^2$ and $\hat{\lambda}_{K, N_p}$. Specifically, $\tilde{\tau}_{K, N_p}^2 - \tau^2 = (\hat{b}_{K, N_p}^2 - \theta_{N_p}^2) - \frac{1}{N_p}(\hat{w}_{K, N_p}^2 - \sigma^2)$, hence by the TI, $\sup_{K \in \mathcal{G}_B} |\tilde{\tau}_{K, N_p}^2 - \tau^2| \leq D_b + D_w/N_p$. The same bound is mechanically preserved for the contraction $\hat{\tau}_{K, N_p}^2 = \Pi_{[\tau_m^2, \tau_M^2]}(\tilde{\tau}_{K, N_p}^2)$. Similarly,

$$\sup_{K \in \mathcal{G}_B} |\tilde{\lambda}_{K, N_p} - \lambda_0| = \sup_{K \in \mathcal{G}_B} \left| \frac{\tau^2(\hat{w}_{K, N_p}^2 - \sigma^2) - \sigma^2(\hat{\tau}_{K, N_p}^2 - \tau^2)}{\tau^2 \hat{\tau}_{K, N_p}^2} \right| \leq \frac{1}{\tau_m^2} \left[\left(1 + \frac{\lambda_M}{N_p}\right) D_w + \lambda_M D_b \right] \leq L(D_w + D_b)$$

with $L := (1 + \lambda_M)/\tau_m^2$. The first inequality follows from $\hat{\tau}^2 \geq \tau_m^2$ (clipping) and $\sigma^2/\tau^2 = \lambda_0 \leq \lambda_M$ (compactness), and the second inequality from $N_p \geq 1$. Clipping of $\tilde{\lambda}$ preserves the bound. Therefore,

$$\mathbb{P}\left(|\hat{\lambda}_{K_{pB}^\pi, N_p} - \lambda_0| \geq \varepsilon\right) \leq \mathbb{P}\left(\sup_{K \in \mathcal{G}_B} |\hat{\lambda}_{K, N_p} - \lambda_0| \geq \varepsilon\right) \leq \mathbb{P}\left(D_w \geq \frac{\varepsilon}{2L}\right) + \mathbb{P}\left(D_b \geq \frac{\varepsilon}{2L}\right)$$

The first inequality is crucial as it discharges the complications arising from ACS's random stopping time. Since $K_{pB}^{\text{ACS}} \in \mathcal{G}_B$ a.s., the random index is dominated by the supremum, which is why the window ratio C_0 is assumed to be bounded. Now, observe that since $\hat{\lambda}_{K_{pB}^\pi, N_p}$ and λ_0 both lie in $[\lambda_m, \lambda_M]$, the probability on the LHS is zero for $\varepsilon > \Lambda := \lambda_M - \lambda_m$, so it suffices to establish the inequality for $\varepsilon \in (0, \Lambda]$. To do so, I evaluate the probability bounds of the primitives on $u = \varepsilon/2L$ with $u \in (0, u_M)$, $u_M := \Lambda/2L$. I note that on a bounded range, the Bernstein exponent can be uniformly lower-bounded by a quadratic function of u , such that

$$\min\{au^2, bu\} \geq u^2 \cdot \min\left\{a, \frac{b}{u_M}\right\}$$

so each of the three exponential bounds are bounded by $2 \exp(-C_j K_B \varepsilon^2)$, for positive finite constants (C_w, C_B, C_M) which absorb and depend only on the sub-exponential parameters, the window width C_0 , Λ and L . Setting $C_1 := \min\{C_w, C_B, C_M\}$ delivers the statement in the Lemma

with a prefactor of 6 for all $\varepsilon \in (0, \Lambda]$ satisfying $\varepsilon \geq \varepsilon_B := 6Lr_B$ (as per the requirement in bounding D_b). Finally, one can show that small deviations—i.e. $\varepsilon \in (0, \varepsilon_B)$ —are trivial for large B . Since $\varepsilon_B = O(1/K_{B-})$ and $K_{B-} \xrightarrow{B} \infty$, B_0 can be selected such that $C_1 K_{B-} \varepsilon_B^2 \leq \ln 6$ for all $B \geq B_0$. Then, for every $\varepsilon < \varepsilon_B$,

$$6 \exp(-C_1 K_{B-} \varepsilon^2) \geq 6 \exp(-C_1 K_{B-} \varepsilon_B^2) \geq 6e^{-\ln 6} = 1$$

so the probability statement holds trivially. To derive the expectation bound, simply integrate over the tail, i.e

$$\mathbb{E} \left[\left(\hat{\lambda}_{K_{pB}^\pi, N_p} - \lambda_0 \right)^2 \right] = \int_0^\infty \mathbb{P} \left(\left| \hat{\lambda}_{K_{pB}^\pi, N_p} - \lambda_0 \right| > \sqrt{t} \right) dt \leq \int_0^\infty 6 \exp(-C_1 K_{B-} t) dt = \frac{6}{C_1 K_{B-}}$$

■

Remark C.6. *Role of Clipping.* Clipping plays two distinct roles in Lemma C.5. First, it bounds the estimation error by Λ , so that only deviations $\varepsilon \leq \Lambda$ carry probability and all Bernstein regimes collapse into the quadratic one, i.e. on its effective range, the clipped estimator is sub-gaussian. Second, it bounds the denominator $\hat{\tau}^2$ away from zero. Both roles require the parameter space to be bounded away from degeneracy, i.e. $\tau_m^2 > 0$, as per Assumption 3.2. On the other hand, the unclipped ratio $\tilde{\lambda}$ need not admit sub-exponential tails as it is a ratio of sample moments and its extreme deviations are governed by a small-denominator event.

Proof of Theorem 5.4. I first argue that both algorithms satisfy the assumptions entering Lemma C.5. First, both algorithms set $N_{pB}^\pi = N_m$ fixed. For EtC-CS, $K_{pB}^\pi = K_M$ deterministically, so the window is degenerate $K_{B-} = K_{B+}$ and $C_0 = 1$. For ACS, $K_{pB}^\pi = \max\{s-1, K_M\} \in [K_M, K_m]$ a.s., therefore, for all large B , the assumption is satisfied for $C_0 = K_m/K_M \leq 2h(\lambda_m)/h(\lambda_M)$ not depending on B . Finally, both EtC-CS and ACS trivially sample $K_{pB}^\pi \geq K_M \geq Bh(\lambda_M) - 1 \xrightarrow{B} \infty$.

Next, fix $\pi \in \{\text{EtC-CS}, \text{ACS}\}$. Under the usual assumptions, the policy implements an optimal map with $\lambda = \hat{\lambda}_{K_{pB}^\pi, N_m}$. Using the representation of the excess variance in (8) and the upper bound in the proof of Lemma 5.2

$$\mathbb{E}_\pi [\Delta_\pi] \leq \frac{\tau_M^2 \sqrt{FV}}{4\lambda_m^{3/2}} \cdot \frac{1}{B} \cdot \mathbb{E} \left[\left(\hat{\lambda}_{K_{pB}^\pi, N_p} - \lambda_0 \right)^2 \right] \leq \frac{\tau_M^2 \sqrt{FV}}{4\lambda_m^{3/2}} \cdot \frac{6}{C_1 B K_M}$$

where the second inequality follows from Lemma C.5. Recall $K_M = \Theta(B)$, so the right hand side is $O(B^{-2})$. Absorbing constants, $\mathbb{E}_\pi [\Delta_\pi] \leq C_\pi / B^2$ with C_π depending on $(\tau_M^2, \lambda_m, \lambda_M, F, V, N_m, C_0, C_1) \perp B$. Setting $C_\pi = C_{\text{EtC-CS}}$ for EtC-CS and $C_\pi = C_{\text{ACS}}$ for ACS completes the proof with constants differing only in the window width C_0 . □

C.3 Lower Bounds – Proof of Theorem 5.5

Policy Class. I first define the relevant competitor policy class Π . The policy class is fairly general and contains most reasonable policies as special cases. In particular, I consider policies which, in every period t , select a cluster $k_t = k$ and sample at least one unit $N_t^{k_t} \geq 1$ until they run out of budget. As a result, the game contains at most $T \leq \lfloor B/V \rfloor$ rounds with strictly positive sampling. If no additional observation can be sampled without violating the remaining budget, the algorithm terminates. Importantly, the same cluster k can be revisited several times. Let $N_T^k = \sum_{t=1}^T \mathbb{1}(k_t = k)$

$k)N_t^{k_t}$ be the number of samples from cluster k up to period T . Define as well $K_T = \sum_k \mathbb{1}(N_T^k > 0)$ and $\mathcal{K}_T = \{k : N_T^k > 0\}$. All policies must satisfy the budget constraint almost surely, i.e.

$$\mathbb{P}_\pi \left(\sum_{k \in \mathcal{K}_T} (F + VN_T^k) \leq B \right) = 1$$

Data Generating Process. Fix an instance $(\tau_0^2, \lambda_0) \in \text{int}(\Omega)$ and set $\sigma^2 = \lambda_0 \tau_0^2$. Consider two DGPs (y_-, y_+) indexed by $\omega_- = (\tau_-^2, \lambda_-)$ and $\omega_+ = (\tau_+^2, \lambda_+)$, where $\lambda_\pm = \lambda_0 \pm \delta_B$ for some small positive $\delta_B \rightarrow 0$. $\tau_\pm^2 = \sigma^2 / \lambda_\pm$, so that the individual-level variance σ^2 is common to both environments and only the cluster-level variance differs. For B large enough $(\tau_\pm^2, \lambda_\pm) \in \Omega$. The additive errors model is preserved as in (1),

$$y_{\pm ki} = \mu + u_{\pm k} + \varepsilon_{ki}$$

with $u_{\pm k} \sim N(0, \tau_\pm^2)$ and $\varepsilon_{ki} \sim N(0, \sigma^2)$.⁵⁰

Separating Events. As is common practice in lower bound derivations, I characterize a separating event. The separating event is written in terms of the harmonic mean N_T^H . By the linearity of the cost structure $K_T F + \sum_{k \in \mathcal{K}_T} VN_T^k = K_T(F + V\bar{N}_T)$ where $\bar{N}_T = \frac{1}{K_T} \sum_{k \in \mathcal{K}_T} N_T^k$ is the arithmetic mean. Recall that $N_T^H \leq \bar{N}_T$, hence $\frac{1}{K_T} \geq \frac{F + VN_T^H}{B}$. It follows

$$W(\mathbf{N}_T) = \frac{\tau^2}{K_T(\mathbf{N}_T)} \left(1 + \frac{\lambda}{N_T^H(\mathbf{N}_T)} \right) \geq \frac{\tau^2(F + VN_T^H(\mathbf{N}_T))}{B} \left(1 + \frac{\lambda}{N_T^H(\mathbf{N}_T)} \right) = W^H(N_T^H)$$

Next, let $N_\pm = N^*(\lambda_\pm) = \sqrt{\lambda_\pm F/V}$ and $N_0 = N^*(\lambda_0)$, so that $N_- < N_0 < N_+$. Define the separating event $E = \{N_T^H \leq N_0\}$. In environment ω_+ , $W^H(\cdot; \omega_+)$ is strictly convex with minimum at $N_+ > N_0$, hence decreasing on $(0, N_+]$. It follows that

$$W(\mathbf{N}_T; \omega_+) - W(N_+; \omega_+) \geq W^H(N_T^H; \omega_+) - W^H(N_+; \omega_+) \underset{\text{on } E}{\geq} W^H(N_0; \omega_+) - W^H(N_+; \omega_+)$$

where the first inequality uses the harmonic mean relaxation and the fact that $W(N_+; \omega_+) = W^H(N_+; \omega_+)$ —equality follows from the balancedness result in Appendix A—, and the second inequality uses the monotonicity argument above. On E^c , $W^H(\cdot; \omega_-)$ is increasing on $[N_-, \infty)$ and $N_T^H > N_0 > N_-$, so the same inequality runs for ω_- . The same computation as in Equation (7)—with truth $\lambda_\pm$ and input λ_0 —yields

$$W^H(N_0; \omega_\pm) - W^H(N_\pm; \omega_\pm) = \frac{\tau_\pm^2 \sqrt{FV}}{B} \cdot \frac{(\sqrt{\lambda_0} - \sqrt{\lambda_\pm})^2}{\sqrt{\lambda_0}}$$

Finally, both branches can be uniformly bounded in terms of δ_B and common parameters by noting: (i) $(\sqrt{\lambda_0} - \sqrt{\lambda_\pm})^2 = \delta_B^2 / (\sqrt{\lambda_0} + \sqrt{\lambda_\pm})^2 \geq \delta_B^2 / (4\lambda_M)$, (ii) $\lambda_0^{-1/2} \geq \lambda_M^{-1/2}$, and (iii) $\sigma^2 / \lambda_M \leq$

⁵⁰Gaussian errors are exactly subgaussian, so the lower bound is established over a strict subclass of the environment considered in Theorem 5.5, thus, the lower bound carries over to the broad environment class. Gaussianity is only invoked to derive closed-form solutions to the KL-divergence below.

$$\min\{\tau_-^2, \tau_+^2\}$$

$$W^H(N_T^H; \omega_\pm) - W^H(N_\pm; \omega_\pm) \geq \frac{\sigma^2 \sqrt{FV}}{4\lambda_M^{5/2} B} \delta_B^2 \quad \text{on } E \text{ (for } \omega_+) \text{ and } E^c \text{ (for } \omega_-)$$

The derivation matches the lower bound in the proof of Lemma 5.2 by lower bounding $\tau_\pm^2 \geq \sigma^2/\lambda_M$.

Augmented Experiment and Divergence Decomposition. For some policy $\pi \in \Pi$, let $(\tilde{\mathbb{P}}_{\pi-}, \tilde{\mathbb{P}}_{\pi+})$ be the probability measures induced by the interaction between the policy π and the environments y_- and y_+ , respectively. Now, consider the augmented experiment where, every time the learner samples cluster k , she observes both $(y_{ki})_{i \leq N_k}$ **and** u_k . Let $(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+})$ be the probability measures induced by the interaction between the policy π and the augmented environments. By data processing,

$$D_{\text{KL}}(\tilde{\mathbb{P}}_{\pi-}, \tilde{\mathbb{P}}_{\pi+}) \leq D_{\text{KL}}(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+})$$

The augmented game simplifies the analysis of the KL-divergence significantly, as outcomes remain *iid* conditional on the history and revisiting the cluster does not contribute further to the divergence.⁵¹ In addition, unit-level errors share the same distribution across environments, so the KL-divergence analysis is limited to the cluster-level, i.e.

$$D_{\text{KL}}(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+}) = \mathbb{E}_{\pi-}[K_T] D_{\text{KL}}(N(0, \tau_-^2), N(0, \tau_+^2))$$

Solving for D_{KL} explicitly,

$$\begin{aligned} D_{\text{KL}}(N(0, \tau_-^2), N(0, \tau_+^2)) &= \frac{1}{2} \left[\frac{\tau_-^2}{\tau_+^2} - 1 - \ln \left(\frac{\tau_-^2}{\tau_+^2} \right) \right] = \frac{1}{2} \left[\frac{\lambda_+}{\lambda_-} - 1 - \ln \left(\frac{\lambda_+}{\lambda_-} \right) \right] \\ &= \frac{1}{2} \phi \left(\frac{\lambda_+}{\lambda_-} \right) = \frac{1}{2} \phi \left(1 + \frac{2\delta_B}{\lambda_0 - \delta_B} \right) \end{aligned}$$

where the third equality serves as definition of $\phi(x) = x - 1 - \ln(x)$. $\phi(x)$ is monotonically increasing in $x \geq 1$ because $\phi'(x) = 1 - 1/x$, so for all sufficiently large B choose $\delta_B \leq \lambda_0/2$, and using the fact that $\phi(1+x) \leq x^2/2$

$$\frac{1}{2} \phi \left(\frac{\lambda_+}{\lambda_-} \right) \leq \frac{1}{2} \phi \left(1 + \frac{4\delta_B}{\lambda_0} \right) \leq \frac{4\delta_B^2}{\lambda_0^2}$$

Finally, observe that for all policies $\pi \in \Pi$, $FK_T + \sum_k VN_T^k \leq B$ a.s., thus $K_T \leq B/F$ a.s.. Bringing both terms together

$$D_{\text{KL}}(\tilde{\mathbb{P}}_{\pi-}, \tilde{\mathbb{P}}_{\pi+}) \leq D_{\text{KL}}(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+}) \leq \frac{B}{F} \frac{4\delta_B^2}{\lambda_0^2} = C_{\text{KL}} B \delta_B^2$$

⁵¹For a complete derivation of the chain rule under history-conditional *iid* outcomes see Lemma 15.1 in [Lattimore and Szepesvári \(2020\)](#).

Collecting Terms and BH Inequality. The argument is now completed through a standard Bretagnolle-Huber argument⁵²

$$\begin{aligned}
& \tilde{\mathbb{E}}_{\pi-} [W(\mathbf{N}; \omega_-)] - W(K^*(\lambda_-), N^*(\lambda_-); \omega_-) + \tilde{\mathbb{E}}_{\pi+} [W(\mathbf{N}; \omega_+)] - W(K^*(\lambda_+), N^*(\lambda_+); \omega_+) \geq \\
& \tilde{\mathbb{E}}_{\pi-} [W^H(N_T^H; \omega_-)] - W^H(N_-; \omega_-) + \tilde{\mathbb{E}}_{\pi+} [W^H(N_T^H; \omega_+)] - W^H(N_+; \omega_+) \geq \\
& \tilde{\mathbb{P}}_{\pi-}(E^c) \cdot \left(\tilde{\mathbb{E}}_{\pi-} [W^H(N_T^H; \omega_-) \mid E^c] - W^H(N_-; \omega_-) \right) + \tilde{\mathbb{P}}_{\pi+}(E) \cdot \left(\tilde{\mathbb{E}}_{\pi+} [W^H(N_T^H; \omega_+) \mid E] - W^H(N_+; \omega_+) \right) \geq \\
& \frac{\sigma^2 \sqrt{FV}}{4B\lambda_M^{5/2}} \delta_B^2 \left(\tilde{\mathbb{P}}_{\pi-}(E^c) + \tilde{\mathbb{P}}_{\pi+}(E) \right) \geq \frac{\sigma^2 \sqrt{FV}}{4B\lambda_M^{5/2}} \delta_B^2 \frac{1}{2} \exp \left(-D_{\text{KL}}(\tilde{\mathbb{P}}_{\pi-}, \tilde{\mathbb{P}}_{\pi+}) \right) \geq \\
& \frac{\sigma^2 \sqrt{FV}}{4B\lambda_M^{5/2}} \delta_B^2 \frac{1}{2} \exp \left(-D_{\text{KL}}(\mathbb{P}_{\pi-}, \mathbb{P}_{\pi+}) \right) \geq \frac{\sigma^2 \sqrt{FV}}{8B\lambda_M^{5/2}} \delta_B^2 \exp \left(-C_{\text{KL}} \delta_B^2 B \right)
\end{aligned}$$

Where the first inequality follows from the variance relaxation under the harmonic mean, the second inequality is a simple event partition, the third inequality follows from the uniform lower bound on excess variance, the fourth inequality follows from Bretagnolle-Huber inequality, the fifth inequality from a data processing argument, and the final inequality from the derivations above. Finally, by selecting $\delta_B = \sqrt{1/(C_{\text{KL}}B)} \xrightarrow{B} 0$,

$$\frac{\sigma^2 \sqrt{FV}}{8B\lambda_M^{5/2}} \frac{1}{C_{\text{KL}}B} \exp(-1) \geq C_2/B^2$$

for some finite and strictly positive C_2 not depending on (π, B) . The statement of the theorem follows from applying the inequality $\max\{a, b\} \geq (a+b)/2$ and absorbing it into a constant C . $\square$

D Design-Robust Inference for Asymptotically Even Policies

This section is a self-contained technical note on inference for adaptive designs under symmetry. Its content is that, under a symmetry assumption on the errors, inference on sample averages is unaffected by conditioning on designs induced by even or asymptotically even policies. The result applies well beyond the clustered sampling framework studied in the main text. The notation follows the multi-armed adaptive-design literature: N is the horizon (indexed by i) and K the number of arms (indexed by k). It should not be conflated with the K versus N notation of the main text, where both are design choices.

	Main text (Section 6)	This Appendix
Asymptotic index	Budget B	Horizon N
Summands	Clusters $k \leq K_B^\pi$	Pulls $s \leq N_N^k$
Groups	One population	K arms
Depth (design choice)	N_B^π	—

D.1 Setup and Definitions

Definition D.1. *Random Sequences.* For each arm $k \in [K] = \{1, \dots, K\}$ define an infinite iid error sequence $(\varepsilon_s^k)_{s \geq 1}$, where s counts the times arm k has been played. Let $(\zeta_s^k)_{k \geq 1, s \geq 1}$ be an iid Rademacher

⁵²See Theorem 14.2 in [Lattimore and Szepesvári \(2020\)](#).

array, independent from the errors, used to break ties in the sign decomposition of Definition D.9, and let $(U_i)_{i \geq 1}$ be an iid $\text{Unif}[0, 1]$ sequence, also independent from the errors, available to the policy as an independent randomization device.⁵³

Definition D.2. *Sampling Protocol.* In each period $i = 1, 2, \dots, N$: (i) the learner selects an arm $k_i \in [K]$ according to a history-adapted policy π ; (ii) the pull counts update, $N^k(i) = N^k(i-1) + \mathbb{1}(k_i = k)$ with $N^k(0) = 0$; and (iii) the learner observes

$$X_i = \mu_{k_i} + \varepsilon_{N^{k_i}(i)}^{k_i}$$

Write $N_N^k := N^k(N)$ for the terminal count, also referred to as terminal design. It is convenient to index arm-level statistics by pulls rather than periods, so that $X_s^k = \mu_k + \varepsilon_s^k$ denotes the s -th observation from arm k . Moreover, for $M \geq 1$ set

$$\hat{\mu}_M^k = \frac{1}{M} \sum_{s=1}^M X_s^k, \quad \bar{\varepsilon}_M^k = \frac{1}{M} \sum_{s=1}^M \varepsilon_s^k, \quad Z_M^k = \sqrt{M}(\hat{\mu}_M^k - \mu_k) = \frac{1}{\sqrt{M}} \sum_{s=1}^M \varepsilon_s^k$$

with $Z_0^k = 0$. The terminal average of arm k is $\hat{\mu}_{N_N^k}^k$, defined on $\{N_N^k > 0\}$.

Assumption D.3. *Errors.*

i) **Moments.** $\mathbb{E}[\varepsilon_s^k] = 0$ and $\mathbb{E}[(\varepsilon_s^k)^2] = \sigma_k^2 \in (0, \infty)$ for all k, s .

ii) **Independence.** The array $(\varepsilon_s^k)_{k,s}$ has mutually independent entries, and the three collections $(\varepsilon_s^k)_{k,s}$, $(\zeta_s^k)_{k,s}$, $(U_i)_i$ are mutually independent.

iii) **Symmetry.** $\varepsilon_s^k \stackrel{d}{=} -\varepsilon_s^k$ for all k, s .

Remark D.4. *Independence across Both Indices.* Part (ii) emphasizes independence across arms and pulls. Every draw therefore carries its own sign, which is what makes the array factorization of Lemma D.11 available and permits symmetry to be imposed coordinatewise in part (iii). The clustered specialization of the main text with errors $M_{k,N}$ does not share this structure as it is only independent across clusters k but not across the depth index N .

Definition D.5. *Even σ -algebra.* The (full) even σ -algebra is $\mathcal{E}_\infty := \sigma((\varepsilon_s^k)^2)_{k,s}, (U_i)_i$.

D.2 The Gaussian Benchmark

Before turning to symmetry, I record what exact Gaussianity buys. Each arm's sample average is independent of its demeaned data, so any design driven by residuals—sample variances included—can be conditioned on at no cost. As a result, the relevant conditioning object is not the even σ -algebra but the residual σ -algebra, generated by the within-arm demeaned observations.

Definition D.6. *Residual σ -algebra and Nested Designs.* Fix deterministic pilot lengths $(m^k)_{k \in [K]}$ with $m^k \geq 2$, not depending on the data. Define

$$\mathcal{R} := \sigma\left((\varepsilon_s^k - \bar{\varepsilon}_{m^k}^k)_{s \leq m^k, k \in [K]}, (U_i)_i\right)$$

⁵³ $(U_i)_i$ is absent in every argument below. It is carried so that policies employing external randomization, like a randomized tie-breaking or random arm selection, remain inside the even class of Definition D.17, and so that an asymptotically even surrogate may share the randomization of the policy it couples to.

A design $(N_N^k)_{k \in [K]}$ is nested at $(m^k)_k$ if $N_N^k \geq m^k$ almost surely for every k .

By considering deterministic pilot lengths, the exposition is kept tidy. Random commitment stages are accommodated by stopping the residual filtration, as it is the case of ACS in the cluster specialization of Appendix E.1.

Proposition D.7. *Exact Conditional Normality under Gaussian Errors.* Let $\varepsilon_s^k \sim N(0, \sigma_k^2)$, independent across (k, s) , and let π induce a design $(N_N^k)_{k \in [K]}$ that is $\mathcal{R}$ -measurable and nested at $(m^k)_k$. Then, for every horizon N and every event $A \in \mathcal{R}$ with $\mathbb{P}(A) > 0$,

$$\mathcal{L} \left(\left(Z_{N_N^k}^k / \sigma_k \right)_{k \in [K]} \middle| A \right) = N(\mathbf{0}, I) \quad \text{exactly}$$

Proof of Proposition D.7. *Step 1. Orthogonality.* Fix an arm k and any $M^k \geq m^k$. For every $s \leq m^k$,

$$\text{Cov}(\bar{\varepsilon}_{M^k}^k, \varepsilon_s^k - \bar{\varepsilon}_{m^k}^k) = \frac{\sigma_k^2}{M^k} - \frac{1}{m^k} \sum_{j \leq m^k} \text{Cov}(\bar{\varepsilon}_{M^k}^k, \varepsilon_j^k) = \frac{\sigma_k^2}{M^k} - \frac{1}{m^k} \cdot m^k \cdot \frac{\sigma_k^2}{M^k} = 0$$

where I used $m^k \leq M^k$. All the variables involved are jointly Gaussian, so zero covariance implies independence, $\bar{\varepsilon}_{M^k}^k \perp (\varepsilon_s^k - \bar{\varepsilon}_{m^k}^k)_{s \leq m^k}$. Independence across arms holds by assumption, and $(U_i)_i$ is independent of the errors. Since $Z_{M^k}^k = \sqrt{M^k} \bar{\varepsilon}_{M^k}^k$, it follows that for every deterministic vector $(M^k)_k$ with $M^k \geq m^k$,

$$(Z_{M^k}^k)_{k \in [K]} \perp \mathcal{R}, \quad (Z_{M^k}^k / \sigma_k)_{k \in [K]} \sim N(\mathbf{0}, I)$$

Step 2. Cell-wise factorization. The terminal counts take values in the finite grid $\mathcal{M} = \{(M^k)_k \in \mathbb{N}^K : \sum_k M^k = N, M^k \geq m^k\}$, and $\{(N_N^k)_k = (M^k)_k\} \in \mathcal{R}$ by $\mathcal{R}$ -measurability of the design. Hence for $A \in \mathcal{R}$ and any $(t_k)_k \in \mathbb{R}^K$, writing $A_{(M^k)_k} = A \cap \{(N_N^k)_k = (M^k)_k\} \in \mathcal{R}$,

$$\begin{aligned} \mathbb{E} \left[\exp \left(i \sum_k t_k \frac{Z_{N_N^k}^k}{\sigma_k} \right) \mathbb{1}_A \right] &= \sum_{(M^k)_k \in \mathcal{M}} \mathbb{E} \left[\exp \left(i \sum_k t_k \frac{Z_{M^k}^k}{\sigma_k} \right) \mathbb{1}_{A_{(M^k)_k}} \right] \\ &= \sum_{(M^k)_k \in \mathcal{M}} \exp(-\|t\|^2/2) \mathbb{P}(A_{(M^k)_k}) = \exp(-\|t\|^2/2) \mathbb{P}(A) \end{aligned}$$

where the second equality applies Step 1 cell by cell, and the nesting condition guarantees $M^k \geq m^k$ throughout. The proof concludes by dividing by $\mathbb{P}(A) > 0$ and applying Lévy's continuity theorem. ■

Remark D.8. *Five Observations on Gaussianity vs Symmetry.* First, $\mathcal{R}$ and $\mathcal{E}_\infty$ are incomparable. The residuals $\varepsilon_s^k - \bar{\varepsilon}_{m^k}^k$ change sign under $\varepsilon \mapsto -\varepsilon$, so they are not $\mathcal{E}_\infty$ -measurable; conversely, $\mathcal{E}_\infty$ records $(\varepsilon_s^k)^2$ at every s , while $\mathcal{R}$ sees only the first m^k draws of each arm, so neither contains the other.

Second, the two frameworks trade distributional assumptions against policy generality. Under Gaussianity (i) the design may depend arbitrarily on the residuals—including through odd functionals such as the sign of a residual contrast—and (ii) exact normality is preserved exactly at every horizon (Proposition D.7). Under symmetry, a substantial weakening of the error distribution class, the conditioning σ -algebra contracts from the residuals to their magnitudes. Thus, the invariant policy class

contracts accordingly from any mapping of the residuals to (asymptotically) even policies.

Third, neither framework covers policies that condition on the level of the data. The sample average is precisely the direction demeaning removes, so a design that reads a running mean—a futility stop, a threshold rule on $\hat{\mu}$, standard bandit rules—induces events outside $\mathcal{R}$ and $\mathcal{E}_\infty$ alike, and conditional normality fails even under Gaussian errors. This is the boundary at which the adjustment tradition reviewed in the main text becomes necessary. Fourth, nesting is the general-theory analogue of the implementability condition of the main text: the observations used to choose the design must be a subset of the observations entering the terminal average.

Finally, exactness is confined to the known-variance standardization. Replacing σ_k by an estimate breaks it, since any variance estimate is itself a function of the residuals generating $\mathcal{R}$, and its conditional law is therefore affected. Exact studentized statements require the specific two-stage weighting of [Stein \(1945\)](#), and alternative studentized results remain purely asymptotic.

D.3 Signs and Magnitudes

For the remainder of this appendix, I leave behind the assumption on exact Gaussianity and focus on the weaker symmetry assumption.

Definition D.9. *Signs and Magnitudes.* For each (k, s) , write $\varepsilon_s^k = \chi_s^k R_s^k$ with

$$\chi_s^k = \mathbb{1}(\varepsilon_s^k \neq 0) \operatorname{sign}(\varepsilon_s^k) + \mathbb{1}(\varepsilon_s^k = 0) \zeta_s^k, \quad R_s^k = |\varepsilon_s^k|$$

Lemma D.10. *Signs are Independent of Magnitudes.* Under Assumption [D.3](#), for each (k, s) , $\chi_s^k \sim \operatorname{Rad}$ and $\chi_s^k \perp\!\!\!\perp R_s^k$.

Proof of Lemma D.10. Fix (k, s) and drop indices. For any Borel $C \subseteq [0, \infty)$,

$$\mathbb{P}(\chi = 1, R \in C) = \mathbb{P}(\varepsilon > 0, |\varepsilon| \in C) + \frac{1}{2} \mathbb{P}(\varepsilon = 0) \mathbb{1}(0 \in C)$$

where the second term uses $\zeta \sim \operatorname{Rad}$, independent of ε . By symmetry, $\mathbb{P}(\varepsilon > 0, |\varepsilon| \in C) = \mathbb{P}(-\varepsilon > 0, |-\varepsilon| \in C) = \mathbb{P}(\varepsilon < 0, |\varepsilon| \in C)$. Since $\{\varepsilon < 0, |\varepsilon| \in C\}$ and $\{\varepsilon > 0, |\varepsilon| \in C\}$ partition $\{|\varepsilon| \in C \setminus \{0\}\}$, each has probability $\frac{1}{2} \mathbb{P}(|\varepsilon| \in C \setminus \{0\})$. Adding the tie-break term gives $\mathbb{P}(\chi = 1, R \in C) = \frac{1}{2} \mathbb{P}(R \in C)$. The same argument holds for $\chi = -1$. The joint law factorizes, delivering both claims. $\square$

Lemma D.11. *Signs are Independent of the Even σ -algebra.* Under Assumption [D.3](#), $(\chi_s^k)_{k \geq 1, s \geq 1} \perp\!\!\!\perp \mathcal{E}_\infty$.

Proof of Lemma D.11. The pairs (χ_s^k, R_s^k) are functions of $(\varepsilon_s^k, \zeta_s^k)$, which by Assumption [D.3\(ii\)](#) are independent of $(U_i)_i$ and independent across both k and s ([Remark D.4](#)). Each pair factorizes by [Lemma D.10](#), so the joint law of the array factorizes as well with $(\chi_s^k)_{k,s} \perp\!\!\!\perp (R_s^k)_{k,s}$. Since

$$\mathcal{E}_\infty = \sigma((\varepsilon_s^k)^2)_{k,s}, (U_i)_i = \sigma((R_s^k)^2)_{k,s}, (U_i)_i = \sigma((R_s^k)_{k,s}, (U_i)_i)$$

and $(\zeta_s^k)_{k,s} \perp\!\!\!\perp \mathcal{E}_\infty$, it follows that $(\chi_s^k)_{k,s} \perp\!\!\!\perp \mathcal{E}_\infty$. $\square$

D.4 Conditional CLT for a Fixed Design

Throughout this subsection I fix a single arm k and suppress it from subscripts and superscripts. I further replace the possibly random count N_N^k by a deterministic M . Assumption D.3 is imposed throughout.

Lemma D.12. *Frozen Conditions. Let $(R_s)_s$ be iid with $\mathbb{E}[R_s^2] = \sigma^2 < \infty$. Then, there is an almost sure event $\Omega_0 \in \mathcal{E}_\infty$ on which, for every $\delta > 0$,*

$$(i) \quad \frac{1}{M} \sum_{s=1}^M R_s^2 \xrightarrow[M]{a.s.} \sigma^2, \quad (ii) \quad \frac{1}{M} \sum_{s=1}^M R_s^2 \mathbb{1}\{R_s > \delta\sqrt{M}\} \xrightarrow[M]{a.s.} 0$$

Remark D.13. *Why “Frozen”? The squared errors become frozen once the even σ -algebra is conditioned upon: $\mathbb{E}[\varepsilon_s^2 \mid \mathcal{E}_\infty] = R_s^2$. Lemma D.12 verifies that the two Lindeberg–Feller inputs hold pathwise on the frozen array.*

Proof of Lemma D.12. Part (i) follows directly from the SLLN, applicable since $\mathbb{E}[R_s^2] < \infty$. For part (ii), the Lindeberg condition, fix $C > 0$. For every $M \geq C^2/\delta^2$ we have $\delta\sqrt{M} \geq C$, hence $\mathbb{1}\{R_s > \delta\sqrt{M}\} \leq \mathbb{1}\{R_s > C\}$ and

$$\limsup_{M \rightarrow \infty} \frac{1}{M} \sum_{s=1}^M R_s^2 \mathbb{1}\{R_s > \delta\sqrt{M}\} \leq \limsup_{M \rightarrow \infty} \frac{1}{M} \sum_{s=1}^M R_s^2 \mathbb{1}\{R_s > C\} = \mathbb{E}[R_1^2 \mathbb{1}\{R_1 > C\}]$$

where the equality follows from the SLLN applied to the integrable iid summands $R_s^2 \mathbb{1}\{R_s > C\}$. Letting $C \rightarrow \infty$, the bound is sent to zero by dominated convergence. Taking the intersection over rational $\delta > 0$ and integer C —and extending to all $\delta > 0$ by monotonicity—, and intersecting with the almost sure event of part (i), defines Ω_0 . $\square$

Theorem D.14. *L^1 Convergence of the Conditional Characteristic Function. Let $\phi_M(t) := \mathbb{E}[\exp(itZ_M) \mid \mathcal{E}_\infty]$ denote the conditional characteristic function and $\phi(t) = \exp(-\sigma^2 t^2/2)$ the characteristic function of $N(0, \sigma^2)$. Then*

$$\mathbb{E}|\phi_M(t) - \phi(t)| \rightarrow 0 \quad \forall t \in \mathbb{R}$$

Proof of Theorem D.14. *Step 1. Pathwise Lindeberg–Feller.* Fix ω in the almost sure event Ω_0 and $t \in \mathbb{R}$. Write $Y_{s,M} = \varepsilon_s/\sqrt{M} = \chi_s R_s/\sqrt{M}$, so $Z_M = \sum_{s \leq M} Y_{s,M}$. By Lemma D.11, conditional on $\mathcal{E}_\infty$ the magnitudes R_s are fixed and the signs χ_s remain iid Rademacher; hence, conditionally, the $Y_{s,M}$ are independent, mean zero, with variance R_s^2/M . $\phi_M(t)$ then reduces to an ordinary characteristic function of a triangular array with frozen magnitudes. Conditional independence gives

$$\text{Var}(Z_M \mid \mathcal{E}_\infty) = \sum_{s \leq M} \text{Var}(Y_{s,M} \mid \mathcal{E}_\infty) = \frac{1}{M} \sum_{s \leq M} R_s(\omega)^2 \xrightarrow[M]{a.s.} \sigma^2$$

by Lemma D.12(i), and the conditional Lindeberg condition holds by Lemma D.12(ii).

$$\sum_{s \leq M} \mathbb{E}[Y_{s,M}^2 \mathbb{1}\{|Y_{s,M}| > \delta\} \mid \mathcal{E}_\infty] = \frac{1}{M} \sum_{s \leq M} R_s(\omega)^2 \mathbb{1}\{R_s(\omega) > \delta\sqrt{M}\} \xrightarrow[M]{a.s.} 0$$

The Lindeberg–Feller CLT for independent, non-identically distributed triangular arrays therefore

applies along $M \rightarrow \infty$, giving $\phi_M(t)(\omega) \rightarrow \exp(-\sigma^2 t^2/2)$. As this holds for every $\omega \in \Omega_0$, convergence is almost sure. *Step 2. L^1 convergence.* Characteristic functions are bounded by one, so $|\phi_M(t) - \phi(t)| \leq 2$. Almost sure convergence plus uniform boundedness gives $\mathbb{E}|\phi_M(t) - \phi(t)| \rightarrow 0$ by dominated convergence. $\square$

Corollary D.15. *Uniform Mixing. Under the conclusion of Theorem D.14,*

$$\sup_{A \in \mathcal{E}_\infty} |\mathbb{E}[\phi_M(t) \mathbb{1}_A] - \phi(t) \mathbb{P}(A)| \rightarrow 0, \quad \forall t \in \mathbb{R}$$

that is, $Z_M \Rightarrow N(0, \sigma^2)$ $\mathcal{E}_\infty$ -uniformly mixing.

Proof of Corollary D.15. For any $A \in \mathcal{E}_\infty$,

$$\begin{aligned} |\mathbb{E}[\exp(itZ_M) \mathbb{1}_A] - \phi(t) \mathbb{P}(A)| &= |\mathbb{E}[\mathbb{E}[\exp(itZ_M) \mathbb{1}_A \mid \mathcal{E}_\infty]] - \mathbb{E}[\phi(t) \mathbb{1}_A]| \\ &= |\mathbb{E}[(\phi_M(t) - \phi(t)) \mathbb{1}_A]| \leq \mathbb{E}|\phi_M(t) - \phi(t)| \rightarrow 0, \end{aligned}$$

the second equality using $\mathcal{E}_\infty$ -measurability of $\mathbb{1}_A$. The final bound does not depend on A , so the convergence is uniform over $A \in \mathcal{E}_\infty$. $\square$

Remark D.16. *Is Uniform Mixing Necessary? Uniformity is what allows the conditioning event to move with the index, $(A_M)_M$, in a way that simple pointwise mixing cannot accommodate. This is essential below, where the events of interest are terminal design events that move alongside the horizon.*

D.5 Even Policies and Terminal Events

Definition D.17. *Policy Class. A policy π is*

- (i) **Even** if the terminal design $(N_N^k)_{k \in [K]}$ is $\mathcal{E}_\infty$ -measurable;
- (ii) **Non- $\mathcal{K}$ -starving** if $\mathbb{P}(\nu_N^{\mathcal{K}} \geq c \mathbf{1}) \rightarrow 1$ for some $c > 0$, where $\nu_N^{\mathcal{K}} = (N_N^k/N)_{k \in \mathcal{K}}$ for a subset $\mathcal{K} \subseteq [K]$.

Remark D.18. *Reading Definition D.17. Evenness is the substantive restriction. Were the policy able to select the next arm k_i on the basis of the raw errors $(\varepsilon_s^k)_{k, s < i}$, the terminal design and its associated events would not be $\mathcal{E}_\infty$ -measurable, so the results in this appendix would not apply. Non-starvation is required because asymptotic statements can only concern arms played infinitely often, and nothing in evenness prevents a policy from stopping play of an arm. Weaker divergence conditions, not necessarily linear in N , would suffice for most results in this note, but linear non-starvation becomes non-negotiable when passing to asymptotically even policies. Divergence is then stated in linear fashion throughout for consistency. Finally, note that none of these restrictions make a policy admissible in the bandit sense, as an even, non- $\mathcal{K}$ -starving policy could still pick into the future. Adaptivity must therefore be checked separately, yet, all policies in this note are understood to satisfy adaptivity according to the sampling protocol in Definition D.2.*

D.6 Conditional CLT for Random Even Designs

Theorem D.19. *Mixing CLT Conditional on the Design. Write $\phi^k(t) = \exp(-\sigma_k^2 t^2/2)$ and $\phi_M^k(t) = \mathbb{E}[\exp(itZ_M^k) \mid \mathcal{E}_\infty]$. For a subset $\mathcal{K} \subseteq [K]$, let $(N_N^k)_{k \in [K]}$ be a terminal design induced by an even,*

non- $\mathcal{K}$ -starving policy, and let $(A_N^{\mathcal{K}})_N$ with $A_N^{\mathcal{K}} = \{(N_N^k)_{k \in \mathcal{K}} \in C\} \in \mathcal{E}_\infty$ be any sequence of design events. Then

$$\left| \mathbb{E} \left[\exp \left(i \sum_{k \in \mathcal{K}} t_k Z_{N_N^k}^k \right) \mathbb{1}_{A_N^{\mathcal{K}}} \right] - \mathbb{P}(A_N^{\mathcal{K}}) \prod_{k \in \mathcal{K}} \phi^k(t_k) \right| \rightarrow 0 \quad \forall (t_k)_k \in \mathbb{R}^{|\mathcal{K}|}$$

Moreover, for any such sequence with $\liminf_N \mathbb{P}(A_N^{\mathcal{K}}) > 0$,

$$\mathcal{L} \left(\left(Z_{N_N^k}^k \right)_{k \in \mathcal{K}} \mid A_N^{\mathcal{K}} \right) \implies N(\mathbf{0}, \text{diag}(\sigma_k^2)_{k \in \mathcal{K}}).$$

Proof of Theorem D.19. *Step 1. Decomposition of the joint conditional characteristic function.* For any $(t_k)_k \in \mathbb{R}^{|\mathcal{K}|}$,

$$\begin{aligned} \mathbb{E} \left[\exp \left(i \sum_{k \in \mathcal{K}} t_k Z_{N_N^k}^k \right) \mathbb{1}_{A_N^{\mathcal{K}}} \right] &= \mathbb{E} \left[\mathbb{E} \left[\exp \left(i \sum_{k \in \mathcal{K}} t_k Z_{N_N^k}^k \right) \mathbb{1}_{A_N^{\mathcal{K}}} \mid \mathcal{E}_\infty \right] \right] \\ &= \mathbb{E} \left[\mathbb{E} \left[\sum_{(M^k)_{k \in \mathcal{K}}} \mathbb{1}\{(N_N^k)_{k \in \mathcal{K}} = (M^k)_{k \in \mathcal{K}}\} \exp \left(i \sum_{k \in \mathcal{K}} t_k Z_{M^k}^k \right) \mid \mathcal{E}_\infty \right] \mathbb{1}_{A_N^{\mathcal{K}}} \right] \\ &= \mathbb{E} \left[\sum_{(M^k)_{k \in \mathcal{K}}} \mathbb{1}\{(N_N^k)_{k \in \mathcal{K}} = (M^k)_{k \in \mathcal{K}}\} \mathbb{E} \left[\exp \left(i \sum_{k \in \mathcal{K}} t_k Z_{M^k}^k \right) \mid \mathcal{E}_\infty \right] \mathbb{1}_{A_N^{\mathcal{K}}} \right] \\ &= \mathbb{E} \left[\sum_{(M^k)_{k \in \mathcal{K}}} \mathbb{1}\{(N_N^k)_{k \in \mathcal{K}} = (M^k)_{k \in \mathcal{K}}\} \prod_{k \in \mathcal{K}} \phi_{M^k}^k(t_k) \mathbb{1}_{A_N^{\mathcal{K}}} \right] = \mathbb{E} \left[\prod_{k \in \mathcal{K}} \phi_{N_N^k}^k(t_k) \mathbb{1}_{A_N^{\mathcal{K}}} \right] \end{aligned}$$

The second equality follows by $\mathcal{E}_\infty$ -measurability of $A_N^{\mathcal{K}}$, the third by $\mathcal{E}_\infty$ -measurability of the design indicators together with the finiteness of the grid of count vectors. The fourth equality is the load-bearing step: after conditioning on $\mathcal{E}_\infty$ the magnitudes entering each $Z_{M^k}^k$ are frozen and the signs remain *iid* and independent across arms, so the conditional expectation factorizes over k .

Step 2. L^1 convergence along the diverging counts. Step 1 of Theorem D.14 established $|\phi_M^k(t) - \phi^k(t)| \xrightarrow[M]{a.s.} 0$ on Ω_0 . Non-starvation gives $N_N^k \xrightarrow{P} \infty$ for each $k \in \mathcal{K}$, so Lemma B.1 yields $|\phi_{N_N^k}^k(t) - \phi^k(t)| \xrightarrow{P} 0$. A union bound over the finite set $\mathcal{K}$ delivers the vector statement. Uniform boundedness upgrades convergence in probability to L^1 .

Step 3. Uniform mixing. By Corollary D.15, L^1 convergence of the conditional characteristic functions gives uniform mixing over $\mathcal{E}_\infty$, and $(A_N^{\mathcal{K}})_N$ is $\mathcal{E}_\infty$ -measurable by evenness. Finally $\mathbb{E} |\prod_k \phi_{N_N^k}^k(t_k) - \prod_k \phi^k(t_k)| \rightarrow 0$, since each factor converges in L^1 and all factors are bounded by one. This establishes the first claim. The second claim follows by dividing by $\mathbb{P}(A_N^{\mathcal{K}})$, which is eventually bounded away from zero by the theorem's hypothesis, and applying Lévy's continuity theorem. $\square$

Lemma D.20. *Variance Consistency.* For a deterministic diverging index M , the within-arm variance estimator

$$\hat{\sigma}_{k,M}^2 := \frac{1}{M} \sum_{s=1}^M (X_s^k - \hat{\mu}_M^k)^2 \quad \text{satisfies} \quad \hat{\sigma}_{k,M}^2 \xrightarrow[M]{a.s.} \sigma_k^2$$

Proof of Lemma D.20. Since $X_s^k - \hat{\mu}_M^k = \varepsilon_s^k - \bar{\varepsilon}_M^k$,

$$\hat{\sigma}_{k,M}^2 = \frac{1}{M} \sum_{s=1}^M (\varepsilon_s^k)^2 - (\bar{\varepsilon}_M^k)^2 \xrightarrow[M]{\text{a.s.}} \sigma_k^2$$

by the SLLN applied to $((\varepsilon_s^k)^2)_s$ and to $(\varepsilon_s^k)_s$ under *iid*-ness and finite second moments, followed by the continuous mapping theorem. $\square$

Corollary D.21. *Studentized Conditional Inference.* Let $\sigma_k^2 > 0$ for all $k \in \mathcal{K}$, let $(N_N^k)_{k \in [K]}$ be a design induced by an even, non- $\mathcal{H}$ -starving policy, and let $(A_N^{\mathcal{K}})_N$ be a sequence of $\mathcal{E}_\infty$ -measurable design events with $\liminf_N \mathbb{P}(A_N^{\mathcal{K}}) > 0$. Then

$$\mathcal{L} \left(\left(\frac{\sqrt{N_N^k} (\hat{\mu}_{N_N^k}^k - \mu_k)}{\hat{\sigma}_{k,N_N^k}} \right)_{k \in \mathcal{K}} \middle| A_N^{\mathcal{K}} \right) \Rightarrow N(\mathbf{0}, I)$$

Proof of Corollary D.21. Decompose

$$\left(\frac{Z_{N_N^k}^k}{\hat{\sigma}_{k,N_N^k}} \right)_{k \in \mathcal{K}} = \left(\frac{Z_{N_N^k}^k}{\sigma_k} \right)_{k \in \mathcal{K}} \odot \left(\frac{\sigma_k}{\hat{\sigma}_{k,N_N^k}} \right)_{k \in \mathcal{K}}$$

Lemma D.20 gives $\hat{\sigma}_{k,M}^2 \rightarrow \sigma_k^2$ almost surely along deterministic M and non-starvation gives $N_N^k \xrightarrow{p} \infty$, so Lemma B.1 transfers this to $\mathbb{P}(|\hat{\sigma}_{k,N_N^k}^2 - \sigma_k^2| \geq \eta) \rightarrow 0$ for every $\eta > 0$. Consequently, for every $\delta > 0$,

$$\begin{aligned} \mathbb{P} \left(\left\| (\sigma_k / \hat{\sigma}_{k,N_N^k})_{k \in \mathcal{K}} - \mathbf{1} \right\| > \delta \mid A_N^{\mathcal{K}} \right) &= \frac{\mathbb{P} \left(\left\| (\sigma_k / \hat{\sigma}_{k,N_N^k})_{k \in \mathcal{K}} - \mathbf{1} \right\| > \delta \right) \cap A_N^{\mathcal{K}}}{\mathbb{P}(A_N^{\mathcal{K}})} \\ &\leq \frac{\mathbb{P} \left(\left\| (\sigma_k / \hat{\sigma}_{k,N_N^k})_{k \in \mathcal{K}} - \mathbf{1} \right\| > \delta \right)}{\mathbb{P}(A_N^{\mathcal{K}})} \rightarrow 0 \end{aligned}$$

with the ratio being well defined since $\mathbb{P}(\min_{k \in \mathcal{K}} \hat{\sigma}_{k,N_N^k} > 0) \rightarrow 1$ and $\liminf_N \mathbb{P}(A_N^{\mathcal{K}}) > 0$. By Theorem D.19, $\mathcal{L}((Z_{N_N^k}^k / \sigma_k)_{k \in \mathcal{K}} \mid A_N^{\mathcal{K}}) \Rightarrow N(\mathbf{0}, I)$, and the vector Slutsky theorem applied under the conditional law delivers the result. $\square$

Remark D.22. *Studentization Does Not Require Evenness.* The variance estimators $(\hat{\sigma}_{k,N_N^k}^2)_{k \in \mathcal{K}}$ are not $\mathcal{E}_\infty$ -measurable because they demean by the arm sample means $\hat{\mu}^k$ rather than by the true means μ_k , so the resulting cross terms depend on relative signs. The corollary nonetheless goes through, because $\hat{\sigma}^2$ enters only through convergence in probability to a constant, which Slutsky absorbs regardless of measurability.

D.7 Asymptotically Even Designs

Even policies are generally infeasible as the learner does not observe the true errors, but only the residuals. So, at best, she may emulate even policies by acting on functions of squared residuals, such as the within-arm sample variance $\hat{\sigma}_{k,M}^2$. Sample variances demean by the sample mean in a way that a policy tracking a sequence of sample variances could recover relative signs, therefore inducing a design outside the even σ -algebra. The example below summarizes one such policy.

Example D.23. *Sample Variances Leak Relative Signs.* Fix a single arm and drop its index. For $m \geq 1$ let $\bar{\varepsilon}_m = m^{-1} \sum_{s \leq m} \varepsilon_s$ and $\hat{\sigma}_m^2 = m^{-1} \sum_{s \leq m} \varepsilon_s^2 - \bar{\varepsilon}_m^2$. Suppose the designer observes the sequence $(\hat{\sigma}_m^2)_m$ alongside the squared errors $(\varepsilon_s^2)_s \in \mathcal{E}_\infty$. She may then recover $\bar{\varepsilon}_m^2 = m^{-1} \sum_{s \leq m} \varepsilon_s^2 - \hat{\sigma}_m^2$ for every m , and squaring the identity $(m+1)\bar{\varepsilon}_{m+1} = m\bar{\varepsilon}_m + \varepsilon_{m+1}$ gives

$$\bar{\varepsilon}_m \varepsilon_{m+1} = \frac{(m+1)^2 \bar{\varepsilon}_{m+1}^2 - m^2 \bar{\varepsilon}_m^2 - \varepsilon_{m+1}^2}{2m}$$

The left-hand side reveals $\text{sign}(\bar{\varepsilon}_m) \text{sign}(\varepsilon_{m+1})$ for every m , which are not $\mathcal{E}_\infty$ -measurable by Lemma D.11. A policy built on $(\hat{\sigma}_m^2)_m$ may therefore induce a non- $\mathcal{E}_\infty$ -measurable design, and Theorem D.19 fails.

Enlarging the conditioning σ -algebra to contain the relative signs does not help. Knowing all relative signs together with all magnitudes pins down the error path up to a single global flip, so Z_M becomes two-valued rather than Gaussian. The randomness that made the CLT possible has been conditioned away. In what follows, asymptotic evenness will not be defined in terms of leakage, but the intuition remains helpful. The leaked component is asymptotically negligible relative to the even one: for the sample variance, $\bar{\varepsilon}_M^2 = O_p(1/M)$ while the even part $M^{-1} \sum_{s \leq M} \varepsilon_s^2 = O_p(1)$.

Definition D.24. *Asymptotically Even Policies.* Let $\Xi_N^k = (N_N^k - N_{0N}^k)/\sqrt{N}$ for a deterministic centering sequence $(N_{0N}^k)_N$, and write $\Xi_N^{\mathcal{K}} = (\Xi_N^k)_{k \in \mathcal{K}}$. A policy π with terminal design $(N_N^k)_{k \in [K]}$ is asymptotically even if there exists an $\mathcal{E}_\infty$ -measurable design $(\tilde{N}_N^k)_{k \in [K]}$, with $\tilde{\Xi}_N^{\mathcal{K}} = ((\tilde{N}_N^k - N_{0N}^k)/\sqrt{N})_{k \in \mathcal{K}}$, such that

$$\|\Xi_N^{\mathcal{K}} - \tilde{\Xi}_N^{\mathcal{K}}\| \xrightarrow[N]{p} 0$$

Remark D.25. *The Surrogate Is Unconstrained.* The definition leaves both policies quite unrestricted. The surrogate $(\tilde{N}_N^k)_k$ need be neither feasible nor related to π in any substantive way, and π itself may condition on events well outside $\mathcal{E}_\infty$, raw residuals included. However, in practice, the asymptotically even policy will be some functional of squared residuals, and the even surrogate, a functionally equivalent policy with the squared errors replacing the squared residuals. Asymptotic evenness is then verified by showing convergence of the two designs at the $\sqrt{N}$ scale.

Remark D.26. *A Reduced Event Conditioning Class and the $\sqrt{N}$ -Scale.* For exactly even policies the event conditioning class is unrestricted. Theorem D.19 is a uniform mixing statement over $\mathcal{E}_\infty$, so any $\mathcal{E}_\infty$ -measurable event, at any scale and terminal or not, leaves the limiting law intact. The design events induced by asymptotically even policies are no longer $\mathcal{E}_\infty$ -measurable, so the asymptotic normality argument is then constructed through an even surrogate. The class of conditioning events for asymptotically even policies is then restricted to events that agree asymptotically under the two policies, in the sense $\mathbb{P}(A_N^{\mathcal{K}} \triangle \tilde{A}_N^{\mathcal{K}}) \rightarrow 0$. This pairing forces the events to be expressed (at most) on the same scale at which the two designs are coupled.

In the applications motivating this note, terminal counts fluctuate on the $\sqrt{N}$ -scale around the deterministic optimal allocations. In this context, the $\sqrt{N}$ normalization is the finest natural scale at which the feasible and surrogate designs can generally be coupled while retaining non-degeneracy of the conditioning events. This makes conditional and unconditional inference meaningfully distinct and permits conditioning on the events of interest such as the quantiles of N^k . At coarser scales $\nu_N^{\mathcal{K}} = (N_N^k/N)_k$ the coupling is trivial but the events are vacuous: the fractions converge in probability,

so every conditioning event has limiting probability in $\{0, 1\}$ and the conditional law degenerates to the unconditional one—a regime in which a random-index central limit theorem in the spirit of [Anscombe \(1952\)](#) suffices and symmetry is never needed. At the unit-scale coupling is usually not possible and requires stronger convergence than granted by [Definition D.24](#). The main text discusses this trichotomy in greater detail.

Lemma D.27. *Asymptotically Even Events.* Let $A_N^{\mathcal{K}} = \{\Xi_N^{\mathcal{K}} \in C\}$ for a Borel set $C \subseteq \mathbb{R}^{|\mathcal{K}|}$, generated by an asymptotically even, non- $\mathcal{K}$ -starving policy π , and suppose C satisfies the margin condition

$$\lim_{\eta \downarrow 0} \limsup_N \mathbb{P}(\text{dist}(\Xi_N^{\mathcal{K}}, \partial C) \leq \eta) = 0 \quad (17)$$

Then the event $\tilde{A}_N^{\mathcal{K}} = \{\tilde{\Xi}_N^{\mathcal{K}} \in C\}$ belongs to $\mathcal{E}_\infty$ and satisfies $\mathbb{P}(A_N^{\mathcal{K}} \triangle \tilde{A}_N^{\mathcal{K}}) \rightarrow 0$.

Proof of Lemma D.27. Measurability of $\tilde{A}_N^{\mathcal{K}}$ is immediate, since $(\tilde{N}_N^k)_k$ is $\mathcal{E}_\infty$ -measurable and the centering and scaling are deterministic. If the two indicators disagree at a point, that means that one of $\Xi_N^{\mathcal{K}}, \tilde{\Xi}_N^{\mathcal{K}}$ lies in C and the other does not, so the segment joining them meets ∂C and

$$A_N^{\mathcal{K}} \triangle \tilde{A}_N^{\mathcal{K}} = \{\mathbb{1}_C(\Xi_N^{\mathcal{K}}) \neq \mathbb{1}_C(\tilde{\Xi}_N^{\mathcal{K}})\} \subseteq \{\text{dist}(\Xi_N^{\mathcal{K}}, \partial C) \leq \|\Xi_N^{\mathcal{K}} - \tilde{\Xi}_N^{\mathcal{K}}\|\}.$$

Introducing a separating $\eta > 0$,

$$\mathbb{P}(A_N^{\mathcal{K}} \triangle \tilde{A}_N^{\mathcal{K}}) \leq \mathbb{P}(\|\Xi_N^{\mathcal{K}} - \tilde{\Xi}_N^{\mathcal{K}}\| > \eta) + \mathbb{P}(\text{dist}(\Xi_N^{\mathcal{K}}, \partial C) \leq \eta)$$

The first term vanishes for every fixed η by asymptotic evenness, so $\limsup_N \mathbb{P}(A_N^{\mathcal{K}} \triangle \tilde{A}_N^{\mathcal{K}}) \leq \limsup_N \mathbb{P}(\text{dist}(\Xi_N^{\mathcal{K}}, \partial C) \leq \eta)$ for every $\eta > 0$. The left-hand side does not depend on η , so letting $\eta \downarrow 0$ and invoking (17) sends it to zero. $\blacksquare$

Remark D.28. *Reading the Margin Condition.* Condition (17) is an anti-concentration requirement on the scaled counts near ∂C : it forbids the design from piling up on the boundary, which is exactly where the feasible and surrogate designs can disagree. In applications it is usually verified through a central limit theorem for the underlying statistics governing the design. Assume $\Xi_N^{\mathcal{K}} \implies \Xi_\infty^{\mathcal{K}}$ for a well-defined limit law $\Xi_\infty^{\mathcal{K}}$ with $\mathbb{P}(\Xi_\infty^{\mathcal{K}} \in \partial C) = 0$, then since the set $\{x : \text{dist}(x, \partial C) \leq \eta\}$ is closed the Portmanteau theorem gives

$$\limsup_N \mathbb{P}(\text{dist}(\Xi_N^{\mathcal{K}}, \partial C) \leq \eta) \leq \mathbb{P}(\text{dist}(\Xi_\infty^{\mathcal{K}}, \partial C) \leq \eta)$$

and the right-hand side decreases to $\mathbb{P}(\Xi_\infty^{\mathcal{K}} \in \partial C) = 0$ as $\eta \downarrow 0$. The same weak convergence delivers $\liminf_N \mathbb{P}(A_N^{\mathcal{K}}) > 0$ for any C whose interior receives positive limit mass.

Remark D.29. *CLT and Fourth Moments in Asymptotically Even Policies.* Condition (17) and [Remark D.28](#) are usually made concrete by establishing a central limit theorem over the underlying statistic governing the design, rather than a law of large numbers. The intuition boils down again to the scale of the class of conditioning events. The margin condition is a statement about anti-concentration, and a law of large numbers delivers exactly the opposite. Under a law of large numbers the scaled design would converge to a point mass, which either sits off ∂C —in which case $\mathbb{P}(A_N^{\mathcal{K}}) \rightarrow \{0, 1\}$ and conditioning is vacuous—or sits on it, in which case no control is available at all. What is required is a limit law that spreads mass, and the $\sqrt{N}$ centering in [Definition D.24](#) is chosen precisely so that

one exists.

The second point concerns which statistic the limit theorem applies to. Inference here is about first moments, and the central limit theorem behind Theorem D.14 needs only finite variances. But the design is usually driven by second moments, so certifying the margin condition requires a central limit theorem for the pilot variances, whose own variance is $\mu_{4,k} - \sigma_k^4$. This is why finite fourth moments are imposed in applications concerning asymptotically even policies including Lemma D.36 and Theorem 6.10 and nowhere else in this note. Finite fourth moments are a requirement of the design's limit theory, not of the estimator's.

Lemma D.30. *Increment Vanishes at Ratio-close Indices.* Fix a single arm and drop its index. Let $(M_N)_N$ and $(\tilde{M}_N)_N$ be $\{0, 1, \dots, N\}$ -valued random indices with $|M_N/N - \tilde{M}_N/N| \xrightarrow{P} 0$ and $\mathbb{P}(M_N/N \wedge \tilde{M}_N/N \geq c) \rightarrow 1$ for some $c > 0$. Then

$$|Z_{M_N} - Z_{\tilde{M}_N}| \xrightarrow{P} 0$$

Proof of Lemma D.30. Work on the event $\{\tilde{M}_N \leq M_N\}$. Writing $\Sigma_M = \sum_{s \leq M} \varepsilon_s$, so that $Z_M = \Sigma_M/\sqrt{M}$,

$$|Z_{M_N} - Z_{\tilde{M}_N}| \leq \left| \sqrt{\frac{\tilde{M}_N}{M_N}} - 1 \right| \cdot |Z_{\tilde{M}_N}| + \frac{|\Sigma_{M_N} - \Sigma_{\tilde{M}_N}|}{\sqrt{M_N}}$$

On $\{\tilde{M}_N \geq cN\}$ we have $|Z_{\tilde{M}_N}| \leq \max_{cN \leq m \leq N} |\Sigma_m|/\sqrt{cN} = O_p(1)$, by Kolmogorov's maximal inequality applied to the *iid* mean-zero, finite-variance errors of Assumption D.3. Convergence in probability of the scaled indices together with linear non-starvation gives $|\sqrt{\tilde{M}_N/M_N} - 1| \xrightarrow{P} 0$, so the first product is $o_p(1)$.

For the second term, fix $\delta \in (0, 1)$ and intersect $\{\tilde{M}_N \leq M_N\}$ with $\{|M_N - \tilde{M}_N| \leq \delta N\} \cap \{\tilde{M}_N \geq cN\}$, whose probability tends to one under the hypotheses. There,

$$\frac{|\Sigma_{M_N} - \Sigma_{\tilde{M}_N}|}{\sqrt{M_N}} \leq \max_{\substack{cN \leq \tilde{m} \leq m \leq N \\ |m - \tilde{m}| \leq \delta N}} \frac{|\Sigma_m - \Sigma_{\tilde{m}}|}{\sqrt{m}} \leq \frac{1}{\sqrt{c}} \max_{\substack{0 \leq \tilde{m} \leq m \leq N \\ |m - \tilde{m}| \leq \delta N}} \frac{|\Sigma_m - \Sigma_{\tilde{m}}|}{\sqrt{N}} \leq \frac{1}{\sqrt{c}} \sup_{\substack{0 \leq s \leq t \leq 1 \\ |t - s| \leq \delta}} |\Psi_N(t) - \Psi_N(s)|$$

where $\Psi_N(t) = \Sigma_{\lfloor Nt \rfloor}/\sqrt{N}$ is the rescaled partial-sum process. Under Assumption D.3, by Donsker's functional central limit theorem and continuity of the Brownian limit, $(\Psi_N)_N$ is asymptotically uniformly equicontinuous (tight), which is equivalent to the local modulus-of-continuity property

$$\lim_{\delta \downarrow 0} \limsup_{N \rightarrow \infty} \mathbb{P} \left(\sup_{\substack{0 \leq s \leq t \leq 1 \\ |t - s| \leq \delta}} |\Psi_N(t) - \Psi_N(s)| \geq \eta \right) = 0 \quad \forall \eta > 0$$

Fix $\eta, \epsilon > 0$ and choose δ so that

$$\limsup_{N \rightarrow \infty} \mathbb{P} \left(\sup_{\substack{0 \leq s \leq t \leq 1 \\ |t - s| \leq \delta}} |\Psi_N(t) - \Psi_N(s)| \geq \eta\sqrt{c} \right) \leq \epsilon/2$$

Since $\mathbb{P}(|M_N - \tilde{M}_N| > \delta N) \rightarrow 0$, for all N large

$$\mathbb{P}\left(\frac{|\Sigma_{M_N} - \Sigma_{\tilde{M}_N}|}{\sqrt{M_N}} \geq \eta\right) \leq \mathbb{P}(|M_N - \tilde{M}_N| > \delta N) + \mathbb{P}\left(\sup_{\substack{0 \leq s \leq t \leq 1 \\ |t-s| \leq \delta}} |\Psi_N(t) - \Psi_N(s)| \geq \eta\sqrt{c}\right) \leq \epsilon$$

As η and ϵ were arbitrary the second term is $o_p(1)$, and the claim follows for $\{\tilde{M}_N \leq M_N\}$. The converse case $\{\tilde{M}_N > M_N\}$ is completely symmetrical and so the unconditional statement follows. ■

Remark D.31. *Lemma D.30 Is Measurability-Free. Lemma D.30 is a statement about the random walk alone. It does not condition on $\mathcal{E}_\infty$ and imposes no measurability on M_N or $\tilde{M}_N$. That is exactly what makes it usable below, where it bounds the distance between $Z_{N_N^k}^k$ and $Z_{\tilde{N}_N^k}^k$ with the feasible count non-measurable and the surrogate measurable. This is exactly where linear non-starvation enters the analysis: tightness of the sum process only manifests at indices which are negligible at the scale of counts. Asymptotic evenness controls $|N_N^k - \tilde{N}_N^k|$ only to order $o_p(\sqrt{N})$, which is negligible at the scale of the counts precisely because the counts are of order N . Slower forms of divergence will compromise convergence in probability between the scaled sums of the exactly even and asymptotically even policies.*

Theorem D.32. *Inference for Asymptotically Even Designs. Let π be an asymptotically even, non- $\mathcal{K}$ -starving policy, and let $(A_N^{\mathcal{K}})_N$, $A_N^{\mathcal{K}} = \{\Xi_N^{\mathcal{K}} \in C\}$, be a sequence of scaled design events satisfying the margin condition (17). Then*

$$\left| \mathbb{E} \left[\exp \left(i \sum_{k \in \mathcal{K}} t_k Z_{N_N^k}^k \right) \mathbb{1}_{A_N^{\mathcal{K}}} \right] - \mathbb{P}(A_N^{\mathcal{K}}) \prod_{k \in \mathcal{K}} \phi^k(t_k) \right| \rightarrow 0 \quad \forall (t_k)_k \in \mathbb{R}^{|\mathcal{K}|}$$

If in addition $\liminf_N \mathbb{P}(A_N^{\mathcal{K}}) > 0$, then

$$\mathcal{L} \left((Z_{N_N^k}^k)_{k \in \mathcal{K}} \mid A_N^{\mathcal{K}} \right) \Rightarrow N(\mathbf{0}, \text{diag}(\sigma_k^2)_{k \in \mathcal{K}}), \quad \mathcal{L} \left(\left(\frac{\sqrt{N_N^k} (\hat{\mu}_{N_N^k}^k - \mu_k)}{\hat{\sigma}_{k, N_N^k}} \right)_{k \in \mathcal{K}} \mid A_N^{\mathcal{K}} \right) \Rightarrow N(\mathbf{0}, I)$$

Remark D.33. *Not Uniform Mixing. The first statement cannot be phrased as $\mathcal{E}_\infty$ -uniform mixing, since $A_N^{\mathcal{K}}$ is not $\mathcal{E}_\infty$ -measurable to start with. In fact, convergence only holds pointwise along the design event sequence. For the same reason the conditional characteristic function of the feasible design does not reduce to $\phi_{N_N^k}^k$. Yet, the essence of the result is akin that of Theorem D.19 and subsequent corollaries.*

Proof of Theorem D.32. *Step 0. The surrogate inherits non-starvation.* Since $\tilde{\nu}_N^k - \nu_N^k = (\tilde{\Xi}_N^k - \Xi_N^k)/\sqrt{N}$, for any $\eta > 0$

$$\mathbb{P}(\tilde{\nu}_N^{\mathcal{K}} < (c - \eta)\mathbf{1}) \leq \mathbb{P}(\|\Xi_N^{\mathcal{K}} - \tilde{\Xi}_N^{\mathcal{K}}\| > \sqrt{N}\eta) + \mathbb{P}(\nu_N^{\mathcal{K}} < c\mathbf{1}) \rightarrow 0$$

so $\tilde{\pi}$ is non- $\mathcal{K}$ -starving with constant $c - \eta$.

Step 1. Decomposition. Writing $\tilde{A}_N^{\mathcal{K}}$ for the surrogate event of Lemma D.27,

$$\begin{aligned}
& \left| \mathbb{E} \left[e^{i \sum_k t_k Z_{N_N^k}^k} \mathbb{1}_{A_N^{\mathcal{K}}} \right] - \mathbb{P}(A_N^{\mathcal{K}}) \prod_{k \in \mathcal{K}} \phi^k(t_k) \right| \\
& \leq \underbrace{\left| \mathbb{E} \left[e^{i \sum_k t_k Z_{N_N^k}^k} \mathbb{1}_{A_N^{\mathcal{K}}} \right] - \mathbb{E} \left[e^{i \sum_k t_k Z_{\tilde{N}_N^k}^k} \mathbb{1}_{\tilde{A}_N^{\mathcal{K}}} \right] \right|}_{[1]} + \underbrace{\left| \mathbb{E} \left[e^{i \sum_k t_k Z_{\tilde{N}_N^k}^k} \mathbb{1}_{\tilde{A}_N^{\mathcal{K}}} \right] - \mathbb{P}(\tilde{A}_N^{\mathcal{K}}) \prod_{k \in \mathcal{K}} \phi^k(t_k) \right|}_{[2]} \\
& \quad + \underbrace{\left| \prod_{k \in \mathcal{K}} \phi^k(t_k) \right| \cdot \left| \mathbb{P}(A_N^{\mathcal{K}}) - \mathbb{P}(\tilde{A}_N^{\mathcal{K}}) \right|}_{[3]}
\end{aligned}$$

Term [3] vanishes because $|\mathbb{P}(A_N^{\mathcal{K}}) - \mathbb{P}(\tilde{A}_N^{\mathcal{K}})| \leq \mathbb{P}(A_N^{\mathcal{K}} \triangle \tilde{A}_N^{\mathcal{K}}) \rightarrow 0$ by Lemma D.27, the characteristic functions being bounded by one. Term [2] vanishes by Theorem D.19 applied to $((\tilde{N}_N^k)_{k \in \mathcal{K}}, (\tilde{A}_N^{\mathcal{K}})_N)$: the surrogate design and its events are $\mathcal{E}_\infty$ -measurable by construction, and the surrogate counts diverge (linearly) with probability tending to one by Step 0, so the freezing identity $\mathbb{E}[e^{i \sum_k t_k Z_{\tilde{N}_N^k}^k} | \mathcal{E}_\infty] = \prod_k \phi_{\tilde{N}_N^k}^k(t_k)$ is available.

Step 2. The coupling term. Split

$$[1] \leq \left| \mathbb{E} \left[\left(e^{i \sum_k t_k Z_{N_N^k}^k} - e^{i \sum_k t_k Z_{\tilde{N}_N^k}^k} \right) \mathbb{1}_{A_N^{\mathcal{K}}} \right] \right| + \left| \mathbb{E} \left[e^{i \sum_k t_k Z_{\tilde{N}_N^k}^k} (\mathbb{1}_{A_N^{\mathcal{K}}} - \mathbb{1}_{\tilde{A}_N^{\mathcal{K}}}) \right] \right|$$

The second piece is bounded by $\mathbb{P}(A_N^{\mathcal{K}} \triangle \tilde{A}_N^{\mathcal{K}}) \rightarrow 0$, the integrand having modulus one. For the first, $|e^{ia} - e^{ib}| \leq \min\{2, |a - b|\}$ and subadditivity of $x \mapsto \min\{2, x\}$ give

$$\mathbb{E} \left| e^{i \sum_k t_k Z_{N_N^k}^k} - e^{i \sum_k t_k Z_{\tilde{N}_N^k}^k} \right| \leq \sum_{k \in \mathcal{K}} \mathbb{E} \left[\min\{2, |t_k| \cdot |Z_{N_N^k}^k - Z_{\tilde{N}_N^k}^k|\} \right]$$

For each $k \in \mathcal{K}$, Step 0 gives $\mathbb{P}(N_N^k/N \wedge \tilde{N}_N^k/N \geq c') \rightarrow 1$ for some $c' > 0$, and asymptotic evenness gives $|N_N^k/N - \tilde{N}_N^k/N| = o_p(\sqrt{N})/N \xrightarrow{p} 0$, so the hypotheses of Lemma D.30 hold and each increment is $o_p(1)$. Uniform boundedness of the integrand by 2 upgrades this to L^1 convergence, and the sum runs over finitely many arms. Hence $[1] \rightarrow 0$, establishing the first statement.

Step 3. The conditional limits. Dividing by $\mathbb{P}(A_N^{\mathcal{K}})$, eventually bounded away from zero by hypothesis, and applying Lévy's continuity theorem gives the first conditional statement. Studentization requires only $\hat{\sigma}_{k, N_N^k}^2 \xrightarrow{p} \sigma_k^2$, which holds under non-starvation by Lemma D.20 and Lemma B.1, together with $\liminf_N \mathbb{P}(A_N^{\mathcal{K}}) > 0$; the argument of Corollary D.21 then applies verbatim. Neither the measurability of N_N^k nor that of $A_N^{\mathcal{K}}$ was invoked at any point. $\square$

Remark D.34. Decision Boundaries. Allocation rules built on second moments are typically composed with decision boundaries—rounding, thresholds, clipping, arg max operations. When the limiting statistic sits at a positive distance from every such boundary, the feasible and surrogate decisions coincide with probability tending to one and Definition D.24 is satisfied. Disagreements in early periods are immaterial, since the conditioning events concern suitably scaled terminal, thus asymptotic, designs.

D.8 Policy Examples

Proposition D.35. Two-Stage Neyman Allocation Is Asymptotically Even. Let $\sigma^2 = (\sigma_k^2)_k > \mathbf{0}$ and

let $0 < d < \min_k f_k(\sigma)$ be known to the designer, with $dN \in \mathbb{N}$, $f_k(\sigma) = \sigma_k/\Sigma_\sigma$ and $\Sigma_\sigma = \sum_l \sigma_l$. Consider the policy π that samples uniformly for the first dKN periods—so every arm receives dN pulls—computes the pilot variances $\hat{\sigma}_{k,dN}^2 = (dN)^{-1} \sum_{s \leq dN} (X_s^k - \hat{\mu}_{dN}^k)^2$, and allocates the remaining periods so that the terminal counts equal $N_N^k = \lfloor N f_k(\hat{\sigma}) \rfloor$, with leftovers assigned by an arbitrary deterministic rule.⁵⁴ Define the surrogate counts $\tilde{N}_N^k = \lfloor N f_k(q) \rfloor$ under the same leftover rule, with $q_k^2 = (dN)^{-1} \sum_{s \leq dN} (\varepsilon_s^k)^2$. Then π is non- $[K]$ -starving with constant d and asymptotically even for $[K]$.

Proof of Proposition D.35. *Surrogate evenness.* The pilot allocation is deterministic, hence so are the pilot counts. Each q_k^2 is a function of $((\varepsilon_s^k)^2)_{s \leq dN} \in \mathcal{E}_\infty$, and the rounding and leftover rules are deterministic, so $(\tilde{N}_N^k)_k$ is $\mathcal{E}_\infty$ -measurable. *Non-starvation.* Every arm receives at least dN pulls in the pilot, which are later included within the terminal counts, so $N_N^k \geq dN$ almost surely. *Asymptotic evenness.* The pilot variance demeanes by the pilot mean, so $q_k^2 - \hat{\sigma}_{k,dN}^2 = (\varepsilon_{dN}^k)^2 = O_p(1/N)$, using $\bar{\varepsilon}_{dN}^k = O_p(N^{-1/2})$. Since $\sigma_k > 0$, the square root is Lipschitz on a neighbourhood of σ_k^2 with probability tending to one, whence $|\hat{\sigma}_k - q_k| = O_p(1/N)$; and since $\Sigma_\sigma > 0$, f is Lipschitz on a neighbourhood of σ , whence $|f_k(\hat{\sigma}) - f_k(q)| = O_p(1/N)$. Using $|\lfloor Nx \rfloor - \lfloor Ny \rfloor| \leq N|x - y| + 1$ and allowing up to K units from the leftover rule,

$$|N_N^k - \tilde{N}_N^k| \leq N|f_k(\hat{\sigma}) - f_k(q)| + 1 + K = O_p(1)$$

so $|N_N^k - \tilde{N}_N^k|/\sqrt{N} \xrightarrow{P} 0$ and π is asymptotically even with surrogate $(\tilde{N}_N^k)_k$. $\blacksquare$

Lemma D.36. *Central Limit Theorem for the Scaled Counts.* Assume in addition $\mu_{4,k} := \mathbb{E}[(\varepsilon_s^k)^4] < \infty$ and $\mu_{4,k} > \sigma_k^4$ for every k . Centre $\Xi_N^{[K]}$ at the oracle allocation $N_{0N}^k = N f_k(\sigma)$. Then

$$\Xi_N^{[K]} \implies G, \quad \tilde{\Xi}_N^{[K]} \implies G, \quad G = Jz$$

where $z \sim N(\mathbf{0}, \text{diag}(v_k)_k)$ with $v_k = (\mu_{4,k} - \sigma_k^4)/(4d\sigma_k^2) > 0$, and J is the Jacobian of f at σ , namely $J_{kl} = (\mathbb{1}\{k = l\}\Sigma_\sigma - \sigma_k)/\Sigma_\sigma^2$.

Proof of Lemma D.36. Consider the surrogate first. The variables $((\varepsilon_s^k)^2)_s$ are iid with mean σ_k^2 and variance $\mu_{4,k} - \sigma_k^4 < \infty$, and the pilot size dN is deterministic, so $\sqrt{N}(q_k^2 - \sigma_k^2) \implies N(0, (\mu_{4,k} - \sigma_k^4)/d)$, jointly across arms with independent coordinates by cross-arm independence. The delta method applied to $x \mapsto \sqrt{x}$, whose derivative at σ_k^2 is $1/(2\sigma_k)$, gives $\sqrt{N}(q_k - \sigma_k) \implies N(0, v_k)$; applied to f it gives $\sqrt{N}(f(q) - f(\sigma)) \implies Jz = G$. Rounding and the leftover rule perturb $\tilde{\Xi}_N^k$ by at most $(1 + K)/\sqrt{N} \rightarrow 0$, so $\tilde{\Xi}_N^{[K]} \implies G$ by Slutsky. Finally $\|\Xi_N^{[K]} - \tilde{\Xi}_N^{[K]}\| = O_p(1)/\sqrt{N} \xrightarrow{P} 0$ by Proposition D.35, so $\Xi_N^{[K]} \implies G$ as well. $\blacksquare$

Remark D.37. *One Caution on the Limit Law.* G is singular on $\mathbb{R}^K$. Because the allocation fractions sum to one, every column of J sums to zero— $\sum_k J_{kl} = (\Sigma_\sigma - \sum_k \sigma_k)/\Sigma_\sigma^2 = 0$ —so G is supported entirely on the hyperplane $\{x \in \mathbb{R}^K : \sum_k x_k = 0\}$. Conditioning sets over all K arms must respect this singularity. Conditioning on a strict subset of arms avoids the deterministic adding-up constraint.

Corollary D.38. *Design-Conditional Inference for Two-Stage Neyman.* Let $A_N^{[K]} = \{\Xi_N^{[K]} \in C\}$ for a Borel $C \subseteq \mathbb{R}^K$ with $\mathbb{P}(G \in \partial C) = 0$ and $\mathbb{P}(G \in C) > 0$, where G is as in Lemma D.36. Then, under

⁵⁴Since $d < \min_k f_k(\sigma)$ strictly, $\mathbb{P}(\lfloor N f_k(\hat{\sigma}) \rfloor \geq dN \forall k) \rightarrow 1$, so the target allocation is implementable with probability approaching one and the complementary event has vanishing probability which affects none of the statements below.

the assumptions of Proposition D.35 and Lemma D.36,

$$\mathcal{L} \left(\left(\frac{\sqrt{N_N^k} (\hat{\mu}_{N_N^k}^k - \mu_k)}{\hat{\sigma}_{k, N_N^k}} \right)_{k \in [K]} \middle| A_N^{[K]} \right) \Rightarrow N(\mathbf{0}, I)$$

Proof of Corollary D.38. By Proposition D.35, π is asymptotically even and non- $[K]$ -starving. By Lemma D.36 and Remark D.28, $\mathbb{P}(G \in \partial C) = 0$ delivers the margin condition (17), and $\mathbb{P}(A_N^{[K]}) \rightarrow \mathbb{P}(G \in C) > 0$ by weak convergence, so $\liminf_N \mathbb{P}(A_N^{[K]}) > 0$. Theorem D.32 applies and delivers the conditional limit law and the studentized statements. The corollary remains valid for full-vector conditioning sets C provided the boundary has zero probability under the singular law G , as was cautioned against in the remark. $\square$

E Proofs of Section 6

This section translates the general theory of Appendix D into the clustered sampling problem and, in doing so, proves the results stated in Section 6. There are, however, two features of the cluster problem that make the translation not mechanical. First, the problem carries *two* error sequences, $(u_k)_k$ and $(\varepsilon_{ki})_{k,i}$, and *two* degrees of freedom (K, N) in the terminal design, only one of which diverges with the budget. I handle the first by establishing N -consistency and extending the results depth by depth. Second, establishing asymptotic evenness is straightforward for EtC-CS and delicate for ACS. The proof structure for EtC-CS follows closely the constructive approach for Neyman allocations in Section D: the policy commits at the deterministic stage K_M on the strength of a single statistic $\hat{\lambda}_p$, so coupling it to an even surrogate reduces to replacing residuals by errors in that one statistic. ACS instead commits at a random stopping time governed by the whole sequence $(\hat{\lambda}_{k, N_p})_k$. Coupling two designs evaluated at two different random stopping times would require controlling the stopping times themselves at the $\sqrt{B}$ scale, which is not available. The argument therefore avoids the natural “even twin” altogether and gives the surrogate a deterministic stopping stage $\tilde{s} = K_{0B} = \lfloor K^*(\lambda_0) \rfloor$. The proof proceeds in three steps: (i) localize the ACS commitment stage to an $O_p(\sqrt{B})$ window around K_{0B} ; (ii) replace residuals by errors entering the λ mapping at cost $O_p(1/B)$; and (iii) move the even statistic from the random stage to K_{0B} across the commitment window at cost $O_p(B^{-3/4})$. Transmitted through the Lipschitz map $h(\cdot)$, these become width deviations of order $O_p(1)$ and $O_p(B^{1/4})$ respectively, both $o_p(\sqrt{B})$, which is exactly what Definition 6.8 requires.

Definitions and lemmas below are held to a minimum. Connections to the general theory are made throughout, so I recommend reading this section in light of the results in the standalone Appendix D.

E.1 Exact Gaussianity

Proposition 6.2 is the cluster instance of Proposition D.7 with two adaptations. Orthogonality must be established at two layers—unit residuals within each cluster and cluster residuals across clusters—and the commitment stage of ACS is random, which requires stopping the residual filtration. I first start by formally introducing the relevant σ -algebras and σ -fields sketched in the main text.

Lemma E.1. *Residual Filtration and Pilot σ -algebras.* Let $\mathcal{R}_j$, $j \geq 2$, be as in Definition 6.1. Then,

(i) Filtration. $\mathcal{R}_k \subseteq \mathcal{R}_j$ for all $2 \leq k \leq j$, since the k -demeaned cluster residuals can be recovered from the j -demeaned ones through

$$M_{l,N_p} - \bar{M}_{k,N_p} = (M_{l,N_p} - \bar{M}_{j,N_p}) - \frac{1}{k} \sum_{l' \leq k} (M_{l',N_p} - \bar{M}_{j,N_p}), \quad l \leq k \leq j$$

(ii) Measurability. For every $2 \leq k \leq j$, the statistics

$$\hat{w}_{k,N_p}^2 = \frac{1}{k} \sum_{l \leq k} \frac{1}{N_p - 1} \sum_{i \leq N_p} (\varepsilon_{li} - \bar{\varepsilon}_{l,N_p})^2, \quad \hat{b}_{k,N_p}^2 = \frac{1}{k-1} \sum_{l \leq k} (M_{l,N_p} - \bar{M}_{k,N_p})^2$$

and hence $\hat{\lambda}_{k,N_p}$, are $\mathcal{R}_j$ -measurable.

(iii) Stopping. The commitment stages $T^{\text{EtC-CS}} = K_M$ and $T^{\text{ACS}} = s$ are stopping times for $(\mathcal{R}_j)_j$. The first is deterministic so inclusion is trivial. For the second note

$$\{s = j\} = \{j+1 > K^*(\hat{\lambda}_{j,N_p})\} \cap \bigcap_{l=K_M}^{j-1} \{l+1 \leq K^*(\hat{\lambda}_{l,N_p})\} \in \mathcal{R}_j$$

Define the pilot σ -algebras as $\mathcal{P}^{\text{EtC-CS}} := \mathcal{R}_{K_M}$ and the stopped σ -algebra

$$\mathcal{P}^{\text{ACS}} := \mathcal{R}_s = \{A : A \cap \{s = j\} \in \mathcal{R}_j \text{ for every } j \geq 2\}$$

For $\pi \in \{\text{EtC-CS}, \text{ACS}\}$, the terminal design (K_B^π, N_B^π) is $\mathcal{P}^\pi$ -measurable by construction.

Proof of Proposition 6.2. *Step 1. Two-layer orthogonality.* Fix $j \geq 2$ and let $K' \geq j$, $N \geq N_p$. Since $\hat{\mu}_{K',N} - \mu = \bar{u}_{K'} + (K')^{-1} \sum_{k \leq K'} \bar{\varepsilon}_{k,N}$, and $u \perp \varepsilon$ with independence across clusters, it suffices to verify orthogonality layer by layer against the generators of $\mathcal{R}_j$. At the unit layer, for $i \leq N_p$ and $k \leq j$, only the term k contributes to the covariance, i.e.

$$\text{Cov}(\bar{\varepsilon}_{k,N}, \varepsilon_{ki} - \bar{\varepsilon}_{k,N_p}) = \frac{\sigma^2}{N} - \frac{1}{N_p} \cdot N_p \cdot \frac{\sigma^2}{N} = 0$$

both evaluations using $N_p \leq N$. At the cluster layer, for $k \leq j$,

$$\text{Cov}(\bar{u}_{K'}, u_k - \bar{u}_j) = \frac{\tau^2}{K'} - \frac{1}{j} \cdot j \cdot \frac{\tau^2}{K'} = 0$$

both evaluations using $j \leq K'$. The remaining piece of $M_{k,N_p} - \bar{M}_{j,N_p}$, namely $\bar{\varepsilon}_{k,N_p} - j^{-1} \sum_{l \leq j} \bar{\varepsilon}_{l,N_p}$, is also orthogonal to $(K')^{-1} \sum_k \bar{\varepsilon}_{k,N}$ by the same argument as in the unit layer. All variables are jointly Gaussian, so zero covariance implies independence: $\hat{\mu}_{K',N} \perp \mathcal{R}_j$ whenever $j \leq K'$ and $N_p \leq N$.

Step 2. Nesting. $T^\pi \leq K_B^\pi$ and $N_p \leq N_B^\pi$ almost surely. For **EtC-CS** the result follows because $T^{\text{EtC-CS}} = K_M \leq K_B^{\text{EtC-CS}}$, and for **ACS** because $s \leq K^*(\hat{\lambda}_{\max\{s-1, K_M\}, N_p}) = K_B^{\text{ACS}}$. The hypotheses of Step 1 then hold on every cell of the design grid. The attentive reader will note that nesting is the inference analogue of implementability.

Step 3. Cell-wise factorization. The triple $(T^\pi, K_B^\pi, N_B^\pi)$ takes finitely many values. Fix $A \in \mathcal{P}^\pi$ with $\mathbb{P}(A) > 0$ and write $A_{(j,K,N)} := A \cap \{(T^\pi, K_B^\pi, N_B^\pi) = (j, K, N)\}$. Since (K_B^π, N_B^π) is $\mathcal{P}^\pi$ -measurable, $A \cap \{(K_B^\pi, N_B^\pi) = (K, N)\} \in \mathcal{P}^\pi$, and intersecting with $\{T^\pi = j\}$ places $A_{(j,K,N)}$ in $\mathcal{R}_j$ by

the stopped σ -algebra construction of Lemma E.1(iii). By Steps 1 and 2, $\sqrt{K}(\hat{\mu}_{K,N} - \mu)/\theta_N \sim N(0, 1)$ independently of $\mathcal{R}_j \ni A_{(j,K,N)}$ on every cell, so

$$\mathbb{E} \left[\exp \left(\frac{itZ_B^\pi}{\theta_{N_B}^\pi} \right) \mathbb{1}_A \right] = \sum_{(j,K,N)} e^{-t^2/2} \mathbb{P}(A_{(j,K,N)}) = e^{-t^2/2} \mathbb{P}(A)$$

Dividing by $\mathbb{P}(A)$ and applying Lévy's continuity theorem concludes the proof. $\blacksquare$

E.2 Random Sequences in the Cluster Problem

Definition E.2. *Cluster Error Sequences.* For a fixed depth N , define $M_{k,N} = u_k + N^{-1} \sum_{i \leq N} \varepsilon_{ki}$, indexed by the cluster k . Let $(\zeta_k)_{k \geq 1}$ be an iid Rademacher sequence, used to break ties in Lemma 6.4, and $(U_k)_{k \geq 1}$ an iid $\text{Unif}[0, 1]$ sequence available to the policy as an independent randomization device, both independent of $(C_k)_k$.

Remark E.3. *What Carries Over?* For each fixed N , Assumptions 3.1 and 6.3 make $(M_{k,N})_{k \geq 1}$ an iid, mean-zero, symmetric sequence with $\mathbb{E}[M_{k,N}^2] = \theta_N^2 < \infty$. At that fixed depth it therefore plays the role of the error sequence of Assumption D.3, and every result of Appendix D applies with the single “arm” being the population, the pull index replaced by the cluster index, and the horizon N replaced by the diverging cluster count K_B . Nonetheless, the depth margin does not behave as the second index entering Assumption D.3. In particular, cluster average errors $(M_{k,N})$ are certainly not independent across nesting depths N, N' .

E.3 Signs and Magnitudes

Proof of Lemma 6.4. Fix N . By Remark E.3, $(M_{k,N})_k$ is iid, mean zero and symmetric, so Lemma D.10 applies coordinatewise with ε_s^k replaced by $M_{k,N}$ and the tie-breaker ζ_s^k by ζ_k . Notably, independence is only established across clusters. $\square$

Proof of Corollary 6.5. Fix N and let $r = (r_k)_k \in \{\pm 1\}^\infty$. Define the cluster-flip map $g_r : (C_k)_k \mapsto (r_k C_k)_k$. By Assumption 6.3 and cross-cluster independence, g_r is measure preserving. It leaves every generator of $\mathcal{G}_\infty$ invariant, since

$$S_{k,N_p}^2(r_k C_k) = S_{k,N_p}^2(C_k), \quad M_{k,N'}^2(r_k C_k) = (r_k M_{k,N'}(C_k))^2 = M_{k,N'}^2(C_k) \quad \forall N' \geq 1$$

while sending $\chi_{k,N} \mapsto r_k \chi_{k,N}$. Hence for bounded measurable f and bounded $\mathcal{G}_\infty$ -measurable h , $\mathbb{E}[f((\chi_{k,N})_k)h] = \mathbb{E}[f((r_k \chi_{k,N})_k)h]$ at every fixed r . Averaging over r drawn iid Rademacher independently of everything, $(r_k \chi_{k,N})_k$ is iid Rademacher whatever the value of $(\chi_{k,N})_k$, so the inner expectation is the constant $\mathbb{E}[f(\text{Rad})]$ and

$$\mathbb{E}[f((\chi_{k,N})_k)h] = \mathbb{E}[f(\text{Rad})] \mathbb{E}[h]$$

which delivers both claims. $\square$

E.4 Conditional CLT for a Fixed Design

This subsection establishes uniform mixing of Z_B^π over $\mathcal{G}_\infty$ at a *deterministic* design. Consider $(K_{0B}, N_{0B}) \in \mathbb{N}^2$ with $K_{0B}, N_{0B} \geq 2$, $K_{0B} \rightarrow \infty$ and $N_{0B} \rightarrow N_0$. Since N_{0B} is integer valued,

there is B_0 with $N_{0B} = N_0$ for all $B \geq B_0$, so every statement can be understood at the fixed depth N_0 .

Corollary E.4. *Frozen Conditions in the Cluster Problem.* *There is an almost sure event $\Omega_0 \in \mathcal{G}_\infty$ on which, for every $\delta > 0$,*

$$(i) \quad \frac{1}{K_{0B}} \sum_{k=1}^{K_{0B}} R_{k,N_{0B}}^2 \xrightarrow[B]{a.s.} \theta_{N_0}^2 = \tau^2 + \frac{\sigma^2}{N_0}, \quad (ii) \quad \frac{1}{K_{0B}} \sum_{k=1}^{K_{0B}} R_{k,N_{0B}}^2 \mathbb{1}\{R_{k,N_{0B}} > \delta \sqrt{K_{0B}}\} \xrightarrow[B]{a.s.} 0$$

Proof of Corollary E.4. For $B \geq B_0$ the depth is fixed at N_0 , so both statements reduce to Lemma D.12 applied to the *iid* sequence $(R_{k,N_0})_k$, which has $\mathbb{E}[R_{k,N_0}^2] = \theta_{N_0}^2 < \infty$ by Assumption 3.1. Take Ω_0 to be the corresponding almost sure event. $\square$

Corollary E.5. *L^1 Convergence of the Conditional Characteristic Function.* *Let $Z_{0B} := K_{0B}^{-1/2} \sum_{k \leq K_{0B}} M_{k,N_{0B}}$, let $\phi_{0B}(t) := \mathbb{E}[\exp(itZ_{0B}) \mid \mathcal{G}_\infty]$, and let $\phi(t) = \exp(-\theta_{N_0}^2 t^2/2)$. Then*

$$|\phi_{0B}(t) - \phi(t)| \xrightarrow[B]{a.s.} 0 \quad \text{and} \quad \mathbb{E}|\phi_{0B}(t) - \phi(t)| \rightarrow 0 \quad \forall t \in \mathbb{R}$$

Proof of Corollary E.5. Immediate from Theorem D.14 in light of Corollary E.4, the signs $(\chi_{k,N_0})_k$ being *iid* Rademacher and independent of $\mathcal{G}_\infty$ by Corollary 6.5 at the fixed depth N_0 . $\square$

E.5 Conditional CLT for Random Even Designs

Proof of Proposition 6.7. The argument mirrors Theorem D.19 and Corollary D.21. The only difference is that the depth margin must be frozen before the width asymptotics can be run.

Step 1. For fixed B , the decomposition of the conditional characteristic function in Step 1 of Theorem D.19 goes through by summing over the finite grid of budget-feasible designs (K_B^π, N_B^π) , which is $\mathcal{G}_\infty$ -measurable by evenness. On each cell the depth is fixed, so Corollary 6.5 applies and the conditional expectation freezes.

Step 2. Write $\phi_{K,N}(t)$ for the conditional characteristic function at the deterministic design (K, N) , and set $T_B := \sup_{K \geq cB} |\phi_{K,N_0}(t) - \phi(t)|$. On the probability-one frozen event $\phi_{K,N_0}(t) \rightarrow \phi(t)$ along the divergent sequence $K \rightarrow \infty$, hence its tail supremum $T_B \rightarrow 0$ a.s. for any $c \in (0, 1]$ that satisfies the non-starvation condition. Then

$$\mathbb{P}(|\phi_B(t) - \phi(t)| > \eta) \leq \mathbb{P}(N_B^\pi \neq N_0) + \mathbb{P}(K_B^\pi < cB) + \mathbb{P}(T_B > \eta) \rightarrow 0$$

where the first term vanishes by N -consistency and the second by non-starvation. Uniform boundedness upgrades this to L^1 convergence.

Step 3. Corollary D.15 then delivers uniform mixing over $\mathcal{G}_\infty$, the events $(A_B)_B$ being $\mathcal{G}_\infty$ -measurable by evenness. The first conditional statement follows by dividing by $\mathbb{P}(A_B)$, eventually bounded away from zero, and applying Lévy's continuity theorem. For the studentized statement, Lemma 4.1(ii) gives $\hat{b}_{K,N_0}^2 \rightarrow \theta_{N_0}^2$ almost surely along deterministic diverging K ; the same composition argument as in Step 2 transfers this to the random design, and Slutsky concludes exactly as in Corollary D.21. $\blacksquare$

E.6 Asymptotically Even Designs

Corollary E.6. *Asymptotically Even Events in the Cluster Problem.* Let π be asymptotically even, N -consistent policy with surrogate $(\tilde{K}_B, \tilde{N}_B)$, and let $A_B = \{(\Xi_B^\pi, N_B^\pi) \in C\}$ for a Borel $C \subseteq \mathbb{R} \times \mathbb{N}$ satisfying the margin condition (10). Set $\tilde{A}_B = \{(\tilde{\Xi}_B, \tilde{N}_B) \in C\} \in \mathcal{G}_\infty$. Then $\mathbb{P}(A_B \triangle \tilde{A}_B) \rightarrow 0$.

Proof of Corollary E.6. Because N_B^π and $\tilde{N}_B$ are integer valued, $|N_B^\pi - \tilde{N}_B| \xrightarrow{P} 0$ is equivalent to $\mathbb{P}(N_B^\pi = \tilde{N}_B) \rightarrow 1$. The depth coordinate is asymptotically fixed and can contribute to the symmetric difference only on an event of vanishing probability. Hence

$$\mathbb{P}(A_B \triangle \tilde{A}_B) \leq \mathbb{P}(N_B^\pi \neq \tilde{N}_B) + \mathbb{P}\left(\left\{\mathbb{1}_C(\Xi_B^\pi, N_B^\pi) \neq \mathbb{1}_C(\tilde{\Xi}_B, \tilde{N}_B)\right\}\right)$$

where the second term is exactly the object bounded in Lemma D.27, with the width coordinate playing the role of Ξ_N^π and the frozen depth entering only through the section $C_{N_0} = \{x : (x, N_0) \in C\}$. The margin condition and $|\Xi_B^\pi - \tilde{\Xi}_B| \xrightarrow{P} 0$ send it to zero. ■

Lemma E.7. *Increment Vanishes at Ratio-close Designs.* Let (K_B^π, N_B^π) and $(\tilde{K}_B, \tilde{N}_B)$ be designs with bounded depths $2 \leq N_B^\pi, \tilde{N}_B \leq N_M < \infty$ a.s., such that $|\nu_B^\pi - \tilde{\nu}_B| \xrightarrow{P} 0$, $\mathbb{P}(\nu_B^\pi \wedge \tilde{\nu}_B \geq c) \rightarrow 1$ for some $c > 0$, and $\mathbb{P}(N_B^\pi = \tilde{N}_B) \rightarrow 1$. Writing $\tilde{Z}_B = \tilde{K}_B^{-1/2} \sum_{k \leq \tilde{K}_B} M_{k, \tilde{N}_B}$,

$$|Z_B^\pi - \tilde{Z}_B| \xrightarrow{P} 0$$

Proof of Lemma E.7. Split the difference at the common depth N_B^π :

$$|Z_B^\pi - \tilde{Z}_B| \leq \underbrace{\left| \frac{1}{\sqrt{K_B^\pi}} \sum_{k \leq K_B^\pi} M_{k, N_B^\pi} - \frac{1}{\sqrt{\tilde{K}_B}} \sum_{k \leq \tilde{K}_B} M_{k, N_B^\pi} \right|}_{\text{width}} + \underbrace{\left| \frac{1}{\sqrt{\tilde{K}_B}} \sum_{k \leq \tilde{K}_B} (M_{k, N_B^\pi} - M_{k, \tilde{N}_B}) \right|}_{\text{depth}}$$

The depth term vanishes identically on $\{N_B^\pi = \tilde{N}_B\}$, an event of probability tending to one. For the width term, fix $\eta > 0$ and decompose over the finitely many admissible depths

$$\mathbb{P}(\text{width} > \eta) \leq \sum_{N=2}^{N_M} \mathbb{P}\left(\left| \frac{1}{\sqrt{K_B^\pi}} \sum_{k \leq K_B^\pi} M_{k, N} - \frac{1}{\sqrt{\tilde{K}_B}} \sum_{k \leq \tilde{K}_B} M_{k, N} \right| > \eta\right)$$

For each fixed N the sequence $(M_{k, N})_k$ is *iid*, mean zero, with finite variance θ_N^2 , and the indices $K_B^\pi, \tilde{K}_B$ satisfy the hypotheses of Lemma D.30 with the horizon of that lemma replaced by the budget B . Each summand therefore vanishes, and the sum runs over finitely many terms. □

Theorem E.8. *Inference for Asymptotically Even Designs in the Cluster Problem.* Let π be an asymptotically even, non-starving, N -consistent policy, and let $(A_B)_B$ be a sequence of scaled design events as in (9) satisfying the margin condition (10). Then

$$|\mathbb{E}[\exp(itZ_B^\pi) \mathbb{1}_{A_B}] - \phi(t) \mathbb{P}(A_B)| \rightarrow 0 \quad \forall t \in \mathbb{R}$$

with $\phi(t) = \exp(-\theta_{N_0}^2 t^2 / 2)$. If in addition $\liminf_B \mathbb{P}(A_B) > 0$, then

$$\mathcal{L}\left(Z_B^\pi \mid A_B\right) \Rightarrow N(0, \theta_{N_0}^2), \quad \mathcal{L}\left(\frac{\sqrt{K_B^\pi}(\hat{\mu}_{K_B^\pi, N_B^\pi} - \mu)}{\hat{b}_{K_B^\pi, N_B^\pi}} \mid A_B\right) \Rightarrow N(0, 1)$$

Proof of Theorem E.8. The argument is that of Theorem D.32, with two cluster-specific inputs.

Step 0. The surrogate also inherits N -consistency. Non-starvation transfers exactly as in Step 0 of Theorem D.32. For N -consistency, depth being integers gives

$$\mathbb{P}(\tilde{N}_B \neq N_0) \leq \mathbb{P}(\tilde{N}_B \neq N_B^\pi) + \mathbb{P}(N_B^\pi \neq N_0) \rightarrow 0$$

by asymptotic evenness and N -consistency of π respectively. The surrogate is therefore even, non-starving and N -consistent, so Proposition 6.7 applies.

Step 1. Decomposition. Bound the target by the three terms of Step 1 of Theorem D.32, with $(\tilde{K}_B, \tilde{N}_B)$ and $\tilde{A}_B$ in place of the surrogate design and event. The third term is controlled by $|\mathbb{P}(A_B) - \mathbb{P}(\tilde{A}_B)| \leq \mathbb{P}(A_B \triangle \tilde{A}_B) \rightarrow 0$ from Corollary E.6, and the second term by Proposition 6.7 applied to the surrogate. *Step 2. The coupling term.* Split as in Step 2 of Theorem D.32. The event-mismatch piece is bounded by $\mathbb{P}(A_B \triangle \tilde{A}_B) \rightarrow 0$. The increment piece is bounded, using $|e^{ia} - e^{ib}| \leq \min\{2, |a - b|\}$, by $\mathbb{E}[\min\{2, |t| |Z_B^\pi - \tilde{Z}_B|\}]$, which vanishes by Lemma E.7 and uniform boundedness. Lemma E.7 replaces here Lemma D.30 precisely because the cluster design carries a second, non-divergent coordinate. *Step 3. Conditional limits.* Dividing by $\mathbb{P}(A_B)$ and applying Lévy's continuity theorem gives the first conditional statement. Studentization requires only $\hat{b}_{K_B^\pi, N_B^\pi}^2 \xrightarrow{P} \theta_{N_0}^2$, established in Step 3 of the proof of Proposition 6.7, together with $\liminf_B \mathbb{P}(A_B) > 0$. No measurability of (K_B^π, N_B^π) or of A_B was invoked. ■

E.7 Application to EtC-CS and ACS

It remains to verify that both algorithms are asymptotically even, non-starving and N -consistent, and that the relevant terminal design events satisfy the margin condition. Lemma E.10 collects the rates the two propositions need and Lemma E.14 supplies the central limit theorem behind the margin condition. As anticipated in the main text, the margin condition is concretized through a CLT on the design statistic, which requires finite fourth moments. The assumption is strictly weaker than the subgaussian Assumption 5.3 needed for performance.

Assumption E.9. *Fourth Moments.* $\mathbb{E}[u_k^4] < \infty$ and $\mathbb{E}[\varepsilon_{ki}^4] < \infty$.

Throughout, write $\alpha_{k, N_p} = S_{k, N_p}^2 - \sigma^2$ and $\beta_{k, N_p} = M_{k, N_p}^2 - \theta_{N_p}^2$, both *iid* across clusters, mean zero, and of finite variance under Assumption E.9.

Lemma E.10. *Uniform Rates for the Even Surrogate.* Fix N_p and define

$$\tilde{b}_{K, N_p}^2 = \frac{1}{K} \sum_{k \leq K} M_{k, N_p}^2, \quad \tilde{\lambda}_{K, N_p} = \Pi_{[\lambda_m, \lambda_M]} \left(\frac{\hat{w}_{K, N_p}^2}{\Pi_{[\tau_m^2, \tau_M^2]}(\tilde{b}_{K, N_p}^2 - \hat{w}_{K, N_p}^2 / N_p)} \right)$$

Under Assumptions 3.1, 3.2, 3.3 and E.9, and with K ranging over $[K_M, K_m]$,

- i) $\hat{w}_{K,N_p}^2, \tilde{b}_{K,N_p}^2$ and $\tilde{\lambda}_{K,N_p}$ are $\mathcal{G}_\infty$ -measurable for every K ,
- ii) $\sup_K |\hat{b}_{K,N_p}^2 - \tilde{b}_{K,N_p}^2| = O_p(1/B)$, and consequently $\sup_K |\hat{\lambda}_{K,N_p} - \tilde{\lambda}_{K,N_p}| = O_p(1/B)$,
- iii) $\sup_K |\hat{\lambda}_{K,N_p} - \lambda_0| = O_p(B^{-1/2})$ and $\sup_K |\tilde{\lambda}_{K,N_p} - \lambda_0| = O_p(B^{-1/2})$,
- iv) for $K_{0B} = \lfloor K^*(\lambda_0) \rfloor$, any fixed $A < \infty$ and $J = \lceil A\sqrt{B} \rceil$,

$$\max_{|j| \leq J} |\tilde{\lambda}_{K_{0B}+j,N_p} - \tilde{\lambda}_{K_{0B},N_p}| = O_p(B^{-3/4})$$

Proof of Lemma E.10. (i) $\hat{w}_{K,N_p}^2$ and $\tilde{b}_{K,N_p}^2$ are averages of the generators $(S_{k,N_p}^2)_k$ and $(M_{k,N_p}^2)_k$ of $\mathcal{G}_\infty$, and $\tilde{\lambda}_{K,N_p}$ is a measurable map of the two. Result follows.

(ii) Since $\hat{b}_{K,N_p}^2 = (K-1)^{-1} [\sum_{k \leq K} M_{k,N_p}^2 - K \bar{M}_{K,N_p}^2]$,

$$\hat{b}_{K,N_p}^2 - \tilde{b}_{K,N_p}^2 = \frac{1}{K(K-1)} \sum_{k \leq K} M_{k,N_p}^2 - \frac{K}{K-1} \bar{M}_{K,N_p}^2.$$

The first term is $(K-1)^{-1}$ times an average converging to $\theta_{N_p}^2$, hence $O_p(1/B)$ uniformly since $\Theta(B) = K_M \leq K \leq K_m = \Theta(B)$. For the second, Kolmogorov's maximal inequality gives $\max_{K \leq K_m} |\sum_{k \leq K} M_{k,N_p}| = O_p(\sqrt{K_m})$, so $\sup_K \bar{M}_{K,N_p}^2 \leq O_p(K_m)/K_M^2 = O_p(1/B)$, using $K_m/K_M = O(1)$. Hence $\sup_K |\hat{b}^2 - \tilde{b}^2| = O_p(1/B)$. For the transfer to λ , note that $\hat{\tau}_{K,N_p}^2$ and $\tilde{\tau}_{K,N_p}^2 := \Pi_{[\tau_m^2, \tau_M^2]}(\tilde{b}_{K,N_p}^2 - \hat{w}_{K,N_p}^2/N_p)$ share the term $\hat{w}_{K,N_p}^2/N_p$, which cancels, so $|\tilde{\tau}^2 - \hat{\tau}^2| \leq |\tilde{b}^2 - \hat{b}^2|$ by 1-Lipschitzness of the clip. Both clipped denominators exceed τ_m^2 , so

$$\sup_K |\hat{\lambda}_{K,N_p} - \tilde{\lambda}_{K,N_p}| \leq \sup_K \hat{w}_{K,N_p}^2 \frac{|\hat{\tau}_{K,N_p}^2 - \tilde{\tau}_{K,N_p}^2|}{\hat{\tau}_{K,N_p}^2 \tilde{\tau}_{K,N_p}^2} \leq \frac{\sup_K \hat{w}_{K,N_p}^2}{(\tau_m^2)^2} \sup_K |\hat{b}_{K,N_p}^2 - \tilde{b}_{K,N_p}^2| = O_p(1/B)$$

using $\sup_K \hat{w}_{K,N_p}^2 \leq K_M^{-1} \sum_{k \leq K_m} S_{k,N_p}^2 = O_p(1)$ and the 1-Lipschitzness of the outer clip.

(iii) By Kolmogorov's maximal inequality applied to $(\alpha_{k,N_p})_k$ and $(\beta_{k,N_p})_k$, both of finite variance under Assumption E.9,

$$\sup_K |\hat{w}_{K,N_p}^2 - \sigma^2| \leq \frac{1}{K_M} \max_{K \leq K_m} \left| \sum_{k \leq K} \alpha_{k,N_p} \right| = O_p(B^{-1/2}), \quad \sup_K |\tilde{b}_{K,N_p}^2 - \theta_{N_p}^2| = O_p(B^{-1/2})$$

so, together with part (ii), $\sup_K |\hat{b}^2 - \theta_{N_p}^2| = O_p(B^{-1/2})$ as well. On the clipped region the map $g(w, b) = w/(b - w/N_p)$ is Lipschitz, its denominator being bounded below by τ_m^2 , so $\sup_K |\hat{\lambda}_{K,N_p} - \lambda_0| = O_p(B^{-1/2})$, as the clip is 1-Lipschitz and λ_0 is strictly interior by Assumption 3.2. The statement for $\tilde{\lambda}$ follows from part (ii) and the triangle inequality.

(iv) This analysis is more subtle. With probability increasing to one both clips are inactive, so $|\tilde{\lambda}_{K_{0B}+j} - \tilde{\lambda}_{K_{0B}}| \leq L_g (|\hat{w}_{K_{0B}+j}^2 - \hat{w}_{K_{0B}}^2| + |\tilde{b}_{K_{0B}+j}^2 - \tilde{b}_{K_{0B}}^2|)$ for a Lipschitz constant L_g of the g map. The two terms are handled identically, so I discuss $\hat{w}_{K,N_p}^2$ as an illustration. Decompose $\hat{w}_{K,N_p}^2 = \sigma^2 + T_K/K$ with $T_K = \sum_{k \leq K} \alpha_{k,N_p}$. Then, for every $j : |j| \leq J$,

$$\hat{w}_{K_{0B}+j}^2 - \hat{w}_{K_{0B}}^2 = \frac{T_{K_{0B}+j} - T_{K_{0B}}}{K_{0B} + j} - \frac{j T_{K_{0B}}}{K_{0B}(K_{0B} + j)}$$

The first term may be interpreted as new noise, and the latter as a re-dilution term. For new noise, Kolmogorov's maximal inequality applied forwards on $(K_{0B}, K_{0B}+J]$ and backwards on $(K_{0B}-J, K_{0B}]$ around the deterministic centre K_{0B} gives $\max_{|j| \leq J} |T_{K_{0B}+j} - T_{K_{0B}}| = O_p(\sqrt{J}) = O_p(B^{1/4})$. For the re-dilution term, $|T_{K_{0B}}| = O_p(\sqrt{B})$ by Chebyshev and $|j| \leq J = O(\sqrt{B})$. Since $K_{0B} - J = \Theta(B)$ eventually,

$$\max_{|j| \leq J} |\hat{w}_{K_{0B}+j}^2 - \hat{w}_{K_{0B}}^2| = O_p\left(\frac{B^{1/4}}{B}\right) + O_p\left(\frac{\sqrt{B} \cdot \sqrt{B}}{B^2}\right) = O_p(B^{-3/4})$$

The same computation for $\tilde{b}_{K, N_p}^2 = \theta_{N_p}^2 + K^{-1} \sum_{k \leq K} \beta_{k, N_p}$ completes the proof. $\blacksquare$

Proposition E.11. *EtC-CS Is Asymptotically Even. Assume $N^*(\lambda_0) \notin \mathbb{N}$. EtC-CS is asymptotically even, non-starving and N-consistent, with even surrogate $(\tilde{K}_B, \tilde{N}_B) = (K^*(\tilde{\lambda}_{K_M, N_p}), N^*(\tilde{\lambda}_{K_M, N_p}))$ under the rounding convention of the feasible policy. Moreover $|K_B^{\text{EtC-CS}} - \tilde{K}_B| = O_p(1)$.*

Proof of Proposition E.11. Non-starvation and N-consistency are Corollary 4.7 and Lemma 4.4. The commitment stage K_M is deterministic, so $\tilde{\lambda}_{K_M, N_p}$ is $\mathcal{G}_\infty$ -measurable by Lemma E.10(i) and the surrogate design is even, the optimal map being measurable. For the width coupling, observe that h is L_h -Lipschitz on $[\lambda_m, \lambda_M]$ and $K^*(\lambda) = Bh(\lambda)$, so by Lemma E.10(ii)

$$|K_B^{\text{EtC-CS}} - \tilde{K}_B| \leq B L_h |\hat{\lambda}_{K_M, N_p} - \tilde{\lambda}_{K_M, N_p}| + 1 = O_p(1),$$

the 1 accounting for rounding. Therefore, $|K_B^{\text{EtC-CS}} - \tilde{K}_B|/\sqrt{B} \xrightarrow{P} 0$. For the depth, N^* is Lipschitz on $[\lambda_m, \lambda_M]$, so $|N^*(\hat{\lambda}_{K_M, N_p}) - N^*(\tilde{\lambda}_{K_M, N_p})| = O_p(1/B)$, since $N^*(\lambda_0) \notin \mathbb{N}$, the two floors coincide with probability tending to one and $|N_B^{\text{EtC-CS}} - \tilde{N}_B| \xrightarrow{P} 0$. $\blacksquare$

Proposition E.12. *ACS Is Asymptotically Even. ACS is asymptotically even, non-starving and N-consistent, with even surrogate $(\tilde{K}_B, \tilde{N}_B) = (K^*(\tilde{\lambda}_{K_{0B}, N_p}), N^*(\tilde{\lambda}_{K_{0B}, N_p}))$, $K_{0B} = \lfloor K^*(\lambda_0) \rfloor$. Moreover $|K_B^{\text{ACS}} - \tilde{K}_B| = O_p(B^{1/4})$.*

Proof of Proposition E.12. Non-starvation and N-consistency are Corollary 4.7 and Lemma 4.4. The surrogate is even by Lemma E.10(i) and determinism of K_{0B} . For the coupling I proceed in two stages:

Step 1. Localizing the commitment stage. Let $D_B := \sup_K |K^*(\hat{\lambda}_{K, N_p}) - K^*(\lambda_0)| \leq B L_h \sup_K |\hat{\lambda}_{K, N_p} - \lambda_0| = O_p(\sqrt{B})$ by Lemma E.10(iii). Recall $s = \inf\{k \geq K_M : k+1 > K^*(\hat{\lambda}_{k, N_p})\}$ from (5). For any $k \geq K^*(\lambda_0) + D_B$ we have $k+1 > K^*(\lambda_0) + D_B \geq K^*(\hat{\lambda}_{k, N_p})$, so the algorithm stops; and, for any $k \leq K^*(\lambda_0) - D_B - 1$ we have $k+1 \leq K^*(\lambda_0) - D_B \leq K^*(\hat{\lambda}_{k, N_p})$, so the stopping condition is not satisfied. Let the commitment stage be equal to $p_B = \max\{s-1, K_M\} \in [K_M, K_m]$. Then, $|p_B - K_{0B}| \leq D_B + 2 = O_p(\sqrt{B})$, where the constant absorbs the rounding in K_{0B} .

Step 2. Coupling. Fix some constant $A < \infty$ sufficiently large so the event $\{|p_B - K_{0B}| \leq J\}$ with $J = \lceil A\sqrt{B} \rceil$ has probability increasing to 1 (by Step 1). Then

$$\begin{aligned} |K_B^{\text{ACS}} - \tilde{K}_B| &\leq |K^*(\hat{\lambda}_{p_B, N_p}) - K^*(\tilde{\lambda}_{p_B, N_p})| + |K^*(\tilde{\lambda}_{p_B, N_p}) - K^*(\tilde{\lambda}_{K_{0B}, N_p})| + 1 \\ &\leq B L_h |\hat{\lambda}_{p_B, N_p} - \tilde{\lambda}_{p_B, N_p}| + B L_h |\tilde{\lambda}_{p_B, N_p} - \tilde{\lambda}_{K_{0B}, N_p}| + 1 \\ &\leq B L_h \sup_K |\hat{\lambda}_{K, N_p} - \tilde{\lambda}_{K, N_p}| + B L_h \max_{|j| \leq J} |\tilde{\lambda}_{K_{0B}+j, N_p} - \tilde{\lambda}_{K_{0B}, N_p}| + 1 \end{aligned}$$

which is $O_p(1) + O_p(B^{1/4})$ by Lemma E.10(ii) and (iv). Hence $|K_B^{\text{ACS}} - \tilde{K}_B|/\sqrt{B} \xrightarrow{P} 0$. The depth

statement follows as in Proposition E.11, since $|\hat{\lambda}_{p_B, N_p} - \tilde{\lambda}_{K_{0B}, N_p}| \xrightarrow{P} 0$ by the two displayed bounds. ■

Remark E.13. *On the Choice of ACS Surrogate.* The surrogate for ACS exploits decisively the content in Remark D.25. The surrogate need be neither feasible nor the “even twin” of ACS, it only needs to be $\mathcal{G}_\infty$ -measurable. The natural twin policy which runs the same stopping rule on the even sequence $(\tilde{\lambda}_{k, N_p})_k$ with its own stopping time $\tilde{s}$ would be admissible, but would further require coupling two random stopping times at the $\sqrt{B}$ scale. Instead, evaluating the even statistic at the deterministic stage K_{0B} reduces the problem to a single maximal inequality from a fixed center as in Lemma E.10(iv).

Lemma E.14. *Central Limit Theorem for the Scaled Cluster Counts.* Let

$$V_\lambda = \frac{1}{\tau^8} \text{Var} \left(\theta_{N_p}^2 S_{k, N_p}^2 - \sigma^2 M_{k, N_p}^2 \right)$$

finite under Assumption E.9 and strictly positive by assumption. Then, for $\pi \in \{\text{EtC-CS}, \text{ACS}\}$,

$$\Xi_B^\pi \implies N(0, \Lambda_\pi^2), \quad \tilde{\Xi}_B^\pi \implies N(0, \Lambda_\pi^2), \quad \Lambda_\pi^2 := \frac{h'(\lambda_0)^2 V_\lambda}{\nu_\pi}$$

with $\nu_{\text{EtC-CS}} = h(\lambda_M)$ and $\nu_{\text{ACS}} = h(\lambda_0)$.

Proof of Lemma E.14. By Lemma E.10(iii) and strict interiority, both clips are inactive with probability tending to one, so on that event $\hat{\lambda}_{K, N_p} = g(\hat{w}_{K, N_p}^2, \hat{b}_{K, N_p}^2)$ (no clipping) with

$$\nabla g(\sigma^2, \theta_{N_p}^2) = \left(\theta_{N_p}^2 / \tau^4, -\sigma^2 / \tau^4 \right)$$

The pairs $(S_{k, N_p}^2, M_{k, N_p}^2)$ are iid across clusters with finite second moments under Assumption E.9, so the bivariate central limit theorem and the delta method give, along any deterministic $K \rightarrow \infty$, $\sqrt{K}(\hat{\lambda}_{K, N_p} - \lambda_0) \implies N(0, V_\lambda)$ with $V_\lambda = \nabla g' \Sigma \nabla g$ and Σ the covariance of the pair. By Lemma E.10(ii) the feasible and surrogate statistics differ by $O_p(1/B) = o_p(B^{-1/2})$, so they share every $\sqrt{K}$ -scale limit.

For EtC-CS the commitment stage K_M is deterministic with $K_M/B \rightarrow h(\lambda_M)$, giving $\sqrt{B}(\hat{\lambda}_{K_M, N_p} - \lambda_0) \implies N(0, V_\lambda/h(\lambda_M))$. For ACS the commitment stage p_B is random, but $p_B/K_{0B} \xrightarrow{P} 1$ by Step 1 of Proposition E.12 and $K_{0B}/B \rightarrow h(\lambda_0)$. Normalized iid partial sums are uniformly continuous in probability, so Anscombe’s random-index central limit theorem (Anscombe, 1952) applies to the bivariate mean (jointly, via Cramér–Wold) over the random index p_B and the delta method gives $\sqrt{B}(\hat{\lambda}_{p_B, N_p} - \lambda_0) \implies N(0, V_\lambda/h(\lambda_0))$.

Finally, $\Xi_B^\pi = \sqrt{B}(h(\hat{\lambda}_{\cdot, N_p}) - h(\lambda_0)) + O(B^{-1/2})$, with the remainder absorbing the rounding in K_{0B} . One last application of the delta method step through h delivers the limits in the statement. The surrogate shares them because $|\Xi_B^\pi - \tilde{\Xi}_B^\pi| = |K_B^\pi - \tilde{K}_B^\pi|/\sqrt{B} = o_p(1)$, where the numerator uses $|K - \tilde{K}| = O_p(1)$ in EtC-CS and $O_p(B^{1/4})$ in ACS by Propositions E.11 and E.12. ■

Proof of Theorem 6.10. By Propositions E.11 and E.12 both policies are asymptotically even, non-starving and N -consistent. Let $A_B = \{\Xi_B^\pi \in C\}$ for Borel $C \subseteq \mathbb{R}$ with Lebesgue-null boundary and $\mathbb{P}(N(0, \Lambda_\pi^2) \in C) > 0$. Since $\Lambda_\pi^2 > 0$ the limit law is absolutely continuous and assigns zero mass to ∂C , so Lemma E.14 and Remark D.28 guarantee the margin condition (10), and $\mathbb{P}(A_B) \rightarrow \mathbb{P}(N(0, \Lambda_\pi^2) \in C) > 0$. Theorem E.8 delivers both conditional statements. Intersecting A_B

with $\{N_B^\pi = n\}$ is asymptotically innocuous because only $n = N_0$ carries non-vanishing probability under N -consistency, as presented extensively in the scaling discussion. $\square$

Remark E.15. *Reading the Application. Three comments. First, the quantile cells of K_B^π used in Section 7 belong to the relevant class of asymptotically even events: intervals in Ξ_B^π with Lebesgue-null boundary and limiting probability strictly inside $(0, 1)$. Because the cluster design has a single divergent coordinate, the limit law here is a scalar Gaussian and no hyperplane singularity arises, unlike in Remark D.37. Second, the variances of the limiting designs are ranked. Since h is strictly decreasing and $\lambda_0 < \lambda_M$ under strict interiority, $\nu_{ACS} = h(\lambda_0) > h(\lambda_M) = \nu_{ETC-CS}$ and hence $\Lambda_{ACS}^2 < \Lambda_{ETC-CS}^2$. By committing around $K^*(\lambda_0)$ rather than at the worst-case pilot K_M , ACS estimates the variance ratio on a strictly larger sample and induces a strictly less dispersed design in the limit. Third, Anscombe’s theorem enters only for the pilot statistic $\hat{\lambda}_{p_B, N_p}$, where the commitment stage is random. It is silent about the outcome statistic of interest $\hat{\mu}_{K_B^\pi, N_B^\pi}$, whose design-conditional limit is not a random-index problem and is precisely what evenness and symmetry are needed for.*

F Additional Simulation Evidence

F.1 Performance

Figure 13 traces the median and mean terminal design against the the oracle map (3), alongside the true variance ratio. Both algorithms track the oracle closely, with a small tendency to oversample depth and undersample width. This reflects the right skew in $\hat{\lambda}_{p_B}$ documented in Figure 3, since depth is increasing and width decreasing in the imputed ratio. Both algorithms display a similar performance with no clear domination. By definition, the static policies induce the same design throughout.

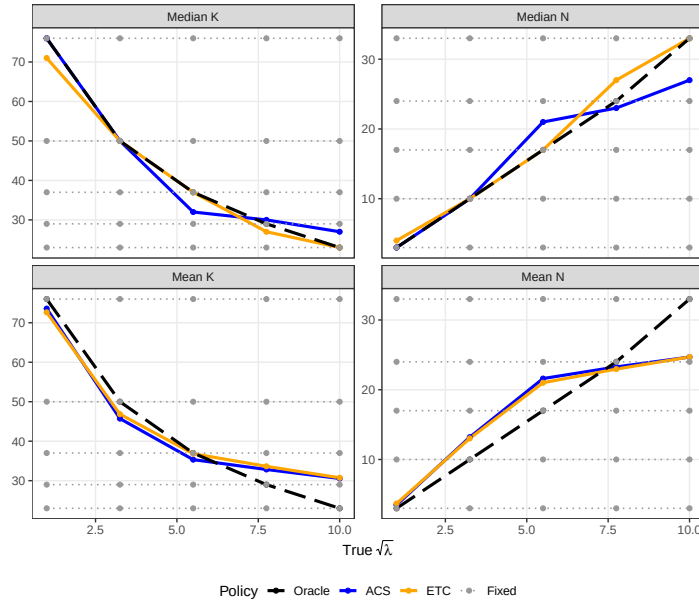


Figure 13: Mean and Median Terminal Design (K_B^π, N_B^π) across λ_0

Notes: Static policies imputed at $\lambda \in \{1, 10, 30, 60, 100\}$. 10^5 replications. $\mu = 0$, $\tau^2 = 1$, $B = 2,000$, $F = 20$, $V = 2$.

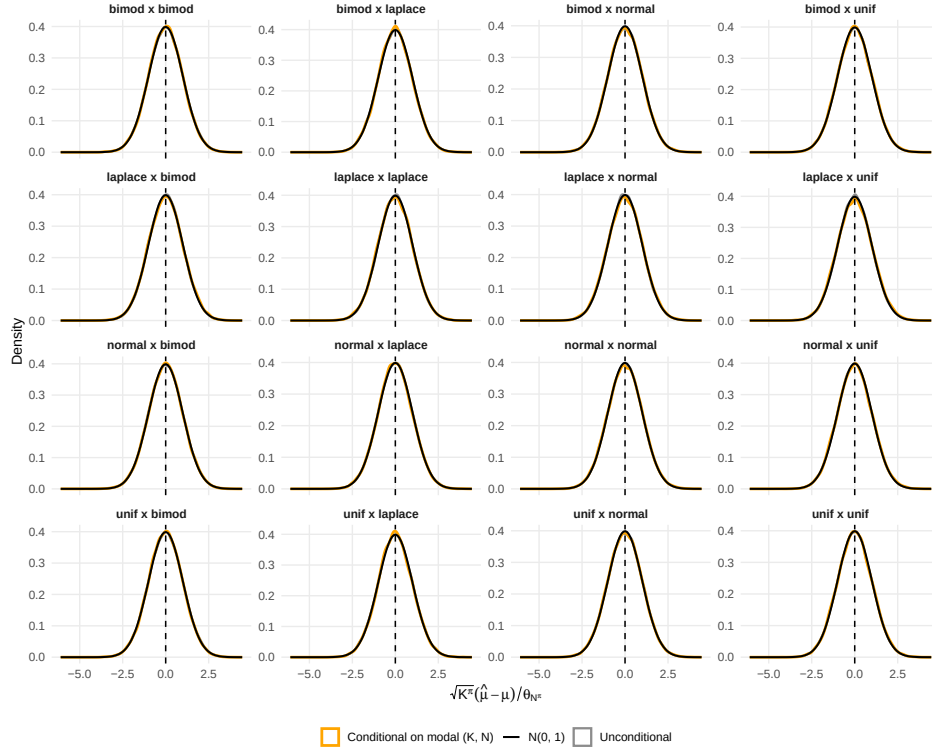


Figure 14: Empirical density of the design-conditional statistic $\sqrt{K_B^\pi}(\hat{\mu}_{K_B^\pi, N_B^\pi} - \mu)/\theta_{N_B^\pi}$ at the modal design, across symmetric error distributions

Notes: Distribution of u_k distribution (left), of ε_{ki} (right). 10^5 replications. $\mu = 0$, $\omega = (1, 5)$. $B = 2,000$, $F = 20$, $V = 2$.

G Appendix to Empirical Application

G.1 Data and Variable Construction

Sampling frame. Fink et al. (2020b) began from a random list of 250 villages—five per administrative block, for 50 of the 53 blocks in Chipata District, Eastern Province, Zambia—with populations reported to comprise between 20 and 100 farming households. After inspection and logistical screening, 175 villages entered the experiment. The smallest of these contained 22 medium-sized farmers, and 22 households were therefore targeted in each village. Realized depth ranged from 5 to 22, averaging 17.9, leading to a baseline sample of 3,139 households, of which 3,028 appear in the year-one survey. Treatment was assigned at the village level after the baseline survey, in three equal arms, by the min-max T -randomization of Bruhn and McKenzie (2009). Details come from Fink et al. (2020b), Fink et al. (2020a), Fink et al. (2024) and Fink et al. (2014), as well as several communications with the authors and the IPA field-implementing team.

Outcome variables. The pre-registration lists harvest value as the primary outcome, while the published paper emphasizes reallocation of informal labor (*ganyu*). I therefore study $\log(1 + \text{total harvest value})$ and $\log(1 + \text{ganyu mandays})$. For the second I use `did_ganyu_mandays_long`, which, despite being available at baseline and in both follow-up rounds, it is not reported in the paper. Instead, FJM report the binary variables `did_ganyu` and `plan_ganyu`. Binary outcomes are not accommodated by the additive-error model and violate the symmetry restriction of Section 6 outright, so a continuous measure is

required. The alternative continuous measures `did_ganyu_hours_yest` and `did_ganyu_hours_short` are unavailable at baseline.

Trimming and residualization. The bottom 0.5% of baseline harvest values (16 observations) is trimmed for implausibly low values. The same absolute threshold (179.5 ZMW) is applied to the year-one data. Both outcomes are then residualized on the covariates of the specification in Fink et al. (2020a), Table 4 Panel B, with three exclusions: the outcome’s own baseline value, the assignment variable `control_gift`, which is not defined pre-assignment, and—for ganyu mandays only—the mechanically related `did_ganyu` and `plan_ganyu`. I take the specification of harvest value for both analyses (Table 5 and Table 6) because ganyu mandays does not enter any of the reported regressions, and because all specifications in the authors’ analysis share nearly identical controls anyway.

G.2 Homogeneity of the Variance Components

Section 8 identifies the design from baseline data alone, so an implicit assumption is imposed all throughout requiring the variance components to be stable across survey rounds. I concretize this simplification in two assumptions

(A1) *Time homogeneity*: $Y_{ki1}(d) \stackrel{d}{=} Y_{ki0}(d)$ for every d .

(A1) is only testable on control units, since $Y_{ki}(1)$ is unobserved at baseline. I therefore test the equality of $(\sigma_0^2, \tau_0^2, \lambda_0)$ across rounds. I run two versions of the test: one using control units in both rounds, which permits a paired bootstrap, and one using all baseline units against year-one controls, which gains precision at the cost of pairing. Secondly,

(A2) *Arm homogeneity at endline*: $(\tau_d^2, \sigma_d^2)_{\text{year } 1} = (\tau^2, \sigma^2)_{\text{year } 1}$ for every d .

(A2) is not required by the framework of Section 3, but it supplies indirect evidence on the untestable half of (A1): if time homogeneity holds for control units and the second moments agree across arms after treatment, it is reasonable to extend time homogeneity to the unobserved $Y_{ki}(1)$. Note that arm homogeneity at baseline holds by randomization, so (A2) allows us to drop subscripts, simplifying the exposition. All tests use a cluster bootstrap on log-ratios.

Figure 15 summarizes the results and Tables 7 and 8 report them in full, with Wald p -values and the number of clusters entering each comparison. Homogeneity is not rejected anywhere except for the time stability of the household-level variance σ^2 , which rises between rounds for both outcomes. This is immaterial for the exercise because the only real input to the algorithms is λ , and time homogeneity of λ cannot be rejected in any specification.

Two qualifications belong on the record. First, $\hat{\tau}^2$ is estimated from roughly 58 villages per arm, so the tests are low-powered against violations in the cluster-level component. Second, a minority of bootstrap draws produced numerically degenerate cluster variances ($\hat{\tau}^2 \leq 10^{-12}$); these draws are discarded uniformly across $\hat{\tau}^2$, $\hat{\sigma}^2$ and $\hat{\lambda}$, and the discarded share, between 0 and 17 percent, is reported in each table.⁵⁵

⁵⁵All estimators ($\hat{\tau}^2, \hat{\sigma}^2, \hat{\lambda}$) use the unclipped expressions of Section 4.1 adapted to unequal village sizes N_k . Using instead the uniform-depth expressions at $N = K^{-1} \sum_k N_k = 17.9$ changes the input estimates negligibly.

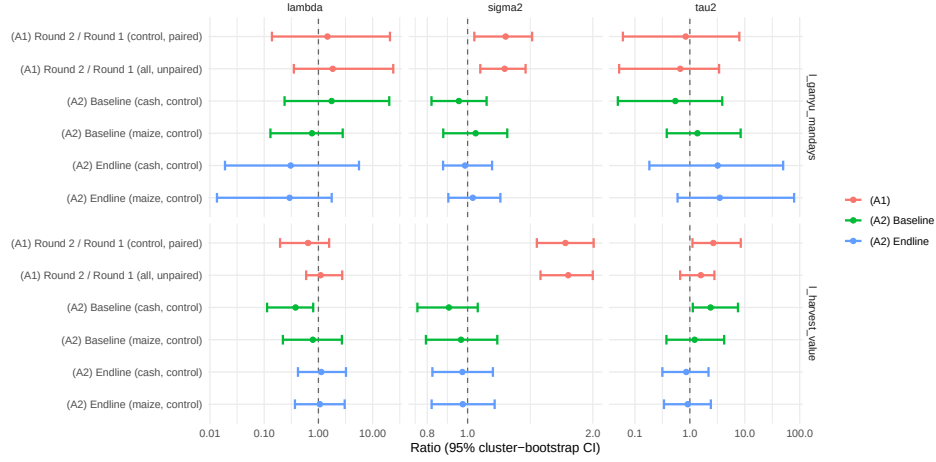


Figure 15: Variance-homogeneity tests

Notes: Cluster bootstrap on log-ratios. Dashed line at 1. 2,000 bootstrap draws.

Outcome	Sample	Estim	Ratio	SE	$CI_{0.025}$	$CI_{0.975}$	p -value	K_1	K_2	Boots	% Exc
l_harvest_value	control	σ^2	1.718	0.080	1.467	2.009	0.000	58	58	1999	0.03
	control	τ^2	2.670	0.512	1.114	8.445	0.055	58	58	1999	0.03
	control	λ	0.644	0.522	0.196	1.574	0.399	58	58	1999	0.03
	all	σ^2	1.745	0.075	1.497	1.999	0.000	58	175	2000	0.00
	all	τ^2	1.587	0.362	0.666	2.798	0.203	58	175	2000	0.00
	all	λ	1.100	0.384	0.594	2.744	0.804	58	175	2000	0.00
l_ganyu_mandays	control	σ^2	1.234	0.078	1.038	1.429	0.007	58	58	1371	16.10
	control	τ^2	0.840	1.188	0.060	7.952	0.884	58	58	1371	16.10
	control	λ	1.469	1.218	0.139	20.869	0.752	58	58	1371	16.10
	all	σ^2	1.228	0.063	1.073	1.378	0.001	58	175	1425	14.37
	all	τ^2	0.669	1.068	0.052	3.403	0.706	58	175	1425	14.37
	all	λ	1.836	1.082	0.352	23.974	0.574	58	175	1425	14.37

Table 7: Time homogeneity of the variance components (A1)

Notes: Bootstrap ratios of year-one to baseline components for the control potential outcome, with 2,000-replication cluster-bootstrap standard errors, percentile intervals and Wald p -values. *Sample* indicates whether the baseline arm is the control villages (paired bootstrap) or all villages (unpaired). K_1, K_2 are village counts, % Exc is the share of replications discarded at the τ^2 boundary.

Outcome	Sample	Estim	Ratio	SE	$CI_{0.025}$	$CI_{0.975}$	p -value	K_1	K_2	Boots	% Exc
1_harvest_value	baseline: cash	σ^2	0.902	0.087	0.757	1.058	0.236	58	58	1993	0.18
	baseline: cash	τ^2	2.383	0.510	1.131	7.511	0.089	58	58	1993	0.18
	baseline: cash	λ	0.378	0.524	0.113	0.802	0.064	58	58	1993	0.18
	baseline: maize	σ^2	0.964	0.097	0.794	1.179	0.709	59	58	1999	0.03
	baseline: maize	τ^2	1.222	0.646	0.374	4.214	0.756	59	58	1999	0.03
	baseline: maize	λ	0.789	0.667	0.221	2.712	0.723	59	58	1999	0.03
	endline: cash	σ^2	0.972	0.082	0.823	1.150	0.727	58	58	2000	0.00
	endline: cash	τ^2	0.860	0.489	0.316	2.186	0.757	58	58	2000	0.00
	endline: cash	λ	1.130	0.511	0.419	3.219	0.811	58	58	2000	0.00
	endline: maize	σ^2	0.974	0.090	0.820	1.161	0.766	59	58	2000	0.00
	endline: maize	τ^2	0.915	0.496	0.338	2.408	0.858	59	58	2000	0.00
	endline: maize	λ	1.064	0.521	0.369	3.052	0.905	59	58	2000	0.00
1_ganyu_mandays	baseline: cash	σ^2	0.953	0.078	0.819	1.111	0.536	58	58	1810	4.85
	baseline: cash	τ^2	0.544	1.075	0.049	3.886	0.572	58	58	1810	4.85
	baseline: cash	λ	1.750	1.086	0.237	20.237	0.606	58	58	1810	4.85
	baseline: maize	σ^2	1.046	0.092	0.874	1.246	0.628	59	58	1962	0.95
	baseline: maize	τ^2	1.373	0.778	0.380	8.404	0.684	59	58	1962	0.95
	baseline: maize	λ	0.762	0.779	0.130	2.803	0.727	59	58	1962	0.95
	endline: cash	σ^2	0.986	0.069	0.873	1.145	0.837	58	58	1355	16.62
	endline: cash	τ^2	3.208	1.370	0.183	49.910	0.395	58	58	1355	16.62
	endline: cash	λ	0.307	1.391	0.019	5.597	0.396	58	58	1355	16.62
	endline: maize	σ^2	1.029	0.074	0.899	1.199	0.699	59	58	1461	13.48
	endline: maize	τ^2	3.500	1.161	0.598	78.969	0.281	59	58	1461	13.48
	endline: maize	λ	0.294	1.179	0.013	1.758	0.299	59	58	1461	13.48

Table 8: Cross-arm homogeneity of the variance components (A2)

Notes: Bootstrap ratios of cash- or maize-arm components to control-arm components at baseline and endline, with 2,000-replication cluster-bootstrap standard errors, percentile intervals and Wald p -values. *Sample* indicates the survey round and treatment arm in the numerator. K_1, K_2 are village counts for the treatment and control arms, % Exc is the share of replications discarded at the τ^2 boundary.

G.3 Error Symmetry

Assumption 6.3 requires the error block $C_k = (u_k, \varepsilon_{k1}, \varepsilon_{k2}, \dots)$ to satisfy $C_k \stackrel{d}{=} -C_k$, which is precisely what the sign-magnitude factorization of Lemma 6.4 uses because the flip must move every coordinate of the block at once. Unfortunately, this assumption is untestable when only cluster average residuals $M_{k,N} = u_k + \bar{\varepsilon}_{k,N}$ are available. Each $M_{k,N}$ is an odd linear functional of C_k , so joint symmetry implies $M_{k,N} \stackrel{d}{=} -M_{k,N}$ at every depth, but the reverse is not true. I am therefore constrained to test a necessary condition. Rejection would falsify Assumption 6.3, but failure to reject is naturally underpowered. Testing across a grid of depths is nonetheless informative. As N grows the symmetry load shifts from the household component toward the village component— $M_{k,N} \rightarrow u_k$ as N diverges, while at small N the household noise dominates. Hence the joint specification test the two error sources in different proportions. In particular, I test at $N \in \{4, 8, 12, 16, 20\}$, restricting each test to villages with $N_k \geq N$ and subsampling N households from each. Figure 16 plots the resulting distributions of the centered residuals $\hat{M}_{k,N} = \hat{\mu}_N^k - \hat{\mu}_{K,N}$ at three depths, which are visibly close to symmetric.⁵⁶

⁵⁶The test is run on residuals rather than on the unobservable $M_{k,N}$. The residual vector inherits symmetry under a global sign flip, but the per-cluster Rademacher flips used to calibrate the null preserve the joint law only up to an

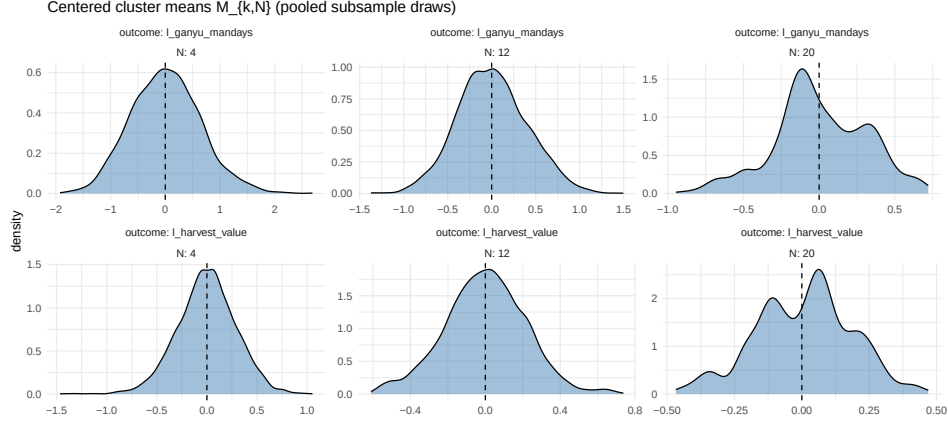


Figure 16: Density of the centered residuals $\hat{M}_{k,N}$

Notes: $N \in \{4, 12, 20\}$. A single random draw of N households per village is plotted (as opposed to the cross-replication density). Only villages with $N_k \geq N$ are included. At $N = 20$ only about 66 villages qualify, which accounts for the coarseness of that panel.

Beyond visual inspection I apply the symmetry test of [Miao et al. \(2006\)](#), calibrating the null by parametric bootstrap. Under symmetry the signs are *iid* Rademacher, so the p -value is read from the empirical distribution of the statistic across sign assignments to the residual magnitudes. Because the choice of N households per village is not unique, I bootstrap over subsampling draws as well and report the dependence-robust aggregate $\min\{1, 2 \times \text{median } p\}$ of [Meinshausen et al. \(2009\)](#). Figure 17 and Table 9 report the results. Symmetry is not rejected for either outcome at any depth, at any conventional level. Although the testing procedure remains limited in nature, taken together, the exercise supports proceeding with standard inference in this application.

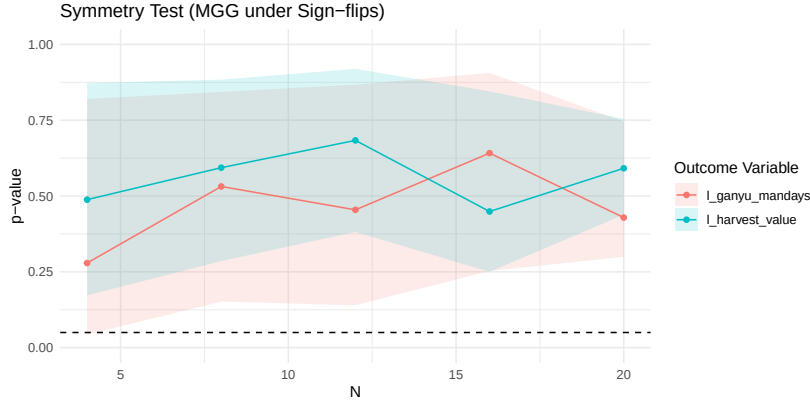


Figure 17: Median p -value of the symmetry test across subsampling draws

Notes: Outer bootstrap over 2,000 sign-flip draws, inner bootstrap over 200 household-selection draws. Ribbon gives the 10th and 90th percentiles of the p -value distribution. The valid bound is twice the median.

$O(1/K)$ term arising from the common demeaning. At $K = 175$ this is immaterial.

Outcome	N	$k : N_k \geq N$	$\text{med}(p)$	Valid p -value	$Q_{0.10}(p)$	$Q_{0.90}(p)$
1_harvest_value	4	175	0.530	1.000	0.110	0.896
	8	174	0.604	1.000	0.229	0.894
	12	166	0.604	1.000	0.358	0.876
	16	141	0.306	0.612	0.141	0.627
	20	66	0.281	0.563	0.212	0.369
1_ganyu_mandays	4	175	0.419	0.839	0.078	0.880
	8	174	0.507	1.000	0.168	0.885
	12	166	0.586	1.000	0.230	0.901
	16	142	0.636	1.000	0.358	0.914
	20	67	0.870	1.000	0.676	0.977

Table 9: Symmetry test of [Miao et al. \(2006\)](#) on the centered residuals $\hat{M}_{k,N}$

Notes: For each depth, villages with $N_k \geq N$ are subsampled to depth N and the statistic is calibrated by a Rademacher sign-flip null. Reported quantities aggregate the sign-flip p -value over subsampling draws. Valid p -value $= \min\{1, 2 \text{med}(p)\}$ is the dependence-robust aggregate of [Meinshausen et al. \(2009\)](#).

G.4 Semi-Synthetic Data Generation

The counterfactual designs of Section 8.3 require a population larger than the realized sample, since any design at the same budget uses either more villages or more households per village than those actually surveyed. I therefore calibrate a semi-synthetic population, separately for each outcome and each survey round (baseline and year 1).

Starting from the realized data, I residualize on the covariates of [Fink et al. \(2020a\)](#), Table 4 Panel B. Year-one data uses the specification exactly. Baseline data omits the outcome’s own lag, the assignment variable `control_gift`, and—for ganyu mandays—`did_ganyu` and `plan_ganyu`, as these variables are either undefined or mechanically related pre-assignment. I then decompose the residuals into an intercept, a village component and a household component, recover $(\hat{\mu}, \hat{\tau}^2, \hat{\sigma}^2)$, and rescale and recenter the two components so that these moments are matched exactly. Synthetic data are generated under the additive model by sampling u_k and ε_{ki} independently and uniformly from the rescaled empirical components. The construction has two limitations. Because it resamples realized residuals it cannot extrapolate beyond their support, so it understates tail mass. Furthermore, because the model is additive it cannot reproduce point masses, so the mass at zero in ganyu mandays is smoothed. For comparison, the tenth percentile is exactly zero in the real data and -0.26 in the synthetic. Tables 10 and 11 compare the synthetic and realized quantile distributions. Away from the zero kink the two track closely.

Outcome	Arm	Source	N	Mean	SD	$Q_{0.10}$	$Q_{0.25}$	$Q_{0.50}$	$Q_{0.75}$	$Q_{0.90}$
1_ganyu_mandays	NA	Simulated	36000	-0.006	1.187	-1.454	-0.839	-0.061	0.766	1.548
		Real	3136	-0.000	1.190	-1.405	-0.856	-0.080	0.793	1.571
1_harvest_value	NA	Simulated	36000	0.004	0.515	-0.638	-0.319	0.026	0.338	0.624
		Real	3123	-0.000	0.513	-0.631	-0.315	0.026	0.328	0.611

Table 10: Validation of the calibrated DGP against the realized baseline sample

Notes: First and second moments are matched by construction.

Outcome	Arm	Source	N	Mean	SD	$Q_{0.10}$	$Q_{0.25}$	$Q_{0.50}$	$Q_{0.75}$	$Q_{0.90}$
1_ganyu_mandays	cash	Simulated	1044	-0.118	1.336	-1.790	-1.114	-0.138	0.792	1.644
		Real	600	-0.182	1.293	-1.921	-1.141	-0.214	0.768	1.499
	control	Simulated	1044	-0.040	1.342	-1.711	-0.997	-0.116	0.892	1.677
		Real	575	0.000	1.305	-1.730	-0.979	-0.060	0.990	1.799
	maize	Simulated	1062	-0.245	1.306	-1.903	-1.229	-0.272	0.667	1.527
		Real	636	-0.210	1.347	-1.983	-1.229	-0.285	0.811	1.561
1_harvest_value	cash	Simulated	1044	0.115	0.643	-0.690	-0.249	0.156	0.517	0.879
		Real	1023	0.112	0.645	-0.737	-0.255	0.172	0.517	0.909
	control	Simulated	1044	-0.040	0.684	-0.884	-0.430	0.013	0.401	0.757
		Real	952	-0.000	0.660	-0.860	-0.414	0.044	0.442	0.791
	maize	Simulated	1062	0.027	0.682	-0.811	-0.404	0.058	0.468	0.895
		Real	1016	0.037	0.657	-0.800	-0.338	0.117	0.475	0.812

Table 11: Validation of the calibrated DGP against the realized year-one sample.

Notes: First and second moments are matched by construction.

H Appendix to Section 9

Assumption H.1. *Assumptions for Heterogeneity Extension Cluster Variances Consider the model with heterogeneous cluster variances in (11)*

(H1) **Errors.** $\mathbb{E}[u_k | X_k] = 0$, $\varepsilon_{ki} \perp\!\!\!\perp X_k$, with $\mathbb{E}[\varepsilon_{ki}] = 0$ and $\mathbb{E}[\varepsilon_{ki}^2] = \sigma^2$; $\mathbb{E}[u_k^2 | X_k] = \tau_k^2 = \ell(X_k, \gamma)$ with $\gamma \in \Gamma$, $\dim(\gamma) < \infty$.

(H2) **Compactness.** $X_k \in [x_m, x_M]$; Γ compact; ℓ continuous and bounded away from zero on $[x_m, x_M] \times \Gamma$; and $\sigma^2 \in [\sigma_m^2, \sigma_M^2]$ with $\sigma_m^2 > 0$. Consequently $\tau_k^2 \in [\tau_m^2, \tau_M^2]$ with $\tau_m^2 > 0$, both globally and conditionally: $\tau_k^2 \in [\tau_{k,m}^2, \tau_{k,M}^2]$ with $\tau_{k,m}^2 = \inf_{\gamma \in \Gamma} \ell(X_k, \gamma)$ and $\tau_{k,M}^2 = \sup_{\gamma \in \Gamma} \ell(X_k, \gamma)$. The implied ratio bounds are $\lambda_m := \sigma_m^2 / \tau_M^2$ and $\lambda_M := \sigma_M^2 / \tau_m^2$, and both aggregate pairs $(\bar{\tau}^2, \bar{\lambda})$ and (τ_h^2, λ_h) lie in the interior of $[\tau_m^2, \tau_M^2] \times [\lambda_m, \lambda_M]$.

(H3) **Interiority.** $\frac{\tau_m^2}{\tau_M^2} \sqrt{\lambda_m F/V} \geq 2$ and $B \geq 2(F + V N_{\max})$ with $N_{\max} = \frac{\tau_M^2}{\tau_m^2} \sqrt{\lambda_M F/V}$.

(H4) **Conditional Subgaussianity.** $u_k | X_k$ is κ_u -subgaussian uniformly in X_k , and ε_{ki} is κ_ε -subgaussian.

Variance of Weighted Averages. Let the weights $(\omega_k)_k$ and design $(K, \{N_k\})$ be X^K -measurable. Under (H1), the cluster average estimator is unbiased for μ — $\mathbb{E}[\hat{\mu}_{N_k}^k | X^K] = \mu$ —, and so it is any weighted combination— $\mathbb{E}[\hat{\mu}_\omega | X^K] = \mu$ —. Therefore mean-squared error minimization reduces to minimizing the expected variance $\mathbb{E}[\text{Var}(\hat{\mu}_\omega | X^K)]$. By the law of total variance, $\mathbb{E}[\text{Var}(\hat{\mu}_\omega | X^K)] = \text{Var}(\hat{\mu}_\omega)$. Conditional on X^K , cluster means are independent with $\text{Var}(\hat{\mu}_{N_k}^k | X^K) = \theta_k^2 := \tau_k^2 + \sigma^2/N_k^k$, so

$$\mathbb{E}[\text{Var}(\hat{\mu}_\omega | X^K)] = \mathbb{E}\left[\frac{1}{(\sum_k \omega_k)^2} \sum_k \omega_k^2 \theta_k^2\right] \quad (18)$$

For weights that depend on the outcomes—such as the feasible weights of Algorithm 3, which plug in estimated variance components—the display is understood conditionally on the design σ -algebra.

Proof of Lemma 9.1 (Unweighted Averages). With $\omega_k = 1/K$, (18) simplifies to $\mathbb{E}[K^{-2} \sum_k \tau_k^2 + \sigma^2 K^{-2} \sum_k 1/N^k]$. Conditional on every X^K , depth does not interact with τ_k^2 , so the rest of the proof tracks closely the generalized problem in Appendix A, where the balanced solution is given by convexity of the map $x \mapsto 1/x$. $N^k = N$ constant with K pinned down by the budget and deterministic. On every realization of X^K the optimal map takes the form

$$N^{k*}(\bar{\lambda}) = N^*(\bar{\lambda}) = \sqrt{\bar{\lambda}F/V}, \quad K^*(\bar{\lambda}) = \frac{B}{F + \sqrt{\bar{\lambda}FV}}$$

where $\bar{\lambda}$ and $\bar{\tau}^2$ are defined in the main text. Plugging-in the optimal maps into the estimator's variance recovers

$$W_u^*(K^*(\bar{\lambda}), N^*(\bar{\lambda})) = \frac{\bar{\tau}^2 \left(\sqrt{\bar{\lambda}V} + \sqrt{F} \right)^2}{B} \quad \square$$

Proof of Corollary 9.2. Under (H1) $(X_k, u_k, (\varepsilon_{ki})_i)$ is iid across k , so unconditionally the cluster errors $M_{k,N}$ are iid with variance $\bar{\tau}^2 + \sigma^2/N$, so the baseline estimators $(\hat{\tau}^2, \hat{\lambda})$ are consistent for $(\bar{\tau}^2, \bar{\lambda})$. The concentration machinery of Appendix C applies verbatim under compactness of the map inputs $(\bar{\tau}^2, \bar{\lambda})$ in (H2) and the (conditional) subexponential guarantees of $M_{k,N}$ in (H6). By Theorem 5.4, EtC-CS therefore attains $\mathbb{E}[\Delta] = O(1/B^2)$ against W_u^* . $\square$

Proof of Proposition 9.3 (GLS Estimator). The proof operates in three steps.

Step 1: the clairvoyant problem. Condition on (K, X^K) and let depths depend on the full vector X^K . This is a relaxation of the adapted class, even for an oracle designer, since feasible depth rules may depend only on $(X_j)_{j \leq k}$, where clusters are indexed by the first time they are explored. Any lower bound for the clairvoyant designer is then a lower bound for the standard oracle designer. By construction of the GLS estimator, $\text{Var}(\hat{\mu}_\omega \mid X^K) = 1/\sum_k \omega_k$. Conditional on (K, X^K) weights are deterministic, so variance minimization is equivalent to maximizing total precision $\sum_k \omega_k = \sum_k N^k/(\tau_k^2 N^k + \sigma^2)$. The objective is strictly increasing in each N^k , so the budget binds and total depth is given by $S = (B - KF)/V$, constant given K . Since $\partial \omega_k / \partial N^k = \sigma^2/(\tau_k^2 N^k + \sigma^2)^2 > 0$ and $\partial^2 \omega_k / \partial N^{2k} < 0$, the first-order conditions equalize $\tau_k^2 N_k + \sigma^2$ across clusters: $N^k = \alpha/\tau_k^2$ for a constant α pinned by the constraint. Solving from the budget constraint recovers the depth profile

$$N^k = \frac{\tau_{h,K}^2}{\tau_k^2} \cdot \bar{N}, \quad \bar{N} = \frac{S}{K} \tag{19}$$

Step 2: exact collapse. At (19), $\tau_k^2 N_k = \bar{N} \tau_{h,K}^2$ for every k , so

$$\sum_k \frac{N^k}{\tau_k^2 N^k + \sigma^2} = \frac{\sum_k N^k}{\bar{N} \tau_{h,K}^2 + \sigma^2} = \frac{\bar{N} K}{\bar{N} \tau_{h,K}^2 + \sigma^2} \implies \text{Var}(\hat{\mu}_\omega \mid K, X^K) = \frac{1}{K} \left(\tau_{h,K}^2 + \frac{\sigma^2}{\bar{N}} \right)$$

Thus the problem of the clairvoyant designer mimics the baseline objective with the realized harmonic

mean in place of τ^2 and $\bar{N}$ in place of N . Minimizing over $\bar{N}$ gives

$$\bar{N}^* = \sqrt{\lambda_{h,K} \frac{F}{\bar{V}}}, \quad \lambda_{h,K} = \sigma^2 / \tau_{h,K}^2$$

Step 3: lower bound. Pathwise, the budget gives $1/K \geq (F + V\bar{N})/B$, hence

$$\frac{1}{K} \left(\tau_{h,K}^2 + \frac{\sigma^2}{\bar{N}} \right) \geq \frac{(F + V\bar{N})(\tau_{h,K}^2 + \sigma^2/\bar{N})}{B} \geq \frac{(\sqrt{F\tau_{h,K}^2} + \sqrt{V\sigma^2})^2}{B}$$

where the last step minimizes over $\bar{N}$ (attained at $\bar{N}^*$). This chain holds realization by realization for any feasible policy, adapted or clairvoyant. Summands $1/\tau_k^2$ are iid and bounded under (H2), and the maps $y \mapsto 1/y$ and $y \mapsto 1/\sqrt{y}$ are convex, so Jensen gives $\mathbb{E}[\tau_{h,K}^2] \geq \tau_h^2$ and $\mathbb{E}[\tau_{h,K}] \geq \tau_h$, where expectations are taken at a deterministic K . It follows that

$$\mathbb{E} \left[\frac{1}{K} \left(\tau_{h,K}^2 + \frac{\sigma^2}{\bar{N}} \right) \right] \geq W_\omega^*$$

Step 4: attainment. Under the pseudo-oracle maps (14) with width adjusted to exhaust the budget along the realized covariate path, the realized objective is (13) evaluated within $O_p(\sqrt{B})$ of $(K_\omega^*, \bar{N}_\omega^*)$. Additionally, a second-order expansion under (H2) delivers $\mathbb{E}[\tau_{h,K}^2] = \tau_h^2 + O(1/K)$ and $\mathbb{E}[\tau_{h,K}] = \tau_h + O(1/K)$, so both Jensen gaps contribute $O(1/K)$. With the realized width linear in B , and the fixed $O(1/B)$ scale entering the variance recovers the $O(1/B^2)$ variation. First-order stationarity of the collapsed objective at the oracle absorbs the $O_p(\sqrt{B})$ width deviations at the same order. $\square$

Proof of Proposition 9.4. Expanding $W_u^* = \bar{\tau}^2(\sqrt{V\bar{\lambda}} + \sqrt{F})^2/B$ using $\bar{\tau}^2\bar{\lambda} = \sigma^2$ and $\bar{\tau}^2\sqrt{V\bar{\lambda}} = \bar{\tau}\sqrt{V\sigma^2}$,

$$B \cdot W_u^* = V\sigma^2 + 2\sqrt{VF\sigma^2}\bar{\tau} + F\bar{\tau}^2, \quad B \cdot W_\omega^* = V\sigma^2 + 2\sqrt{VF\sigma^2}\tau_h + F\tau_h^2$$

and differencing gives (15). Nonnegativity of both terms follows from the arithmetic–harmonic mean inequality, $\bar{\tau}^2 \geq \tau_h^2$, with equality if and only if τ_k^2 is degenerate. Both gaps are increasing in the dispersion of τ_k^2 in the usual convex-order sense. $\square$